

Electrodynamics

Problem Set Solutions – DRAFT

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1.1. Vector calculus

In this problem, we recall a number of standard identities of vector calculus which we will frequently use in electrodynamics.

Definitions/conventions: We commonly write the well-known vectorial differentiation operators grad, div, rot using the vector $\vec{\partial}$ of partial derivatives $\partial_i := \partial/\partial x_i$ as

$$\text{grad } F = \vec{\partial}F, \quad \text{div } \vec{A} = \vec{\partial} \cdot \vec{A}, \quad \text{rot } \vec{A} = \vec{\partial} \times \vec{A}. \quad (1.1)$$

The components of a three-dimensional vector product $\vec{a} \times \vec{b}$ are given by

$$(\vec{a} \times \vec{b})_i = \sum_{j,k=1}^3 \varepsilon_{ijk} a_j b_k. \quad (1.2)$$

Here, ε_{ijk} is the totally anti-symmetric tensor in $\mathbb{R}^3$ with $\varepsilon_{123} = +1$.

a) Show that

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} \quad \text{and} \quad \sum_{i,j=1}^3 \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ijl} = \delta_{kl}. \quad (1.3)$$

Solution: The sum $\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm}$ gets a non-vanishing contribution only if $i \neq j \neq k \neq i$ and $i \neq l \neq m \neq i$. For the case $j = l \wedge k = m$ we have:

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} \sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ijk} = \delta_{jl} \delta_{km} \sum_{i=1}^3 \varepsilon_{ijk}^2 = \delta_{jl} \delta_{km}, \quad (1.S1)$$

For the case $j = m \wedge k = l$ we have:

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jm} \delta_{kl} \sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ikj} = \delta_{jm} \delta_{kl} \sum_{i=1}^3 -\varepsilon_{ijk}^2 = -\delta_{jm} \delta_{kl}, \quad (1.S2)$$

Therefore

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}. \quad (1.S3)$$

The sum $\sum_{i,j=1}^3 \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ijl}$ gets a non-vanishing contribution only if $i \neq j \neq k \neq i$ and at the same time $i \neq j \neq l \neq i$, which implies that the sum vanishes for $k \neq l$. Then

$$\sum_{i,j=1}^3 \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ijl} = \delta_{kl} \frac{1}{2} \sum_{i,j=1}^3 \varepsilon_{ijk}^2 = \delta_{kl} \frac{1}{2} (1^2 + (-1)^2) = \delta_{kl}. \quad (1.S4)$$

b) Show the following identities for arbitrary vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$:

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b}), \\ \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}, \\ (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) &= (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}). \end{aligned} \quad (1.4)$$

Solution: We can show these by relying on the identities of part a):

$$\begin{aligned}
\vec{a} \cdot (\vec{b} \times \vec{c}) &= \sum_{i=1}^3 a_i \sum_{j,k=1}^3 \varepsilon_{ijk} b_j c_k = \sum_{i,j,k=1}^3 \varepsilon_{ijk} a_i b_j c_k \\
&= \sum_{i,j,k=1}^3 \varepsilon_{jki} b_j c_k a_i = \vec{b} \cdot (\vec{c} \times \vec{a}) \\
&= \sum_{i,j,k=1}^3 \varepsilon_{kij} c_k a_i b_j = \vec{c} \cdot (\vec{a} \times \vec{b}), \\
(\vec{a} \times (\vec{b} \times \vec{c}))_i &= \sum_{j,k=1}^3 \varepsilon_{ijk} a_j (\vec{b} \times \vec{c})_k = \sum_{j,k=1}^3 \varepsilon_{ijk} a_j \sum_{l,m=1}^3 \varepsilon_{klm} b_l c_m = \sum_{j,l,m=1}^3 \sum_{k=1}^3 \underbrace{\varepsilon_{kij} \varepsilon_{klm}}_{=\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}} a_j b_l c_m \\
&= b_i \sum_{j,m=1}^3 \delta_{jm} a_j c_m - c_i \sum_{j,l=1}^3 \delta_{jl} a_j b_l = ((\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c})_i, \\
(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) &= \sum_{i=1}^3 \sum_{j,k=1}^3 \varepsilon_{ijk} a_j b_k \sum_{l,m=1}^3 \varepsilon_{ilm} c_l d_m = \sum_{j,k,l,m=1}^3 \underbrace{\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{ilm}}_{=\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}} a_j b_k c_l d_m \\
&= \sum_{j,l=1}^3 \delta_{jl} a_j c_l \sum_{k,m=1}^3 \delta_{km} b_k d_m - \sum_{j,m=1}^3 \delta_{jm} a_j d_m \sum_{k,l=1}^3 \delta_{kl} b_k c_l = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}).
\end{aligned} \tag{1.S5}$$

c) Prove the following identities for arbitrary scalar fields F and vector fields $\vec{A}$:

$$\begin{aligned}
\vec{\partial} \times (\vec{\partial} F) &= 0, \\
\vec{\partial} \cdot (\vec{\partial} \times \vec{A}) &= 0, \\
\vec{\partial} \times (\vec{\partial} \times \vec{A}) &= \vec{\partial}(\vec{\partial} \cdot \vec{A}) - \Delta \vec{A}, \\
\vec{\partial} \cdot (F \vec{A}) &= (\vec{\partial} F) \cdot \vec{A} + F \vec{\partial} \cdot \vec{A}.
\end{aligned} \tag{1.5}$$

Solution: Use some of the previous identities as well as symmetry properties:

$$\begin{aligned}
(\vec{\partial} \times (\vec{\partial} F))_i &= \sum_{j,k=1}^3 \underbrace{\varepsilon_{ijk}}_{\text{alt.}} \underbrace{\partial_j \partial_k}_{\text{symm.}} F = \sum_{j,k=1}^3 (-\varepsilon_{ikj}) \partial_k \partial_j F \\
&= - \sum_{j \leftrightarrow k}^3 \varepsilon_{ijk} \partial_j \partial_k F = -(\vec{\partial} \times (\vec{\partial} F))_i.
\end{aligned} \tag{1.S6}$$

Therefore $\vec{\partial} \times (\vec{\partial} F) = 0$.

$$\begin{aligned}
\vec{\partial} \cdot (\vec{\partial} \times \vec{A}) &= \sum_{i=1}^3 \partial_i \sum_{j,k=1}^3 \varepsilon_{ijk} \partial_j A_k = \sum_{i,j,k=1}^3 \underbrace{\varepsilon_{ijk}}_{\text{alt.}} \underbrace{\partial_i \partial_j}_{\text{symm.}} A_k = \sum_{i,j,k=1}^3 (-\varepsilon_{jik}) \partial_j \partial_i A_k \\
&\stackrel{i \leftrightarrow j}{=} - \sum_{i=1}^3 \partial_i \sum_{j,k=1}^3 \varepsilon_{ijk} \partial_j A_k = -\vec{\partial} \cdot (\vec{\partial} \times \vec{A})
\end{aligned} \tag{1.S7}$$

Therefore $\vec{\partial} \cdot (\vec{\partial} \times \vec{A}) = 0$.

$$\begin{aligned}
 (\vec{\partial} \times (\vec{\partial} \times \vec{A}))_i &= \sum_{j,k=1}^3 \varepsilon_{ijk} \partial_j (\vec{\partial} \times \vec{A})_k = \sum_{j,k,l=1}^3 \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l A_m = \sum_{j,l,m=1}^3 \underbrace{\sum_{k=1}^3 \varepsilon_{kij} \varepsilon_{klm}}_{=\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}} \partial_j \partial_l A_m \\
 &= \partial_i \sum_{j,m=1}^3 \delta_{jm} \partial_j A_m - \underbrace{\sum_{j,m=1}^3 \delta_{jm} \partial_j \partial_m A_i}_{\Delta} = (\vec{\partial}(\vec{\partial} \cdot \vec{A}) - \Delta \vec{A})_i, \\
 \vec{\partial} \cdot (F \vec{A}) &= \sum_{i=1}^3 \partial_i (F A_i) = \sum_{i=1}^3 ((\partial_i F) A_i + F (\partial_i A_i)) = (\vec{\partial} F) \cdot \vec{A} + F \vec{\partial} \cdot \vec{A}.
 \end{aligned} \tag{1.S8}$$

1.2. Gauß's theorem

Consider the following vector fields $\vec{A}_i$ in two dimensions

$$\begin{aligned}
 \vec{A}_1 &= (3xy(y-x), x^2(3y-x)), \\
 \vec{A}_2 &= (x^2(3y-x), 3xy(x-y)), \\
 \vec{A}_3 &= (x/(x^2+y^2), y/(x^2+y^2)) = \vec{x}/\|\vec{x}\|^2.
 \end{aligned} \tag{1.6}$$

- a) Compute the flux of $\vec{A}_i$ through the boundary of the square Q with corners $\vec{x} = (\pm 1, \pm 1)$

$$I_i = \oint_{\partial Q} dx \, \vec{n} \cdot \vec{A}_i. \tag{1.7}$$

Solution: We compute

$$\begin{aligned}
 I_i &= \int_{-1}^1 dx (\vec{e}_y \cdot \vec{A}_i(x, 1) - \vec{e}_y \cdot \vec{A}_i(x, -1)) + \int_{-1}^1 dy (\vec{e}_x \cdot \vec{A}_i(1, y) - \vec{e}_x \cdot \vec{A}_i(-1, y)), \\
 I_1 &= \int_{-1}^1 dx (3x^2 - x^3 + 3x^2 + x^3) + \int_{-1}^1 dy (3y^2 - 3y + 3y^2 + 3y) = 2 \cdot 6 \int_{-1}^1 dx x^2 = 8, \\
 I_2 &= \int_{-1}^1 dx (3x^2 - 3x + 3x^2 + 3x) + \int_{-1}^1 dy (3y - 1 - 3y - 1) = 4 - 4 = 0, \\
 I_3 &= \int_{-1}^1 dx \frac{1 - (-1)}{(x^2 + 1)} + \int_{-1}^1 dy \frac{1 - (-1)}{(1 + y^2)} = 4 \int_{-1}^1 dx \frac{1}{(1 + x^2)} = 2\pi.
 \end{aligned} \tag{1.S9}$$

- b) Compute the divergence of $\vec{A}_i$ and its integral over the area of this square Q

$$I'_i = \int_Q dx^2 \, \vec{\partial} \cdot \vec{A}_i. \tag{1.8}$$

Solution: We compute

$$\begin{aligned}
 \vec{\partial} \cdot \vec{A}_i &= \partial_x A_{i,x} + \partial_y A_{i,y}, \\
 \vec{\partial} \cdot \vec{A}_1 &= 3(y^2 - 2xy + x^2) = 3(x-y)^2, \\
 \vec{\partial} \cdot \vec{A}_2 &= 6xy - 3x^2 + 3x^2 - 6xy = 0, \\
 \text{for all } (x, y) \neq (0, 0) : \vec{\partial} \cdot \vec{A}_3 &= \frac{x^2 + y^2 - 2x^2 + x^2 + y^2 - 2y^2}{(x^2 + y^2)^2} = 0.
 \end{aligned} \tag{1.S10}$$

The result for $\vec{\partial} \cdot \vec{A}_3$ holds for $(x, y) \neq (0, 0)$. Review: Students have struggles with solving this part of the exercise. They just conclude that the integral is zero, since the integrand is zero. Perhaps, some hints would be useful here. I guess a point of the exercise is to precisely notice the resulting discrepancy with Gauß's theorem. It's not that they need to get this right in the first place.

At $(0, 0)$ the divergence $\vec{\partial} \cdot \vec{A}_3$ can be inferred from Gauß's theorem in two dimensions,

$$\int_Q dx^2 \vec{\partial} \cdot \vec{A} = \oint_{\partial Q} dx \vec{n} \cdot \vec{A}. \quad (1.S11)$$

An arbitrary area containing $(0, 0)$ can be chosen, e.g. a (unit) circle around the origin (or the square of this exercise, $I_3 = 2\pi$), which yields

$$\int_C dx^2 \vec{\partial} \cdot \vec{A}_3 = \oint_{\partial C} dx \vec{n} \cdot \vec{A}_3 = \int_0^{2\pi} d\varphi \vec{x} \cdot \frac{\vec{x}}{\|\vec{x}\|^2} = \int_0^{2\pi} d\varphi = 2\pi. \quad (1.S12)$$

As the only non-vanishing contribution to $\int_Q dx^2 \vec{\partial} \cdot \vec{A}_3$ comes from the point $(0, 0)$, we have to conclude

$$\vec{\partial} \cdot \vec{A}_3 = 2\pi\delta^2(x). \quad (1.S13)$$

The integrals then evaluate to

$$\begin{aligned} I'_1 &= \int_{-1}^1 dx \int_{-1}^1 dy \vec{\partial} \cdot \vec{A}_1, \\ I'_1 &= \int_{-1}^1 dx \int_{-1}^1 dy 3(x^2 - 2xy + y^2) = 3 \int_{-1}^1 dx \left(2x^2 + \frac{2}{3} \right) = 3 \left(\frac{4}{3} + \frac{4}{3} \right) = 8, \\ I'_2 &= 0, \\ I'_3 &= \int_{-1}^1 dx \int_{-1}^1 dy 2\pi\delta^2(x) = 2\pi. \end{aligned} \quad (1.S14)$$

1.3. Stokes' theorem

Consider the vector field

$$\vec{A} = (x^2y, x^3 + 2xy^2, xyz). \quad (1.9)$$

- a) Compute the integral along the circle S around the origin in the x, y -plane with radius R

$$I = \oint_S d\vec{x} \cdot \vec{A}. \quad (1.10)$$

Solution: Let $\vec{x} = (x, y, z) = R(\cos \varphi, \sin \varphi, 0)$. Therefore $d\vec{x} = R(-\sin \varphi, \cos \varphi, 0) d\varphi$. Then

$$d\vec{x} \cdot \vec{A} = R^4(-\cos^2 \varphi \sin^2 \varphi + \cos^4 \varphi + 2 \cos^2 \varphi \sin^2 \varphi) = R^4 \cos^2 \varphi. \quad (1.S15)$$

The integral is $I = \int_0^{2\pi} d\varphi R^4 \cos^2 \varphi = \pi R^4$.

- b) Compute the rotation $\vec{B}$ of the vector field $\vec{A}$

$$\vec{B} = \vec{\partial} \times \vec{A}. \quad (1.11)$$

Solution: We obtain by direct calculation $\vec{\partial} \times \vec{A} = (xz, -yz, 2x^2 + 2y^2)$.

- c) Compute the flux of the rotation $\vec{B}$ through the disk S whose boundary is S , $\partial D = S$

$$I' = \int_D dx^2 \vec{n} \cdot \vec{B}. \quad (1.12)$$

Solution: The normal to the disk is $\vec{n} = (0, 0, 1)$ and we obtain the integral

$$I' = \int_0^R r \, dr \int_0^{2\pi} d\varphi \, 2r^2 = \frac{2}{3} R^4 (2\pi) = \pi R^4. \quad (1.S16)$$

1.4. Potential and electric field strength

Four point charges are placed at the corners $(a, 0)$, (a, a) , $(0, a)$, $(0, 0)$ of a square. Find the potential and the electric field strength in the plane of this square. Sketch the field lines and equipotential lines of the following charge distributions:

- a) $+q, +q, +q, +q$;
- b) $-q, +q, -q, +q$;
- c) $+q, +q, -q, -q$.

Hint: Using Mathematica, the commands `ContourPlot` and `StreamPlot` might come in handy.

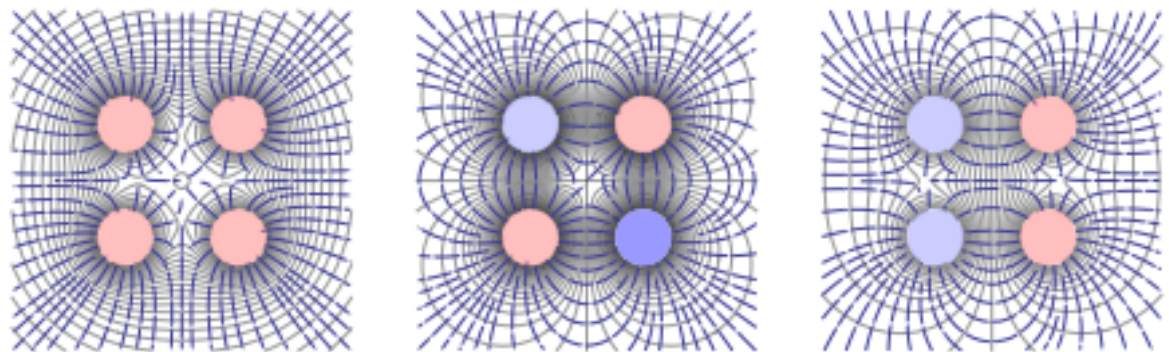
Solution: label of solution wrong: applies to all parts The potential of the four point charges q_i is given by

$$\Phi(x, y) = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{\sqrt{x^2 + (y-a)^2}} + \frac{q_2}{\sqrt{(x-a)^2 + (y-a)^2}} + \frac{q_3}{\sqrt{(x-a)^2 + y^2}} + \frac{q_4}{\sqrt{x^2 + y^2}} \right]. \quad (1.S17)$$

The electric field strength follows easily from $\vec{E} = -\vec{\nabla}\Phi$,

$$\vec{E}(x, y) = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 \begin{pmatrix} x \\ y-a \end{pmatrix}}{[x^2 + (y-a)^2]^{3/2}} + \frac{q_2 \begin{pmatrix} x-a \\ y-a \end{pmatrix}}{[(x-a)^2 + (y-a)^2]^{3/2}} + \frac{q_3 \begin{pmatrix} x-a \\ y \end{pmatrix}}{[(x-a)^2 + y^2]^{3/2}} + \frac{q_4 \begin{pmatrix} x \\ y \end{pmatrix}}{[x^2 + y^2]^{3/2}} \right]. \quad (1.S18)$$

The electric field strength follows easily from $\vec{E} = -\vec{\nabla}\Phi$. It can be visualised as follows in the three cases part a), part b), part c):



(1.S19)

In Mathematica, we define the potential of a point charge and then the potentials for the three cases. (q and a are set to one)

```
PointPot[x_, y_, q_] = q/Sqrt[x^2 + y^2];
Pot[a][x_, y_] = PointPot[x, y - 1, 1] + PointPot[x - 1, y - 1, 1] +
  PointPot[x - 1, y, 1] + PointPot[x, y, 1];
Pot[b][x_, y_] = PointPot[x, y - 1, -1] + PointPot[x - 1, y - 1, 1] +
  PointPot[x - 1, y, -1] + PointPot[x, y, 1];
Pot[c][x_, y_] = PointPot[x, y - 1, -1] + PointPot[x - 1, y - 1, 1] +
  PointPot[x - 1, y, 1] + PointPot[x, y, -1];
```

The electric field is defined as simple as

```
EField[ex_][x_, y_] = -Map[D[Pot[ex][x, y], #] &, {x, y}];
```

The plots are finally realized as

```
plot[ex_] := Show[
  ContourPlot[ Pot[ex][x, y], {x, -1, 2}, {y, -1, 2},
    {ContourShading -> None, Contours -> 40}],
  StreamPlot[EField[ex][x, y], {x, -1, 2}, {y, -1, 2}]];
plot[a]
plot[b]
plot[c]
```

2.1. Stable equilibrium

Two balls, each with charge $+q$, are placed on an insulating plate within the $z = 0$ plane where they can move freely without friction. Under the plate, a third ball with charge $-2q$ is fixed at $\vec{x} = (0, 0, -b)$. Treat the balls as point charges, and find stable positions for the two balls on the plate.

Solution: Due to symmetry a stable position is only achieved if the balls are at equal distance from the z -axis. W.l.o.g. we can assume that they are placed at $\vec{x} = (\pm a, 0, 0)$. Now we only need to determine the distance a . The potential created by the charges at $(-a, 0, 0)$ and $(0, 0, -b)$ is given by

$$\Phi(x, y, z) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{(x+a)^2 + z^2 + y^2}} - \frac{2}{\sqrt{x^2 + y^2 + (z+b)^2}} \right]. \quad (2.S1)$$

We obtain the electric field strength

$$\vec{E}(x, y, z) = -\vec{\nabla}\Phi = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{[(x+a)^2 + y^2 + z^2]^{3/2}} \begin{pmatrix} x+a \\ y \\ z \end{pmatrix} - \frac{2}{[x^2 + y^2 + (z+b)^2]^{3/2}} \begin{pmatrix} x \\ y \\ z+b \end{pmatrix} \right]. \quad (2.S2)$$

Therefore, the third ball at $\vec{x} = (a, 0, 0)$ experiences the force

$$q\vec{E}(a, 0, 0) = -\frac{q^2}{4\pi\epsilon_0} \left[-\frac{1}{[(2a)^2]^{3/2}} \begin{pmatrix} 2a \\ 0 \\ 0 \end{pmatrix} + \frac{2}{[a^2 + b^2]^{3/2}} \begin{pmatrix} a \\ 0 \\ b \end{pmatrix} \right]. \quad (2.S3)$$

In the z -direction, the force is compensated by the plate. The condition that the force vanishes in the x -direction yields the stable position

$$4a^2 = a^2 + b^2 \implies a = \frac{1}{\sqrt{3}} b. \quad (2.S4)$$

2.2. Energy stored in a parallel plate capacitor

Two plates with charges $+Q$ and $-Q$ and area A are placed parallel to each other at a (small) distance d . Find the energy stored in the electric field between them.

Solution: First we calculate the electric field strength E caused by a single plate with charge density σ . Let the plate be located in the $z := x_3 = 0$ plane. We integrate a thin rectangular box B with base area A parallel to the plate. We omit boundary effects and assume that the field is approximately constant on both sides of the plate. Due to symmetry, the electric field is perpendicular to the plate and points to either side in opposite directions (for a finite plate this follows from the ordinary boundary conditions at $x \rightarrow \infty$). Gauß's theorem

$$\oint_{\partial B} dx^2 \vec{n}(x) \cdot \vec{E}(x) = A \Delta E_z = \int_B dx^3 \frac{\rho(x)}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \quad (2.S5)$$

then determines the electric field of the single plate

$$\vec{E}(x) = \frac{\sigma}{2\epsilon_0} \begin{cases} +\vec{e}_z, & z > 0, \\ -\vec{e}_z, & z < 0. \end{cases} \quad (2.S6)$$

The electric field of the parallel plate capacitor is obtained by superposition

$$\vec{E}(x) = \frac{\sigma}{\epsilon_0} \begin{cases} 0, & z > d, \\ \vec{e}_z, & d > z > 0, \\ 0, & 0 > z. \end{cases} \quad (2.S7)$$

The energy is now calculated as

$$W = \frac{\epsilon_0}{2} \int dx^3 \|\vec{E}(x)\|^2 = \frac{\sigma^2}{\epsilon_0^2} \frac{\epsilon_0}{2} Ad = \frac{1}{2} \frac{d}{\epsilon_0 A} Q^2. \quad (2.S8)$$

2.3. Electric field strength in a hollow sphere

A charged ball with homogeneous charge density ρ and radius R_A contains a spherical cavity with radius R_I that is shifted from the centre by the vector $\vec{a}$ with $\|\vec{a}\| < R_A - R_I$. Calculate the electric field strength within the cavity.

Hint: Use Gauß's theorem and the superposition principle to calculate the field strength.

Solution: Due to the superposition principle, the situation is equivalent to a ball without cavity superposed with a smaller ball of the size of the cavity but with opposite charge density $-\rho$. Therefore we first calculate the electric field strength inside a ball with homogeneous charge density ρ and radius R but without cavity. We apply Gauß's theorem to the ball K with radius $r < R$ inside the ball and obtain

$$\oint_{\partial K} dx^2 \vec{n}(x) \cdot \vec{E}(x) = 4\pi r^2 \|\vec{E}(r)\| = \frac{1}{\epsilon_0} \int_K dx^3 \rho(x) = \frac{\rho}{\epsilon_0} \int_0^r 4\pi s^2 ds = \frac{4\pi}{\epsilon_0} \rho \frac{r^3}{3}. \quad (2.S9)$$

Inside a ball with homogeneous charge density centred at the origin we therefore find the electric field

$$\vec{E}_K(x) = \frac{\rho}{3\epsilon_0} \vec{x} \quad \text{for } \|x\| < R. \quad (2.S10)$$

Now we can calculate the electric field strength inside the cavity, using the superposition principle

$$\vec{E}(x) = \vec{E}_K(x) - \vec{E}_{K'}(x - a) = \frac{\rho}{3\epsilon_0} (\vec{x} - (\vec{x} - \vec{a})) = \frac{\rho}{3\epsilon_0} \vec{a}. \quad (2.S11)$$

Inside the cavity, the field is constant and proportional to the eccentricity of the cavity.

2.4. delta-function

The delta-function is often used to describe the charge density of a point charge. It is defined through the following property when integrated over a smooth test function f with compact support:

$$\int_{-\infty}^{\infty} dx f(x) \delta(x - a) = f(a). \quad (2.1)$$

a) Show that it can be written as the limit $\lim_{\epsilon \rightarrow 0} g_\epsilon(x) = \delta(x)$, where g_ϵ is defined as

$$g_\epsilon(x) = \frac{1}{\sqrt{2\pi\epsilon}} e^{-x^2/(2\epsilon)}. \quad (2.2)$$

Solution: In the context of distributions we have to show the following for suitable test functions $f(x)$:

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dx f(x) g_\epsilon(x) &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dx f(x) \frac{1}{\sqrt{2\pi\epsilon}} e^{-x^2/(2\epsilon)} \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dy f(\sqrt{\epsilon} y) \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \\ &= \int_{-\infty}^{\infty} dy \lim_{\epsilon \rightarrow 0} f(\sqrt{\epsilon} y) \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \\ &= f(0) \int_{-\infty}^{\infty} dy \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \\ &= f(0). \end{aligned} \quad (2.S12)$$

b) Show also that $\lim_{\epsilon \rightarrow 0} h_\epsilon(x) = \delta(x)$, where h_ϵ is defined as

$$h_\epsilon(x) = \frac{1}{2\pi i} \left(\frac{1}{x - i\epsilon} - \frac{1}{x + i\epsilon} \right). \quad (2.3)$$

Solution: For suitable test functions $f(x)$ we show:

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dx f(x) h_\epsilon(x) &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dx f(x) \frac{1}{2\pi i} \left(\frac{1}{x - i\epsilon} - \frac{1}{x + i\epsilon} \right) \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dx f(x) \frac{1}{2\pi i} \frac{2i\epsilon}{x^2 + \epsilon^2} \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dy f(\epsilon y) \frac{1}{2\pi i} \frac{2i\epsilon^2}{\epsilon^2(y^2 + 1)} \\ &= \int_{-\infty}^{\infty} dy \lim_{\epsilon \rightarrow 0} f(\epsilon y) \frac{1}{\pi} \frac{1}{y^2 + 1} \\ &= f(0) \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{dy}{y^2 + 1} \\ &= f(0) \frac{1}{\pi} \arctan(y) \Big|_{-\infty}^{+\infty} = f(0). \end{aligned} \quad (2.S13)$$

c) Show, that the derivative of the delta-function satisfies

$$\int_{-\infty}^{\infty} dx f(x) \delta'(x - a) = -f'(a). \quad (2.4)$$

The properties of test functions ensure that there are no asymptotic contributions (boundary terms).

Solution: This property follows immediately from integration by parts:

$$\int_{-\infty}^{\infty} dx f(x) \delta'(x - a) = - \int_{-\infty}^{\infty} dx f'(x) \delta(x - a) = -f'(a). \quad (2.S14)$$

2.5. Green functions in one dimension

Consider the region of space between two conducting, grounded, infinitely extended parallel plates that lie at $x = 0$ and $x = d$. Suppose that a third plate with uniform charge density σ is placed at $x = a$ where $0 < a < d$.

a) Show that finding the potential $\Phi(\vec{x})$ for $0 \leq x \leq d$ is equivalent to computing a Green function in one dimension, i.e. solving the equation

$$\frac{\partial^2}{\partial x^2} G(x, a) = -\delta(x - a) \quad (2.5)$$

with Dirichlet boundary conditions $G(0, a) = G(d, a) = 0$.

Solution: Given that the problem is invariant with respect to translations in the y, z -plane, we have that the potential can be a function of the x -coordinate only, $\Phi(\vec{x}) = \Phi(x)$. The charge density of the plate is described by $\rho(\vec{x}) = \sigma \delta(x - a)$, thus the equation for the potential (in 3D) reads

$$\Delta \Phi(\vec{x}) = -\frac{\rho(\vec{x})}{\epsilon_0} = -\frac{\sigma}{\epsilon_0} \delta(x - a), \quad (2.S15)$$

which reduces to

$$\frac{\partial^2}{\partial x^2} \Phi(x) = -\frac{\sigma}{\epsilon_0} \delta(x - a). \quad (2.S16)$$

Making the dependence on a explicit and reabsorbing the constant by setting

$$\Phi(x) = \frac{\sigma}{\varepsilon_0} G(x, a), \quad (2.S17)$$

gives (2.5).

In order to simplify the solution of the following subproblems, change the reference frame in such a way that the charged plate is at $x = 0$ and the two conductors at $x = -a$ and $x = d - a$ respectively.

- b) Divide the space in two regions, $-a < x < 0$ and $0 < x < d - a$, where there are no charges, and solve the two separate homogeneous Laplace equations for the potential. Integrating Poisson's equation from $x = -\epsilon$ to $x = +\epsilon$ and taking the limit $\epsilon \rightarrow 0$, determine the conditions that connect the two solutions in $x = 0$. Find the potential for the whole range $-a < x < d - a$, using the continuity of the potential at $x = 0$.

Solution: The one-dimensional Laplace equation for the regions without charges reads

$$\Phi_h''(x) = 0. \quad (2.S18)$$

Clearly, it has the solution

$$\Phi_h(x) = mx + q. \quad (2.S19)$$

Thus the potential will read

$$\Phi(x) = \begin{cases} \Phi_-(x) = m_-x + q_- & \text{für } -a < x < 0, \\ \Phi_+(x) = m_+x + q_+ & \text{für } 0 < x < d - a. \end{cases} \quad (2.S20)$$

Poisson's equation in the shifted coordinate system reads

$$\Phi''(x) = -\frac{\sigma}{\varepsilon_0} \delta(x). \quad (2.S21)$$

Integrating it from $-\epsilon$ to ϵ we get

$$\Phi'(\epsilon) - \Phi'(-\epsilon) = -\frac{\sigma}{\varepsilon_0}, \quad (2.S22)$$

and taking the limit $\epsilon \rightarrow 0$ gives

$$\Phi'(0^+) - \Phi'(0^-) = -\frac{\sigma}{\varepsilon_0}. \quad (2.S23)$$

Further, we argue that the potential should be continuous in $x = 0$, because $\Phi'(x)$ is finite close to the origin. Thus we set

$$\begin{cases} \Phi_-(-a) = 0, \\ \Phi_-(0) = \Phi_+(0), \\ \Phi'_+(0) = \Phi'_-(0) - \frac{\sigma}{\varepsilon_0} \\ \Phi_+(d-a) = 0; \end{cases} \quad \Rightarrow \quad \begin{cases} q_- = m_-a, \\ q_- = q_+, \\ m_+ = m_- - \frac{\sigma}{\varepsilon_0}, \\ m_+(d-a) + q_+ = 0; \end{cases} \quad \Rightarrow \quad \begin{cases} m_- = \frac{\sigma}{\varepsilon_0} \left(1 - \frac{a}{d}\right), \\ m_+ = -\frac{\sigma}{\varepsilon_0} \frac{a}{d}, \\ q_- = \frac{\sigma}{\varepsilon_0} \left(1 - \frac{a}{d}\right)a, \\ q_+ = \frac{\sigma}{\varepsilon_0} \left(1 - \frac{a}{d}\right)a. \end{cases} \quad (2.S24)$$

Thus we get the result

$$\Phi(x) = \frac{\sigma d}{\varepsilon_0} \begin{cases} \left(1 - \frac{a}{d}\right) \frac{x}{d} + \left(1 - \frac{a}{d}\right) \frac{a}{d} & -a < x < 0 \\ -\frac{a}{d} \frac{x}{d} + \left(1 - \frac{a}{d}\right) \frac{a}{d} & 0 < x < d - a \end{cases} = \frac{\sigma d}{\varepsilon_0} \left[\frac{x}{d} \theta\left(-\frac{x}{d}\right) - \frac{a}{d} \frac{x}{d} + \left(1 - \frac{a}{d}\right) \frac{a}{d} \right]. \quad (2.S25)$$

- c) Directly integrate the differential equation for the potential and impose boundary conditions at $x = -a$ and $x = d - a$ to re-obtain the same result. In order to compare results, keep in mind the identities

$$\theta(x) = 1 - \theta(-x), \quad 2x\theta(x) = x + |x|. \quad (2.6)$$

Solution: Direct integration yields

$$\begin{aligned} \Phi'(x) &= -\frac{\sigma}{\varepsilon_0} \int^x dx' \delta(x') = -\frac{\sigma}{\varepsilon_0} \theta(x) + m; \\ \Phi(x) &= \int^x dx' \left[-\frac{\sigma}{\varepsilon_0} \theta(x') + m \right] = -\frac{\sigma}{\varepsilon_0} x\theta(x) + mx + q. \end{aligned} \quad (2.26)$$

Boundary conditions then set

$$\begin{cases} \Phi(-a) = 0, \\ \Phi(d-a) = 0; \end{cases} \implies \begin{cases} q = ma, \\ -\frac{\sigma}{\varepsilon_0} (d-a) + m(d-a) + q = 0; \end{cases} \implies \begin{cases} m = \frac{\sigma}{\varepsilon_0} \left(1 - \frac{a}{d}\right), \\ q = \frac{\sigma}{\varepsilon_0} \left(1 - \frac{a}{d}\right) a. \end{cases} \quad (2.27)$$

We thus find

$$\Phi(x) = \frac{\sigma d}{\varepsilon_0} \left[\frac{x}{d} \theta\left(\frac{x}{d}\right) - \left(1 - \frac{a}{d}\right) \left(\frac{x}{d} + \frac{a}{d}\right) \right]. \quad (2.28)$$

Using the identities

$$\theta(x) = 1 - \theta(-x), \quad 2x\theta(x) = x + |x|, \quad (2.29)$$

one finds the same result as before.

- d) Transform the differential equation for the potential to Fourier space, solve it, and carry out the inverse transformation. Convince yourself that you can find a particular solution that is consistent with previous results.
-

Solution: Switching to Fourier space with the convention

$$\hat{\Phi}(k) = \int_{\mathbb{R}} dx e^{-ikx} \Phi(x), \quad \Phi(x) = \int_{\mathbb{R}} \frac{dk}{2\pi} e^{ikx} \hat{\Phi}(k), \quad (2.30)$$

gives

$$-k^2 \hat{\Phi}(k) = -\frac{\sigma}{\varepsilon_0}. \quad (2.31)$$

By dividing by $-k^2$ we obtain a solution, but this is in fact not the most general one. Inspecting the equation more carefully, we realise the equations does not constrain $\hat{\Phi}$ at $k = 0$. The most general solution can be expressed using delta-functions

$$\hat{\Phi}(k) = \frac{\sigma}{\varepsilon_0} \frac{1}{k^2} + 2\pi\alpha\delta(k) + 2\pi i\beta\delta'(k). \quad (2.32)$$

The two undetermined constants α and β essentially correspond to the freedom to add homogeneous solutions. The inverse transformation then formally yields

$$\Phi(x) = \frac{\sigma}{\varepsilon_0} \int_{\mathbb{R}} \frac{dk}{2\pi} \frac{e^{ikx}}{k^2} + \alpha + \beta x. \quad (2.33)$$

The remaining Fourier integral is ill-defined due to the double pole at $k = 0$ which lies right on the integration contour. In order to make sense of the integral, we have to shift the pole away from the real k -axis. Let us shift it down by a tiny amount ϵ

$$\hat{\Phi}(k) = \frac{\sigma}{\varepsilon_0} \frac{1}{(k + i\epsilon)^2} + 2\pi\alpha\delta(k) + 2\pi i\beta\delta'(k). \quad (2.34)$$

Equivalently, we could deform the integration contour up in the neighbourhood of the point $k = 0$. This is clearly a choice as we could as well shift or deform in the opposite direction. Importantly, the result will depend on this choice, however, we shall see later that this is not an issue and also why this is so.

The shift or deformation has regularised the Fourier integral, which is now well-defined. The standard method to perform the integral is by Jordan's lemma and the residue theorem. Suppose we want to perform the Fourier integral

$$\Phi(x) = \int_{\mathbb{R}} \frac{dk}{2\pi} e^{ikx} \hat{\Phi}(k) = \lim_{K \rightarrow \infty} \int_{-K}^{+K} \frac{dk}{2\pi} e^{ikx} \hat{\Phi}(k), \quad (2.S35)$$

where the function $\hat{\Phi}(k)$ is assumed to be rational. We then build a contour $\gamma(K)$ in the complex k -plane which follows the real axis between $-K$ and $+K$ and close it with a semi-circle $C(K)$. The Fourier integral is then expressed as the limit $K \rightarrow \infty$

$$\Phi(x) = \lim_{K \rightarrow \infty} \oint_{\gamma(K)} \frac{dk}{2\pi} e^{ikx} \hat{\Phi}(k) = \lim_{K \rightarrow \infty} \int_{C(K)} \frac{dk}{2\pi} e^{ikx} \hat{\Phi}(k). \quad (2.S36)$$

Jordan's lemma ensures that the integral over the semi-circle $C(K)$ vanishes in the limit $K \rightarrow \infty$ provided that the semi-circle is chosen in the upper complex plane for $x > 0$ and in the lower plane for $x < 0$ such that the factor e^{ikx} is exponentially small in both cases. The remaining integral is then evaluated by the residue theorem.

In our case the pole is at $k = -i\epsilon$ with residue ix (in the limit $\epsilon \rightarrow 0$)

$$\frac{e^{ikx}}{k^2} = \frac{1 + ikx - \frac{1}{2}k^2x^2 + \dots}{k^2} = \frac{1}{k^2} + \frac{ix}{k} - \frac{1}{2}x^2 + \dots \quad (2.S37)$$

The residue theorem determines the Fourier integral. Note that there is no pole in the upper complex half-plane (relevant to $x > 0$) and that the pole in the lower complex half-plane is encircled clockwise

$$\lim_{K \rightarrow \infty} \oint_{\gamma(K)} \frac{dk}{2\pi} \frac{e^{ikx}}{(k + i\epsilon)^2} = \begin{cases} 0 & x > 0, \\ x & x < 0. \end{cases} \quad (2.S38)$$

Altogether we find

$$\Phi(x) = \frac{\sigma}{\epsilon_0} x \theta(-x) + \alpha + \beta x, \quad (2.S39)$$

which is consistent with the above results.

In order to understand the relevance of the above choice we consider the following Fourier integral

$$\lim_{K \rightarrow \infty} \oint_{\gamma(K)} \frac{dk}{2\pi} e^{ikx} \left[\frac{1}{k - i\epsilon} - \frac{1}{k + i\epsilon} \right] = \begin{cases} i & x > 0, \\ -i & x < 0. \end{cases} \quad (2.S40)$$

Interestingly, the result equals the same constant function for $x > 0$ and $x < 0$. The same result is obtained for the delta-function

$$\int_{\mathbb{R}} \frac{dk}{2\pi} e^{ikx} 2\pi i \delta(k) = i. \quad (2.S41)$$

Therefore we can make the following identification, at least for the purposes of the Fourier transformation

$$\frac{1}{k - i\epsilon} - \frac{1}{k + i\epsilon} = 2\pi i \delta(k). \quad (2.S42)$$

The derivative of this expression reads

$$\frac{1}{(k - i\epsilon)^2} - \frac{1}{(k + i\epsilon)^2} = -2\pi i \delta'(k). \quad (2.S43)$$

If we use these expressions for the delta-functions in the general solution, we see that changing the regularisation has the same effect as changing the (undetermined) coefficients α and β . E.g. we might as well write

$$\hat{\Phi}(k) = \frac{\sigma}{\epsilon_0} \frac{1}{(k - i\epsilon)^2} + 2\pi\alpha \delta(k) + 2\pi i \left(\beta - \frac{\sigma}{\epsilon_0} \right) \delta'(k) \quad (2.S44)$$

and

$$\Phi(x) = -\frac{\sigma}{\epsilon_0} x \theta(x) + \alpha + \left(\beta - \frac{\sigma}{\epsilon_0} \right) x. \quad (2.S45)$$

The coefficients of the delta-functions thus depend on the chosen regularisation. Eventually, they are fixed by implementing the desired boundary conditions. Importantly, the regularisation has to be fixed before fixing the coefficients. Then and only then the final result is independent of the choice of regularisation.

3.1. Capacities

A simple capacitor consists of two isolated conductors with opposite charges of equal magnitude $Q_1 = +Q$ and $Q_2 = -Q$. In general both conductors will have different electrical potential and $U = \Phi_1 - \Phi_2$ denotes the potential difference. A characteristic quantity of the capacitor is the capacity C defined by

$$C = \frac{|Q|}{|U|}. \quad (3.1)$$

Calculate the capacities for the following cases:

a) two big parallel planar surfaces with area A and small distance $d \ll \sqrt{A}$;

Solution: First we evaluate the electric field E of a single plate with surface charge density σ . Let the plate be located in the $z := x_3 = 0$ plane. We integrate over a rectangular box B with the same base A parallel to the plate, which completely contains the plate. Neglecting boundary effects, for symmetry reasons the electric field must be perpendicular to the plate, and both oppositely directed and equally sized on the two opposite sides of the plate. From Gauß' theorem

$$\oint_{\partial B} dx^2 \vec{n}_x \cdot \vec{E}(x) = 2EA = \int_B dx^3 \frac{\rho(x)}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} \quad (3.S1)$$

we receive the electric field of a single plate,

$$\vec{E}(x) = \begin{cases} \frac{\sigma}{2\varepsilon_0} \vec{e}_z & z > 0, \\ -\frac{\sigma}{2\varepsilon_0} \vec{e}_z & z < 0. \end{cases} \quad (3.S2)$$

The electric field of the plate capacitor is the result of a superposition with another plate at $z = d$ with surface charge density $-\sigma$,

$$\vec{E}(x) = \begin{cases} 0 & z > d, \\ \frac{\sigma}{\varepsilon_0} \vec{e}_z & d > z > 0, \\ 0 & z < 0. \end{cases} \quad (3.S3)$$

Knowing the electric field, the potential difference results from

$$U = \Phi(z=d) - \Phi(z=0) = - \int_0^d dx^3 \cdot \vec{E} = - \frac{\sigma}{\varepsilon_0} d. \quad (3.S4)$$

Consequently, the capacity of a plate capacitor of area A and distance d is

$$C = \frac{|Q|}{|U|} = \sigma A \Big/ \frac{\sigma d}{\varepsilon_0} = \varepsilon_0 \frac{A}{d}. \quad (3.S5)$$

b) two concentric, conducting spheres with radii $a < b$;

Solution: We evaluate the electric field E between the surfaces of the two spheres by means of Gauß' theorem for a (concentric) sphere with radius $b > r > a$:

$$\oint_{\|x\|=r} dx^2 \vec{n}_x \cdot \vec{E}(x) = 4\pi r^2 E(r) = \frac{Q}{\varepsilon_0} \quad \implies \quad E(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}. \quad (3.S6)$$

The potential difference is thus

$$U = - \int_a^b dr E(r) = - \frac{Q}{4\pi\varepsilon_0} \int_a^b dr \frac{1}{r^2} = \frac{Q}{4\pi\varepsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right), \quad (3.S7)$$

and the capacity is

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}. \quad (3.S8)$$

c) two coaxial, conducting cylindrical surfaces of length L and radii $a < b \ll L$.

Solution: The electric field E between the cylinders results from applying Gauß' theorem to a (coaxial) cylinder with radius r , $b > r > a$:

$$\oint_{\partial Z_r} dx^2 \vec{n}_x \cdot \vec{E}(x) = 2\pi r L E(r) = \frac{Q}{\varepsilon_0} \quad \Rightarrow \quad E(r) = \frac{1}{2\pi\varepsilon_0} \frac{Q}{rL}. \quad (3.S9)$$

Thus the potential difference is

$$U = - \int_a^b \frac{1}{2\pi\varepsilon_0} \frac{Q}{rL} dr = - \frac{1}{2\pi\varepsilon_0} \frac{Q}{L} \log \frac{b}{a}, \quad (3.S10)$$

and the capacity is

$$C = 2\pi\varepsilon_0 \frac{L}{\log(b/a)}. \quad (3.S11)$$

3.2. Imaginary dipoles

Consider two point charges q and q' at a distance d from each other, and a plane perpendicular to the line through q and q' in a distance αd from q .

- a) Show that, in order for the plane to be at constant potential, one must have $q' = -q$ and $\alpha = 1/2$.

Hint: Look at the potential at large distance first.

Solution: Let us use cylindrical coordinates around the axis through the two charges, where r denotes the radial coordinate. For a point on the plane we have

$$U = \frac{1}{4\pi\varepsilon_0} \left(\frac{q}{\sqrt{r^2 + \alpha^2 d^2}} + \frac{q'}{\sqrt{r^2 + (1-\alpha)^2 d^2}} \right). \quad (3.S12)$$

For $r \gg d$ one finds

$$U \sim \frac{1}{4\pi\varepsilon_0} \frac{1}{r} (q + q'), \quad (3.S13)$$

which is constant ($U = 0$) only if $q' = -q$.

Now looking at the potential for general r , we find

$$U(r) = \frac{q}{4\pi\varepsilon_0} \left(\frac{1}{\sqrt{r^2 + \alpha^2 d^2}} - \frac{1}{\sqrt{r^2 + (1-\alpha)^2 d^2}} \right) \stackrel{!}{=} 0, \quad (3.S14)$$

which implies

$$r^2 + \alpha^2 d^2 = r^2 + (1-\alpha)^2 d^2 \quad (3.S15)$$

and therefore $\alpha = 1/2$. The same result is obtained if we demand that $U'(r) = 0$ for all r .

Now consider a point charge at the (cartesian) coordinates $(a, b, 0)$ in the empty region of a space filled with a grounded conductor except for positive x and y .

- b) Argue that introducing two mirror charges – with respect to the planes $x = 0$ and $y = 0$, respectively – is not sufficient to reproduce the boundary conditions of the conductor.
-

Solution: We start by introducing mirror charges $-q$ at $(-a, b, 0)$ and $(a, -b, 0)$ as done for the single conducting plane. This is depicted in the second figure. In this case, we have for any of the two planes one dipole (which generates a constant potential on the plane) plus a single charge (which does not), and thus the potential will depend on the position. To obtain a constant potential, we should add another mirror charge (cf. the third figure and the next exercise) at $(-a, -b, 0)$ with charge $+q$ thereby constructing two dipoles across each plane. The rule of thumb is to mirror successively until you cover all possible mirror charges.



(3.S16)

- c) Exploiting the symmetry of the problem, introduce one more appropriate virtual charge and show explicitly that this makes the potential on the planes constant. Sketch the charge distribution.

Solution: Adding another charge $+q$, which is the image through reflection of any of the two starting negative charged images, we have that the potential on any of the two planes is the sum of two dipole potentials of the form found above, and thus constant. Explicitly, we find

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{(x-a)^2 + (y-b)^2 + z^2}} - \frac{1}{\sqrt{(x+a)^2 + (y-b)^2 + z^2}} - (b \leftrightarrow -b) \right), \quad (3.S17)$$

which is clearly zero for $x = 0$ since the two denominators become the same (and a similar mechanism also works for $y = 0$).

Finally, consider a point charge in the empty region of a space filled with a grounded conductor except for the region of the angle $0 \leq \varphi \leq \pi/n$ (n integer).

- d) Determine graphically the distribution of imaginary charges that reproduces the electric field of this charge in the empty region of space. What is the value of the electric field on the line where the two faces of the conductor meet?

Solution: Given what we have seen up to this point, it is clear that for the potential to be constant on a plane every charge has to be part of a symmetric dipole. Therefore, starting from the original point charge, we iteratively take its image across all planes until we get back to the initial point – which we do without having to place imaginary charges in the “real” space because the angle divides π exactly. This yields n dipoles across the planes at angles $2\pi/n$, which means $2n$ charges. Drawing the charges it is plain to see that the electric field at the meeting point in the corner of the conductor planes is zero, which after all makes sense because it has to be perpendicular to both (warning: this last argument is loose because in the case of a sharp spike it is infinite for the same reason, but the first strictly holds).

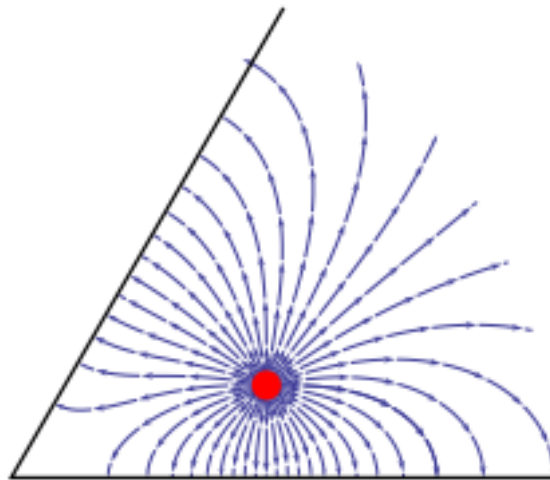


(3.S18)

- e) *optional:* Sketch the electric field lines for the charge distribution of part d).

Solution: By considering a charge at a specific angle and determining the angle of the image charges, one can find the electric field at a give point. Following is the plot of the electric field of a positive charge

placed at an angle $\pi/9$ from the x -axis between two plates with an opening angle $\pi/3$:



(3.S19)

In Mathematica this plot can be generated by the code:

```
EField0[ang_][x_, y_] :=
Module[{xc = Cos[ang], yc = Sin[ang]},
  {x - xc, y - yc}/((x - xc)^2 + (y - yc)^2)^(3/2)]
EField[x_, y_] =
Sum[(-1)^n EField0[(2 Pi/6)*(n + 1/2) - (-1)^n Pi/18][x, y], {n, 0, 5}];
StreamPlot[EField[x, y], {x, 0, 2}, {y, 0, 2},
  RegionFunction -> Function[{x, y, vx, vy, n}, 0 < y < x Tan[Pi/3]]]
```

3.3. Conducting sphere in an external electric field

A conducting sphere, bearing total charge Q , is introduced into a homogeneous electric field $\vec{E} = E\vec{e}_z$. How does the electric field change due to the presence of the sphere? How is the charge distributed on the surface of the sphere?

Hint: Motivate the following ansatz in spherical coordinates

$$\Phi(r, \vartheta, \varphi) = f_0(r) + f_1(r) \cos \vartheta, \quad (3.2)$$

and solve the Poisson equation $\Delta\Phi = 0$ using the Laplace operator

$$\Delta\Phi(r, \vartheta, \varphi) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \Phi}{\partial \vartheta} \right) + \frac{1}{r^2 \sin^2 \vartheta} \frac{\partial^2 \Phi}{\partial \varphi^2}. \quad (3.3)$$

A set of boundary conditions that completely fixes the solution is:

- at distances far from the sphere the homogeneous field has to dominate;
- the surface of the conducting sphere has to be an equipotential surface;
- the electric field has to satisfy Gauß' theorem.

Solution: The problem shows cylindrical symmetry, so the potential does not depend on φ . Due to the superposition principle, the potential can be decomposed into a component from the homogeneous field $\Phi_{\text{hom}} = -Ez = -Er \cos \vartheta$ and a component from the potential of the sphere Φ_K . As the sphere is a conductor, its surface must be equipotential,

$$\Phi_K(r = R, \vartheta) + \Phi_{\text{hom}}(r = R, \vartheta) = \text{const}, \quad (3.S20)$$

so that Φ_K gets only a ϑ -dependent component proportional to $\cos \vartheta$ (which compensates for the ϑ -dependence of Φ_{hom} at $r = R$). Inserting this ansatz for the overall potential into the Laplace equation yields

$$\begin{aligned}\Delta\Phi &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_0}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_1}{\partial r} \right) \cos \vartheta + \frac{f_1}{r^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial}{\partial \vartheta} \cos \vartheta \right), \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_0}{\partial r} \right) + \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_1}{\partial r} \right) - 2 \frac{f_1}{r^2} \right] \cos \vartheta \stackrel{!}{=} 0.\end{aligned}\quad (3.S21)$$

As this must hold for all ϑ , both coefficients vanish:

$$\begin{aligned}0 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_0}{\partial r} \right) = f_0'' + \frac{2}{r} f_0', \\ 0 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f_1}{\partial r} \right) - 2 \frac{f_1}{r^2} = \frac{1}{r^2} (r^2 f_1'' + 2r f_1' - 2f_1).\end{aligned}\quad (3.S22)$$

The general solutions of these two differential equations read

$$f_0(r) = \frac{A}{r} + B, \quad f_1(r) = \frac{C}{r^2} + Dr, \quad (3.S23)$$

and the solution of the potential has the form

$$\Phi = \frac{A}{r} + B + \left(\frac{C}{r^2} + Dr \right) \cos \vartheta. \quad (3.S24)$$

The following comment can be made in class as far as the spherical harmonics have been introduced in the lecture as part of the solution of the Laplace equation: This expression is in line with the general solution of the Laplace equation given by the basis of functions $r^\ell Y_{\ell,m}$ and $r^{-1-\ell} Y_{\ell,m}$. Here only the two spherical harmonics $Y_{00} \sim 1$ and $Y_{10} \sim \cos \vartheta$ contribute.

The constant B has no physical meaning, so we can choose $B = 0$. The remaining constants are obtained from the boundary conditions:

- Far away from the sphere, the potential must be that of the homogeneous field,

$$\lim_{r \rightarrow \infty} \Phi = D(r \cos \vartheta) \stackrel{!}{=} -zE, \quad (3.S25)$$

i.e. $D = -E$.

- The surface of the conducting sphere must be an equipotential surface,

$$\Phi(r, \vartheta) \Big|_{r=R} = \frac{A}{R} + \left(\frac{C}{R^2} - ER \right) \cos \vartheta = \text{const} \quad \text{for all } \vartheta. \quad (3.S26)$$

i.e. $C = ER^3$.

- Gauß' theorem yields the last remaining constant,

$$\oint_{r=R} dx^2 \, \vec{n} \cdot \vec{E}(x) = -R^2 \int_0^\pi d\vartheta \sin \vartheta \int_0^{2\pi} d\varphi \, \partial_r \Phi|_{r=R} = 4\pi A \stackrel{!}{=} \frac{1}{\varepsilon_0} Q, \quad (3.S27)$$

i.e. $A = Q/4\pi\varepsilon_0$.

Thus the solution of the potential is

$$\Phi = \frac{Q}{4\pi\varepsilon_0 r} + E \left(\frac{R^3}{r^2} - r \right) \cos \vartheta. \quad (3.S28)$$

The surface charge density is calculated from the discontinuity of the normal component of the electric field on the surface of the sphere:

$$\sigma(\vartheta) = -\varepsilon_0 \frac{\partial \Phi}{\partial r} \Big|_{r=R} = \left(\frac{Q}{4\pi R^2} + 3\varepsilon_0 E \cos \vartheta \right). \quad (3.S29)$$

3.4. Interior of a sphere with Neumann boundary conditions

The Green function for the interior of a sphere $B := \{x \in \mathbb{R}^3; \|x\| \leq R\}$ with radius R and Neumann boundary conditions is given by

$$G(x, y) = \frac{1}{4\pi\|x - y\|} + \frac{R}{4\pi\|y\|\|x - y'\|} - \frac{1}{4\pi R} \log \frac{\|y\|\|x - y'\| - \vec{y} \cdot (\vec{x} - \vec{y}')}{R^2}, \quad (3.4)$$

where $\vec{y}' := R^2 \vec{y} / \|y\|^2$ is the inversion of the point $\vec{y}$ on the sphere.

- a) Show that it satisfies the Laplace equation almost everywhere as well as the Neumann boundary condition for Green functions.

Solution: Acting with the Laplacian Δ_y on each individual term of the Green function results in a vanishing contribution at $x \neq y$,

$$\Delta_y G(x, y) = 0 \quad \text{at } x \neq y. \quad (3.S30)$$

Only the first term, which corresponds to the contribution of a point charge at $x = y$, is divergent at $x = y$ and causes the Laplacian to act like a delta-function,

$$\Delta_y G(x, y) = -\delta^3(x - y). \quad (3.S31)$$

As for the Neumann boundary conditions, we should verify that $\vec{n}_y \cdot \vec{\partial}_y G = -F(y)$ on the boundary $y \in \partial B$ for some function F that is independent of $x \in B$. Equivalently, we can thus verify that $\vec{\partial}_x (\vec{n}_y \cdot \vec{\partial}_y G) = 0$. On the surface of the sphere we know that,

$$\vec{n}_y \cdot \vec{\partial}_y = \sum_{i=1}^3 \frac{y_i}{\|y\|} \frac{\partial}{\partial y_i}. \quad (3.S32)$$

Due to the symmetry of the problem, we may further choose $y \in \partial B$ as $\vec{y} = (R, 0, 0)$ or $\vec{x} = (x_1, 0, 0)$ (but not both at the same time). The result then follows from direct computations, we obtain

$$\vec{n}_y \cdot \vec{\partial}_y G = -\frac{1}{4\pi R^2} \quad \text{for } y \in \partial B. \quad (3.S33)$$

- b) Show that it is symmetric in its two arguments. Interpret the value of $\vec{n}_y \cdot \vec{\partial}_y G$ on the boundary $y \in \partial B$.

Solution: We show that each individual term in the Green function is symmetric under interchanging x and y . While the symmetry is evident for the first term, for the second term we need to show

$$\|y\|\|x - y'\| = \|x\|\|y - x'\|. \quad (3.S34)$$

We can first expand the square norm and substitute the definition of y' to find

$$\begin{aligned} \|x - y'\|^2 &= (\vec{x} - \vec{y}') \cdot (\vec{x} - \vec{y}') = \|x\|^2 - 2\vec{x} \cdot \vec{y}' + \|y'\|^2 = \|x\|^2 - 2R^2 \frac{\vec{x} \cdot \vec{y}}{\|y\|^2} + \frac{R^4 \|y\|^2}{\|y\|^4} \\ &= \|x\|^2 - 2R^2 \frac{\vec{x} \cdot \vec{y}}{\|y\|^2} + \frac{R^4}{\|y\|^2}. \end{aligned} \quad (3.S35)$$

Multiplying by $\|y\|^2$, we have

$$\|y\|^2 \|x - y'\|^2 = \|y\|^2 \|x\|^2 - 2R^2 \vec{x} \cdot \vec{y} + R^2 = \|x\|^2 \|y - x'\|^2. \quad (3.S36)$$

The square root of the last identity yields the expected result and therefore the second term in the Green function is also symmetric. For the third term, the first part of the argument of the logarithm being symmetric, we only need to show

$$\vec{y} \cdot (\vec{x} - \vec{y}') = \vec{x} \cdot (\vec{y} - \vec{x}'). \quad (3.S37)$$

This is readily obtained by distributing the scalar product and using the definition of y'

$$\vec{y} \cdot (\vec{x} - \vec{y}') = \vec{y} \cdot \vec{x} - \frac{R^2 \vec{y} \cdot \vec{y}}{\|y\|^2} = \vec{y} \cdot \vec{x} - R^2 = \vec{x} \cdot (\vec{y} - \vec{x}'). \quad (3.S38)$$

Due to the above symmetry the function $\vec{n}_y \cdot \vec{\partial}_y G(x, y)$ is not only independent of $x \in B$, but also of $y \in \partial B$. The resulting value $-1/4\pi R^2$ multiplied by the area $A = 4\pi R^2$ of ∂B yields $-1 = -Q$ which must equal minus the unit total charge $Q = 1$ in the compact domain B by Gauß' theorem.

- c) Verify that $G(x, y)/\varepsilon_0$ describes the potential of the sum of a unit charge at y , a charge $q' = R/\|y\|$ at y' as well as a linear charge distribution with homogeneous charge density $\lambda = 1/R$ on the straight ray starting at y' and stretching out radially to infinity.

Hint: Sum and integrate the potential of the given charge distribution. Restrict the linear charge density to a maximal radius L at first. How can the divergent integral be understood or repaired?

Solution: The first two terms of the given Green function clearly correspond to the potential of two point charges as described. By means of the superposition principle, it remains to show that the third term agrees with the given linear charge distribution. We chose our frame of reference such that the points y and y' reside on the positive 1-axis, $y' = (y'_1, 0, 0)$, and we consider y fixed for the time being. The charge distributions extends along the same 1-axis from $(y'_1, 0, 0)$ to infinity

$$\rho(z) = \frac{1}{R} \theta(z_1 - y'_1) \delta(z_2) \delta(z_3), \quad (3.339)$$

which is supported only on the outside of the ball. The potential $\Phi(x)$ resulting from this charge distribution takes the general form

$$\Phi(x) = \int \frac{dz^3 \rho(z)}{4\pi\varepsilon_0 \|x - z\|} = \frac{1}{4\pi\varepsilon_0 R} \int_{y'_1}^{\infty} \frac{dz}{\sqrt{(z - x_1)^2 + x_2^2 + x_3^2}}. \quad (3.340)$$

Given the following elementary integral

$$\int^x \frac{dz}{\sqrt{z^2 + c}} = \log(x + \sqrt{x^2 + c}), \quad (3.341)$$

one finds that the integral diverges. To make sense of the integral, we recall the fact that constant terms in (electrostatic) potentials do not have any physical meaning.

There are two equivalent ways to obtain a finite result: One can add an x -constant but z -dependent term to the integrand such that the integral converges. Alternatively, one can restrict the integral to a finite domain y'_1 to L , and then add an x -constant but L -dependent term to the integral such that the limit $L \rightarrow \infty$ converges. The procedure is called regularisation. Even though we have modified the result, this tweaking merely affects unphysical information and leaves the electrostatic field E unchanged.

To regularise the integral we introduce a regulator $L > y'_1$, perform the integral from y'_1 to L , and add a well-chosen L -dependent x -independent term that cancels the divergency in the limit $L \rightarrow \infty$. Doing so one finds that one needs to subtract the term $\log(L/C)$ with C some constant of dimension length. The regularised potential then reads

$$\begin{aligned} \tilde{\Phi}(x) &= \frac{1}{4\pi\varepsilon_0 R} \lim_{L \rightarrow \infty} \left[\int_{y'_1}^L \frac{dz}{\sqrt{(z - x_1)^2 + x_2^2 + x_3^2}} - \log \frac{L}{C} \right] \\ &= -\frac{1}{4\pi\varepsilon_0 R} \lim_{L \rightarrow \infty} \log \left[\frac{L}{C} \frac{\sqrt{(y'_1 - x_1)^2 + x_2^2 + x_3^2} + (y'_1 - x_1)}{\sqrt{(L - x_1)^2 + x_2^2 + x_3^2} + (L - x_1)} \right] \\ &= -\frac{1}{4\pi\varepsilon_0 R} \log \frac{\sqrt{(y'_1 - x_1)^2 + x_2^2 + x_3^2} + (y'_1 - x_1)}{2C}. \end{aligned} \quad (3.342)$$

Regulating the integrand instead, one should choose a term $-1/z$ to eliminate the logarithmic divergence of the integral

$$\tilde{\Phi}(x) = \frac{1}{4\pi\varepsilon_0 R} \int_{y'_1}^{\infty} dz \left[\frac{1}{\sqrt{(z - x_1)^2 + x_2^2 + x_3^2}} - \frac{1}{z} \right] = \dots \quad (3.343)$$

This regularisation amounts to imposing a vanishing potential at $x = 0$ as the boundary condition for the elementary point charge rather than the conventional asymptotic boundary condition at $x \rightarrow \infty$. It yields the same result as above with the particular constant $C = y'_1$.

To compare with the Green function, we express its third term in the same coordinate system using the formula

$$\|y\| \|x - y'\| - \vec{y} \cdot (\vec{x} - \vec{y}') = y_1 \left(\sqrt{(y'_1 - x_1)^2 + x_2^2 + x_3^2} + (y'_1 - x_1) \right). \quad (3.344)$$

This yields the expression

$$G(x, y) = \dots - \frac{1}{4\pi R} \log \left[\frac{y_1}{R^2} \left(\sqrt{(y'_1 - x_1)^2 + x_2^2 + x_3^2} + (y'_1 - x_1) \right) \right]. \quad (3.S45)$$

Therefore, $G(x, y)/\varepsilon_0$ indeed describes the potential $\tilde{\Phi}$ of the given charge distribution up to a remaining term $(4\pi R)^{-1} \log(2Cy_1/R^2)$. This term is both finite and constant in x and therefore inessential for the consideration of a potential.

4.1. Green functions in electrostatics

In this problem we analyse the electrostatic Green function in more detail. We consider Green functions G on a volume V with Dirichlet and Neumann boundary conditions on the surface ∂V .

- a) Express the difference $G(y, z) - G(z, y)$ in terms of an integral over the surface ∂V . To do so, apply Green's second identity,

$$\int_V dx^3 (\phi \Delta \psi - \psi \Delta \phi) = \oint_{\partial V} dx^2 \vec{n} \cdot (\phi \vec{\partial} \psi - \psi \vec{\partial} \phi) \quad (4.1)$$

with $\phi = G(y, x)$ and $\psi = G(z, x)$. Use that $\Delta_x G(y, x) = -\delta^3(x - y)$.

Solution: From Green's second identity we get

$$\int_V dx^3 [G(y, x) \Delta_x G(z, x) - G(z, x) \Delta_x G(y, x)] = \oint_{\partial V} dx^2 \vec{n}_x \cdot [G(y, x) \vec{\partial}_x G(z, x) - G(z, x) \vec{\partial}_x G(y, x)], \quad (4.S1)$$

using $\Delta_x G(y, x) = -\delta^3(x - y)$ we immediately find

$$G(y, z) - G(z, y) = - \oint_{\partial V} dx^2 \vec{n}_x \cdot [G(y, x) \vec{\partial}_x G(z, x) - G(z, x) \vec{\partial}_x G(y, x)]. \quad (4.S2)$$

- b) Show that a Green function $G_D(x, y)$ with Dirichlet boundary conditions $G_D(x, y) = 0$ for all $y \in \partial V$ must be symmetric in x and y .

Solution: For Dirichlet boundary conditions we can define $G_D(x, y) = 0$ for $y \in \partial V$. Inserting this on the RHS of the previous equation we immediately obtain $G_D(x, y) - G_D(y, x) = 0$.

- c) Argue that $\vec{n}_y \cdot \vec{\partial}_y G_D(x, y) \rightarrow -\delta^2(x - y)$ for $x \rightarrow \partial V$ and $y \in \partial V$. For the case $x \not\rightarrow y$ you can use the Dirichlet boundary condition for $G_D(x, y)$. To understand the special case $x \rightarrow y$, use the above expression of Green's second identity.

Solution: Let f be an arbitrary test function. Using Green's second identity with $\phi = f$ and $\psi = G_D(x, y)$ we find

$$\int_V dy^3 [f(y) \Delta_y G_D(x, y) - G_D(x, y) \Delta_y f(y)] = \oint_{\partial V} dy^2 [f(y) \vec{n}_y \cdot \vec{\partial}_y G_D(x, y) - G_D(x, y) \vec{n}_y \cdot \vec{\partial}_y f(y)], \quad (4.S3)$$

which, using $\Delta_y G_D(x, y) = -\delta^3(x - y)$ and $G_D(x, y) = 0$ for $y \in \partial V$, leads to

$$-f(x) - \int_V dy^3 G_D(x, y) \Delta_y f(y) = \oint_{\partial V} dy^2 [f(y) \vec{n}_y \cdot \vec{\partial}_y G_D(x, y)]. \quad (4.S4)$$

For $x \in \partial V$ we also have $G_D(x \in \partial V, y) = 0$ and we find

$$-f(x) = \oint_{\partial V} dy^2 [f(y) \vec{n}_y \cdot \vec{\partial}_y G_D(x, y)], \quad (4.S5)$$

which proves that $\vec{n}_y \cdot \vec{\partial}_y G_D(x, y) = -\delta^2(x, y)$.

d) Consider the (alternative formulation of the) Neumann boundary condition

$$\vec{\partial}_x [\vec{n}_y \cdot \vec{\partial}_y G_N(x, y)] = 0 \quad \text{for all } y \in \partial V. \quad (4.2)$$

Show that $G_N(x, y)$ is not symmetric in x and y in general. Construct a Green function $\tilde{G}_N(x, y) = G_N(x, y) + H(y) + K(x)$ that is symmetric in x and y . What properties must H and K have such that $\tilde{G}_N$ is again a proper Green function?

Solution: Since $\vec{\partial}_y [\vec{n}_z \cdot \vec{\partial}_z G_N(y, z)] = 0$, we must have that $\vec{n}_z \cdot \vec{\partial}_z G_N(y, z)$ is a function only of z . Let us call this function $-F(z)$. It is specified (up to an integration constant) in terms of the given Green function.

Inserting $\vec{n}_z \cdot \vec{\partial}_z G_N(y, z) = -F(z)$ in the RHS of the equation from the first part of the problem we have:

$$\begin{aligned} G_N(x, y) - G_N(y, x) &= - \oint_{\partial V} dz^2 \left[G_N(x, z) \underbrace{\vec{n}_z \cdot \vec{\partial}_z G_N(y, z)}_{=-F(z)} - G_N(y, z) \underbrace{\vec{n}_z \cdot \vec{\partial}_z G_N(x, z)}_{=-F(z)} \right] \\ &= \underbrace{\oint_{\partial V} dz^2 G_N(x, z) F(z)}_{=: P(x)} - \underbrace{\oint_{\partial V} dz^2 G_N(y, z) F(z)}_{=: P(y)} \end{aligned} \quad (4.S6)$$

Hence, $G_N(x, y)$ is not generally symmetric but for $\tilde{G}_N(x, y) = G_N(x, y) - P(x)$ we have

$$\tilde{G}_N(x, y) - \tilde{G}_N(y, x) = 0. \quad (4.S7)$$

Therefore, we need $K(x) = - \oint dz^2 G_N(x, z) F(z)$. The second function $H(y)$ is not constrained by the boundary condition, but it amounts to a redefinition of $F(z)$. In other words, it can be added to the original Green function but then $K(x)$ has to be computed using the new $F(z)$ in order to ensure the symmetry of $\tilde{G}_N(x, y)$. Note that $H(x)$ must be harmonic, $\Delta H = 0$, in order for the defining equation of the Green function to remain valid.

4.2. Spherical cavity

Consider a spherical cavity with radius R and let the potential on its boundary be specified by an arbitrary function $U(\vartheta, \varphi)$.

a) Show that the potential within the cavity can be expressed as

$$\Phi(x) = \int \sin \vartheta' d\vartheta' d\varphi' \frac{R(R^2 - r^2) U(\vartheta', \varphi')}{4\pi(r^2 + R^2 - 2rR \cos \gamma)^{3/2}}, \quad (4.3)$$

where γ is the angle between x and x' . Determine $\cos \gamma$ in terms of the variables ϑ , φ , ϑ' and φ' .

Hint: Use the Green function obtained with the method of image charges and use spherical coordinates.

Solution: The Green function for the sphere with Dirichlet boundary conditions reads (for derivation see lecture notes; this holds both for the inside and the outside of a sphere):

$$G(x, x') = \frac{1}{4\pi} \left(\frac{1}{\|x - x'\|} - \frac{R/\|x\|}{\|x - x'R^2/\|x'\|^2\|} \right). \quad (4.S8)$$

We write $\vec{x} = r\vec{n}(\vartheta, \varphi)$ and $\vec{x}' = r'\vec{n}(\vartheta', \varphi')$ in spherical coordinates. Therefore, given that γ is the angle between $\vec{x}$ and $\vec{x}'$ (i.e. $\vec{x} \cdot \vec{x}' = rr' \cos \gamma$)

$$G(x, x') = \frac{1}{4\pi} \left(\frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos \gamma}} - \frac{R}{\sqrt{r^2 r'^2 + R^4 - 2rr'R^2 \cos \gamma}} \right). \quad (4.S9)$$

Green's theorem for Dirichlet boundary conditions states

$$\Phi(x) = \frac{1}{\varepsilon_0} \int_V dx'^3 \rho(x') G(x, x') - \oint_{\partial V} dx'^2 U(x') \vec{n}_{x'} \cdot \vec{\partial}_{x'} G(x, x'). \quad (4S10)$$

Since there are no charges inside the cavity the first term vanishes. The surface we have to integrate on is the sphere $\|x'\| = r' = R$, so

$$dx'^2 = R^2 d\varphi' \sin \vartheta' d\vartheta', \quad (4S11)$$

and the only component of the gradient that contributes is the radial one. We find

$$\vec{n}_{x'} \cdot \vec{\partial}_{x'} G(x, x') = \frac{1}{4\pi} \left(-\frac{r' - r \cos \gamma}{[r^2 + r'^2 - 2rr' \cos \gamma]^{3/2}} + \frac{r^2 r' R - r R^3 \cos \gamma}{[r^2 r'^2 + R^4 - 2rr' R^2 \cos \gamma]^{3/2}} \right), \quad (4S12)$$

which on the surface $r' = R$ simplifies to

$$\vec{n}_{x'} \cdot \vec{\partial}_{x'} G(x, x') = \frac{r^2 - R^2}{4\pi R [r^2 + R^2 - 2rR \cos \gamma]^{3/2}}. \quad (4S13)$$

This implies

$$\Phi(x) = \frac{R}{4\pi} \int U(\vartheta', \varphi') \frac{R^2 - r^2}{[r^2 + R^2 - 2rR \cos \gamma]^{3/2}} d\varphi' \sin \vartheta' d\vartheta', \quad (4S14)$$

which is the relation we wanted. The link between γ and the variables (ϑ', φ') is obtained by observing that $\vec{x} \cdot \vec{x}' = rr' \cos \gamma$ on the sphere, and

$$\cos \gamma = \begin{pmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{pmatrix} \cdot \begin{pmatrix} \sin \vartheta' \cos \varphi' \\ \sin \vartheta' \sin \varphi' \\ \cos \vartheta' \end{pmatrix} = \sin \vartheta \sin \vartheta' \underbrace{(\cos \varphi \cos \varphi' + \sin \varphi \sin \varphi')}_{\cos(\varphi - \varphi')} + \cos \vartheta \cos \vartheta'. \quad (4S15)$$

- b) Write down the general solution of the Laplace equation in terms of spherical harmonics $Y_{\ell,m}$. Then use orthonormality relations to determine the coefficients for the given boundary condition.

Solution: The expansion of the solution of the Laplace equation

$$\Delta \Phi(r, \vartheta, \varphi) = 0 \quad (4S16)$$

in terms of spherical harmonics for domains that include the origin $r = 0$ reads (see chapter on solutions to the Laplace equation in spherical coordinates)

$$\Phi(r, \vartheta, \varphi) = \sum_{\ell=0}^{\infty} r^{\ell} \sum_{m=-\ell}^{\ell} c_{\ell,m} Y_{\ell,m}(\vartheta, \varphi). \quad (4S17)$$

We require the boundary condition

$$\Phi(R, \vartheta, \varphi) = \sum_{\ell'=0}^{\infty} \sum_{m'=-\ell'}^{\ell'} R^{\ell'} c_{\ell',m'} Y_{\ell',m'}(\vartheta, \varphi) \stackrel{!}{=} U(\vartheta, \varphi), \quad (4S18)$$

which multiplying by $Y_{\ell,m}^*(\vartheta, \varphi)$ and integrating over the solid angle yields

$$\begin{aligned} \oint \frac{d^2 \Omega}{4\pi} Y_{\ell,m}^*(\vartheta, \varphi) U(\vartheta, \varphi) &= \sum_{\ell'=0}^{\infty} \sum_{m'=-\ell'}^{\ell'} R^{\ell'} c_{\ell',m'} \oint \frac{d^2 \Omega}{4\pi} Y_{\ell,m}^*(\vartheta, \varphi) Y_{\ell',m'}(\vartheta, \varphi) \\ &= \sum_{\ell'=0}^{\infty} \sum_{m'=-\ell'}^{\ell'} R^{\ell'} c_{\ell',m'} \delta_{\ell\ell'} \delta_{mm'} = R^{\ell} c_{\ell,m}. \end{aligned} \quad (4S19)$$

Therefore

$$\Phi(r, \vartheta, \varphi) = \sum_{\ell=0}^{\infty} \frac{r^{\ell}}{R^{\ell}} \sum_{m=-\ell}^{\ell} \frac{1}{4\pi} \left(\oint \frac{d^2 \Omega'}{4\pi} Y_{\ell,m}^*(\vartheta', \varphi') U(\vartheta', \varphi') \right) Y_{\ell,m}(\vartheta, \varphi). \quad (4S20)$$

c) Find the explicit potential $\Phi(x)$ inside the sphere for the boundary condition

$$U(\vartheta, \varphi) = U_0 \cos \vartheta. \quad (4.4)$$

Solution: Since $U(\vartheta, \varphi) \propto Y_{1,0}(\vartheta, \varphi)$ we can exploit the orthonormality relation to find immediately

$$\Phi(r, \vartheta, \varphi) = U_0 \frac{r}{R} \cos \vartheta. \quad (4.S21)$$

4.3. Multipole moments of a cube

Assume that positive and negative point charges $\pm q$ are located on the corners of a cube with side length a . The point of origin is the centre of the cube, and the edges are aligned with the x, y, z -axes. Let the charge at $x, y, z > 0$ be positive. Charges on neighbouring corners have opposite signs.

a) Determine the positions of the charges in cartesian and spherical coordinates.

Solution: All charges are equally distant from the origin (half the long diagonal), namely

$$R = \frac{a}{2} \sqrt{3}. \quad (4.S22)$$

With the labels in figure (4.S24), the charges 1 through 4 all have $z = a/2$ and, accordingly, the same $\tilde{\vartheta}$,

$$\cos \tilde{\vartheta} = \frac{z}{R} = \frac{a/2}{\frac{\sqrt{3}a}{2}} = \frac{1}{\sqrt{3}} \quad \Rightarrow \quad \tilde{\vartheta} = \arccos \frac{1}{\sqrt{3}} \approx 54.74^\circ \approx 0.955 \text{ rad}, \quad (4.S23)$$



$$(4.S24)$$

whereas the charges 5 through 8 have $z = -a/2$ and $\vartheta = \pi - \tilde{\vartheta}$. Together with the obvious φ values, we get the following positions of the eight charges in Cartesian and spherical coordinates:

k	q_k	$\vec{x}_k$	r_k	ϑ_k	φ_k
1	$+q$	$(+\frac{1}{2}a, +\frac{1}{2}a, +\frac{1}{2}a)$	R	$\tilde{\vartheta}$	$\frac{1}{4}\pi$
2	$-q$	$(-\frac{1}{2}a, +\frac{1}{2}a, +\frac{1}{2}a)$	R	$\tilde{\vartheta}$	$\frac{3}{4}\pi$
3	$+q$	$(-\frac{1}{2}a, -\frac{1}{2}a, +\frac{1}{2}a)$	R	$\tilde{\vartheta}$	$\frac{5}{4}\pi$
4	$-q$	$(+\frac{1}{2}a, -\frac{1}{2}a, +\frac{1}{2}a)$	R	$\tilde{\vartheta}$	$\frac{7}{4}\pi$
5	$-q$	$(+\frac{1}{2}a, +\frac{1}{2}a, -\frac{1}{2}a)$	R	$\pi - \tilde{\vartheta}$	$\frac{1}{4}\pi$
6	$+q$	$(-\frac{1}{2}a, +\frac{1}{2}a, -\frac{1}{2}a)$	R	$\pi - \tilde{\vartheta}$	$\frac{3}{4}\pi$
7	$-q$	$(-\frac{1}{2}a, -\frac{1}{2}a, -\frac{1}{2}a)$	R	$\pi - \tilde{\vartheta}$	$\frac{5}{4}\pi$
8	$+q$	$(+\frac{1}{2}a, -\frac{1}{2}a, -\frac{1}{2}a)$	R	$\pi - \tilde{\vartheta}$	$\frac{7}{4}\pi$

b) Determine the charge density in cartesian coordinates and, thereafter, in spherical coordinates, using

$$\delta^3(x - x_0) = \frac{1}{r^2 \sin \vartheta} \delta(r - r_0) \delta(\vartheta - \vartheta_0) \delta(\varphi - \varphi_0), \quad (4.5)$$

as well as $\sin \vartheta = \sin(\pi - \vartheta)$ and $\cos(\pi - \vartheta) = -\cos \vartheta$.

Solution: The charge density in Cartesian coordinates reads

$$\rho(x, y, z) = +q \delta^3(x - x_1) - q \delta^3(x - x_2) + q \delta^3(x - x_3) - q \delta^3(x - x_4) \\ - q \delta^3(x - x_5) + q \delta^3(x - x_6) - q \delta^3(x - x_7) + q \delta^3(x - x_8). \quad (4.S26)$$

Transforming the δ -distributions to spherical coordinates (see identity on the problem sheet), in spherical coordinates we find

$$\rho(r, \vartheta, \varphi) = +q \frac{1}{r^2 \sin \vartheta} \delta(r - R) \delta(\vartheta - \tilde{\vartheta}) \delta(\varphi - \pi/4) \mp \dots \\ = \frac{q}{r^2 \sin \vartheta} \delta(r - R) \left[\delta(\vartheta - \tilde{\vartheta}) - \delta(\vartheta - (\pi - \tilde{\vartheta})) \right] \\ \cdot \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right]. \quad (4.S27)$$

c) Calculate the spherical dipole, quadrupole and octupole moments of this charge configuration, using

$$Q_{\ell, m} = \int d^3x \rho(x) r^\ell Y_{\ell, m}^*(\vartheta, \varphi), \quad m = -\ell, -(\ell - 1), \dots, +(\ell - 1), +\ell. \quad (4.6)$$

Solution: The spherical multipole moments are

$$Q_{\ell, m} = \int d^3x \rho(x) r^\ell Y_{\ell, m}^*(\vartheta, \varphi) \\ = \int_0^\infty dr \int_0^{2\pi} d\varphi \int_0^\pi d\vartheta r^{2+\ell} \sin \vartheta \rho(x) Y_{\ell, m}^*(\vartheta, \varphi) \\ = q \int_0^\infty dr r^\ell \delta(r - R) \int_0^{2\pi} d\varphi \int_0^\pi d\vartheta Y_{\ell, m}^*(\vartheta, \varphi) \left[\delta(\vartheta - \tilde{\vartheta}) - \delta(\vartheta - (\pi - \tilde{\vartheta})) \right] \\ \cdot \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right]. \quad (4.S28)$$

Symmetry considerations directly show that a large majority of the requested multipole moments are zero:

$$Q_{\ell, 0} = \int d^3x \rho(x) r^\ell Y_{\ell, 0}^*(\vartheta) : \quad \int_0^{2\pi} d\varphi \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right] \\ = 1 - 1 + 1 - 1 = 0, \\ Q_{\ell, \pm 1} = \int d^3x \rho(x) r^\ell Y_{\ell, \pm 1}^*(\vartheta, \varphi) : \quad \int_0^{2\pi} d\varphi e^{\mp i\varphi} \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right] \\ = e^{\mp i\pi/4} - e^{\mp 3i\pi/4} + e^{\mp 5i\pi/4} - e^{\mp 7i\pi/4} \\ = e^{\mp i\pi/4} (1 + \underbrace{e^{\mp i\pi}}_{=-1}) - e^{\mp 3i\pi/4} (1 + \underbrace{e^{\mp i\pi}}_{=-1}) = 0, \\ Q_{\ell, \pm 2} = \int d^3x \rho(x) r^\ell Y_{\ell, \pm 2}^*(\vartheta, \varphi) : \quad \int_0^{2\pi} d\varphi e^{\mp 2i\varphi} \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right] \\ = e^{\mp i\pi/2} - e^{\mp 3i\pi/2} + e^{\mp 5i\pi/2} - e^{\mp 7i\pi/2} \\ = e^{\mp i\pi/2} (1 - \underbrace{e^{\mp i\pi}}_{=-1}) + \underbrace{e^{\mp 2i\pi}}_{=+1} e^{\mp i\pi/2} (1 - \underbrace{e^{\mp i\pi}}_{=-1}) = 4 e^{\mp i\pi/2}, \\ Q_{\ell, \pm 3} = \int d^3x \rho(x) r^\ell Y_{\ell, \pm 3}^*(\vartheta, \varphi) : \quad \int_0^{2\pi} d\varphi e^{\mp 3i\varphi} \left[\delta(\varphi - \frac{1}{4}\pi) - \delta(\varphi - \frac{3}{4}\pi) + \delta(\varphi - \frac{5}{4}\pi) - \delta(\varphi - \frac{7}{4}\pi) \right] \\ = e^{\mp 3i\pi/4} - e^{\mp 9i\pi/4} + e^{\mp 15i\pi/4} - e^{\mp 21i\pi/4} \\ = e^{\mp 3i\pi/4} \left[1 - e^{\mp 3i\pi/2} + \underbrace{e^{\mp 3i\pi}}_{=-1} (1 - e^{\mp 3i\pi/2}) \right] = 0. \quad (4.S29)$$

Considering the φ -integration only, up to the octupole moments, all but $Q_{2, \pm 2}$ and $Q_{3, \pm 2}$ vanish due to the underlying symmetry with respect to φ .

The ϑ -integration vanishes—obviously from $\sin \vartheta = \sin(\pi - \vartheta)$ —if $Y_{\ell,m}(\vartheta, \varphi)$ depends only on $\sin \vartheta$ (is an odd function in ϑ). Thus, also $Q_{2,\pm 2} = 0$.

For the only non-vanishing (with $\ell \leq 3$) moments $Q_{3,\pm 2}$ we find

$$\begin{aligned} Q_{3,\pm 2} &= q \sqrt{\frac{105}{8}} \int_0^\infty dr r^3 \delta(r - R) \int_0^\pi d\vartheta \cos \vartheta \sin^2 \vartheta \left(\delta(\vartheta - \tilde{\vartheta}) - \delta(\vartheta - (\pi - \tilde{\vartheta})) \right) \\ &\quad \cdot \int_0^{2\pi} d\varphi e^{\mp 2i\varphi} \left[\delta(\varphi - \tfrac{1}{4}\pi) - \delta(\varphi - \tfrac{3}{4}\pi) + \delta(\varphi - \tfrac{5}{4}\pi) - \delta(\varphi - \tfrac{7}{4}\pi) \right] \\ &= q \sqrt{\frac{105}{2}} \underbrace{R^3}_{=3\sqrt{3}a^3/8} \underbrace{\cos \tilde{\vartheta} \sin^2 \tilde{\vartheta}}_{=(1-1/3)/\sqrt{3}} \underbrace{e^{\mp i\pi/2}}_{=\mp i} = \mp i \sqrt{\frac{105}{2}} q a^3. \end{aligned} \quad (4.S30)$$

The required spherical harmonics are:

$$\begin{aligned} Y_{00} &= 1, & Y_{11} &= -\sqrt{\frac{3}{2}} \sin \vartheta e^{i\varphi}, \\ Y_{10} &= \sqrt{3} \cos \vartheta, & Y_{22} &= \sqrt{\frac{15}{8}} \sin^2 \vartheta e^{2i\varphi}, \\ Y_{21} &= -\sqrt{\frac{15}{2}} \cos \vartheta \sin \vartheta e^{i\varphi}, & Y_{20} &= \sqrt{\frac{5}{4}} (3 \cos^2 \vartheta - 1), \\ Y_{33} &= -\sqrt{\frac{35}{16}} \sin^3 \vartheta e^{3i\varphi}, & Y_{32} &= \sqrt{\frac{105}{8}} \cos \vartheta \sin^2 \vartheta e^{2i\varphi}, \\ Y_{31} &= -\sqrt{\frac{21}{16}} (5 \cos^2 \vartheta - 1) \sin \vartheta e^{i\varphi}, & Y_{30} &= \sqrt{\frac{7}{4}} (5 \cos^3 \vartheta - 3 \cos \vartheta). \end{aligned} \quad (4.7)$$

Furthermore: $Y_{\ell,-m} = (-1)^m Y_{\ell,m}^*$, and thus $Q_{\ell,-m} = (-1)^m Q_{\ell,m}^*$.

5.1. Expansion of the potential in Legendre polynomials

Suppose the potential on a spherical shell at radius R depends only on the polar angle ϑ and is given by $\Phi_0(\vartheta)$. Inside and outside the sphere there is empty space.

- a) Find expressions for the potential $\Phi(r, \vartheta)$ inside and outside the sphere and for the charge density $\sigma(\vartheta)$ on the sphere.

Hint: The Legendre polynomials satisfy the following relations:

$$\int_0^\pi P_n(\cos \vartheta) P_m(\cos \vartheta) \sin \vartheta \, d\vartheta = \delta_{n,m} \frac{2}{2n+1}. \quad (5.1)$$

Solution: Since both the outside and inside of the spherical shell are empty space, the potential satisfies the Laplace equation in these regions. Thus the potential inside the sphere has the form

$$\Phi_1 = \sum_{n=0}^{\infty} a_n r^n P_n(\cos \vartheta), \quad (5.S1)$$

while that outside the sphere is

$$\Phi_2 = \sum_{n=0}^{\infty} \frac{b_n}{r^{n+1}} P_n(\cos \vartheta). \quad (5.S2)$$

From continuity of the potential on the inside we find

$$\Phi_0(\vartheta) = \sum_{n=0}^{\infty} a_n R^n P_n(\cos \vartheta). \quad (5.S3)$$

Multiplying both sides by $P_n(\cos \vartheta) \sin \vartheta \, d\vartheta$ and integrating from 0 to π , we obtain

$$a_n R^n = \frac{2n+1}{2} \int_0^\pi \Phi_0(\vartheta) P_n(\cos \vartheta) \sin \vartheta \, d\vartheta. \quad (5.S4)$$

Hence

$$\Phi_1 = \sum_{n=0}^{\infty} \left(\frac{2n+1}{2R^n} \int_0^\pi \Phi_0(\vartheta') P_n(\cos \vartheta') \sin \vartheta' \, d\vartheta' \right) P_n(\cos \vartheta) r^n, \quad (5.S5)$$

and similarly

$$\Phi_2 = \sum_{n=0}^{\infty} \left(\frac{(2n+1)R^{n+1}}{2} \int_0^\pi \Phi_0(\vartheta') P_n(\cos \vartheta') \sin \vartheta' \, d\vartheta' \right) \frac{P_n(\cos \vartheta)}{r^{n+1}}. \quad (5.S6)$$

The charge distribution on the spherical shell is given by the boundary condition for the displacement vector:

$$\begin{aligned} \sigma(\vartheta) &= \varepsilon_0 \left. \frac{\partial \Phi_1}{\partial r} \right|_{r=R} - \varepsilon_0 \left. \frac{\partial \Phi_2}{\partial r} \right|_{r=R} \\ &= \varepsilon_0 \sum_{n=0}^{\infty} \left(\frac{(2n+1)^2}{2R} \int_0^\pi \Phi_0(\vartheta') P_n(\cos \vartheta') \sin \vartheta' \, d\vartheta' \right) P_n(\cos \vartheta), \end{aligned} \quad (5.S7)$$

- b) Evaluate the expressions found for $\Phi(r > R, \vartheta)$, $\Phi(r < R, \vartheta)$ and $\sigma(\vartheta)$ with $\Phi_0(\vartheta) = U \cos^2 \vartheta$.

Hint: The first three Legendre polynomials are given as follows:

$$P_0(\cos \vartheta) = 1, \quad P_1(\cos \vartheta) = \cos \vartheta, \quad P_2(\cos \vartheta) = \frac{3}{2} \cos^2 \vartheta - \frac{1}{2}. \quad (5.2)$$

Solution: From

$$\Phi_0(\vartheta) = U \cos^2 \vartheta = \frac{2}{3} U P_2(\cos \vartheta) + \frac{1}{3} U P_0(\cos \vartheta), \quad (5.S8)$$

we obtain

$$\int_0^\pi \Phi(\vartheta) P_n(\cos \vartheta) \sin \vartheta \, d\vartheta = \begin{cases} \frac{2}{3} U & \text{for } n = 0, \\ \frac{4}{15} U & \text{for } n = 2, \\ 0 & \text{for } n \neq 0, 2. \end{cases} \quad (5.S9)$$

Hence

$$\begin{aligned} \Phi_1 &= \frac{1}{3} U + \frac{2}{3} U P_2(\cos \vartheta) \frac{r^2}{R^2}, \quad r < R, \\ \Phi_2 &= \frac{1}{3} \frac{R}{r} U + \frac{2}{3} U P_2(\cos \vartheta) \frac{R^3}{r^3}, \quad r > R, \end{aligned} \quad (5.S10)$$

and

$$\sigma(\vartheta) = \frac{\varepsilon_0 U}{3R} [1 + 10 P_2(\cos \vartheta)]. \quad (5.S11)$$

Take now a spherical shell of radius R with uniform charge distribution $\sigma = Q/(4\pi R^2)$, except for a spherical cap at the north pole defined by the cone $\vartheta < \alpha$ in which $\sigma = 0$.

c) Show that the potential inside the sphere can be expressed as:

$$\Phi(r, \vartheta) = \frac{Q}{8\pi\varepsilon_0} \sum_{\ell=0}^{\infty} \frac{1}{2\ell+1} [P_{\ell+1}(\cos \alpha) - P_{\ell-1}(\cos \alpha)] \frac{r^\ell}{R^{\ell+1}} P_\ell(\cos \vartheta), \quad (5.3)$$

where for $\ell = 0$, $P_{\ell-1}(x) = -1$. What is the potential outside?

Hint: Use the following identities for the Legendre polynomials:

$$P_n(x) = \frac{1}{2n+1} (P'_{n+1}(x) - P'_{n-1}(x)). \quad (5.4)$$

Solution: Starting from equations (5.S1) and (5.S2) we define:

$$a_n = \frac{\alpha_n}{R^n}, \quad b_n = \beta_n R^{n+1} \quad (5.S12)$$

so that:

$$\begin{aligned} \Phi_1(r, \vartheta) &= \sum_{n=0}^{\infty} \alpha_n \left(\frac{r}{R}\right)^n P_n(\cos \vartheta), \\ \Phi_2(r, \vartheta) &= \sum_{n=0}^{\infty} \beta_n \left(\frac{R}{r}\right)^{n+1} P_n(\cos \vartheta). \end{aligned} \quad (5.S13)$$

We require the potential to be continuous on the surface of the sphere $r = R$, hence

$$\alpha_n = \beta_n. \quad (5.S14)$$

We can compute the coefficients α_n using equation (5.S7):

$$\sigma(\cos \vartheta) = \varepsilon_0 \sum_{n=0}^{\infty} \frac{2n+1}{R} \alpha_n P_n(\cos \vartheta). \quad (5.S15)$$

Multiplying both sides by $P_m(\cos \vartheta)$ and integrating over $\cos \vartheta$ we get:

$$\int_{-1}^1 P_m(\cos \vartheta) \sigma(\cos \vartheta) \, d\cos \vartheta = \varepsilon_0 \sum_{n=0}^{\infty} \frac{2n+1}{R} \alpha_n \int_{-1}^1 P_n(\cos \vartheta) P_m(\cos \vartheta) \, d\cos \vartheta. \quad (5.S16)$$

Using the orthogonality relation for the Legendre polynomials we can extract the α_n :

$$\alpha_n = \frac{R}{2\varepsilon_0} \int_{-1}^1 P_n(\cos \vartheta) \sigma(\cos \vartheta) d \cos \vartheta. \quad (5.S17)$$

In this case we have:

$$\sigma(\cos \vartheta) = \frac{Q}{4\pi R^2} \begin{cases} 1 & \text{if } \cos \alpha < \cos \vartheta, \\ 0 & \text{if } \cos \alpha > \cos \vartheta. \end{cases} \quad (5.S18)$$

Thus we have:

$$\alpha_n = \frac{Q}{8\pi\varepsilon_0 R} \int_{-1}^{\cos \alpha} P_n(\cos \vartheta) d \cos \vartheta. \quad (5.S19)$$

Recalling the identity for the Legendre polynomials provided in the hint

$$P_n(x) = \frac{1}{2n+1} (P'_{n+1}(x) - P'_{n-1}(x)), \quad (5.S20)$$

where we formally define $P_{-1}(x)$ as $P_{-1}(x) = -1$, the integration of (5.S19) is straightforward:

$$\alpha_n = \frac{Q}{8\pi\varepsilon_0 R} \frac{1}{2n+1} [P_{n+1}(\cos \vartheta) - P_{n-1}(\cos \vartheta)]_{\cos \vartheta=-1}^{\cos \vartheta=\cos \alpha} = \frac{Q}{8\pi\varepsilon_0 R} \frac{1}{2n+1} [P_{n+1}(\cos \alpha) - P_{n-1}(\cos \alpha)], \quad (5.S21)$$

where we have used $P_n(-1) = (-1)^n$. We already observed that $\alpha_n = \beta_n$, thus the result above can be used for both the inside and outside of the sphere.

$$\begin{aligned} \Phi_1(r, \vartheta) &= \frac{Q}{8\pi\varepsilon_0 R} \sum_{n=0}^{\infty} \frac{1}{2n+1} [P_{n+1}(\cos \alpha) - P_{n-1}(\cos \alpha)] \left(\frac{r}{R}\right)^n P_n(\cos \vartheta), \\ \Phi_2(r, \vartheta) &= \frac{Q}{8\pi\varepsilon_0 R} \sum_{n=0}^{\infty} \frac{1}{2n+1} [P_{n+1}(\cos \alpha) - P_{n-1}(\cos \alpha)] \left(\frac{R}{r}\right)^{n+1} P_n(\cos \vartheta). \end{aligned} \quad (5.S22)$$

- d) Using the same configuration as in part c), find the magnitude and direction of the electric field at the origin.

Solution: The system is symmetric under rotations around the z -axis, so we expect the field at the origin to point along z . Even if the coordinate system is singular in $r = 0$, the origin is a perfectly regular point for the electric field which is thus continuous there. In order to obtain it, it is therefore sufficient to compute the radial derivative of the potential along the semi-line $\vartheta = 0$ and take the limit $r \rightarrow 0$:

$$E_z(r, \vartheta = 0) = -\partial_r \Phi_1(r, 0) = -\frac{Q}{8\pi\varepsilon_0 R} \sum_{n=1}^{\infty} \frac{n}{2n+1} \frac{r^{n-1}}{R^n} [P_{n+1}(\cos \alpha) - P_{n-1}(\cos \alpha)] P_n(1). \quad (5.S23)$$

Remembering that $P_n(1) = 1$ and taking $r \rightarrow 0$ we find (only the $n = 1$ term does not vanish)

$$E_z = \frac{Q}{8\pi\varepsilon_0 R^2} \frac{1}{3} [P_0(\cos \alpha) - P_2(\cos \alpha)]. \quad (5.S24)$$

Using the expressions for the Legendre polynomials

$$P_0(\cos \alpha) = 1, \quad P_2(\cos \alpha) = \frac{3}{2} \cos^2(\alpha) - \frac{1}{2}, \quad (5.S25)$$

yields:

$$\vec{E} = \frac{Q \sin^2 \alpha}{16\pi\varepsilon_0 R^2} \vec{e}_z. \quad (5.S26)$$

5.2. Rotation gymnastics

The rotation group (in N dimensions) is defined as the set of linear mappings of a vector space which leave the canonical scalar product invariant.

- a) Prove that a linear transformation which leaves the norm of all vectors invariant also conserves the scalar product between two arbitrary vectors. Then show that any matrix preserving the norm of all vectors is orthogonal.

Solution: Given two vectors $\vec{v}, \vec{w}$ we have

$$(\vec{v} + \vec{w})^2 = \vec{v}^2 + 2\vec{v} \cdot \vec{w} + \vec{w}^2, \quad (5.S27)$$

therefore solving with respect to $\vec{v} \cdot \vec{w}$ we see that any transformation preserving the norm also leaves every scalar product unchanged (the converse is trivial since $\vec{v}^2 = \vec{v} \cdot \vec{v}$). This means that for every linear mapping R preserving the norm

$$(R\vec{v}) \cdot (R\vec{w}) = (R\vec{v})^T (R\vec{w}) = \vec{v}^T R^T R \vec{w} \stackrel{!}{=} \vec{v}^T \vec{w} = \vec{v} \cdot \vec{w} \quad \text{for all } v, w \quad (5.S28)$$

which implies $R^T R = I$ with I the identity matrix. In other words, $R^T = R^{-1}$.

- b) Show that the determinant of any orthogonal matrix is either $+1$ or -1 .

Solution: Since $\det(A) = \det(A^T)$ and $\det(AB) = \det(A)\det(B)$, we find

$$\det(R^T R) = [\det(R)]^2 = \det(I) = 1. \quad (5.S29)$$

Orthogonal matrices with negative determinant represent transformations that involve reflection. Since we are interested in rotations, let us restrict ourselves to the group of matrices with positive determinant, i.e. the special orthogonal group $SO(N)$.

- c) Write down the matrices that represent rotations of an infinitesimal angle $\delta\varphi$ around the i -th axis in three dimensions and subtract the identity from each of them. The resulting matrices are the generators of the rotations in 3D. Find a simple expression of the generators in terms of the totally antisymmetric tensor ε_{ijk} .

Solution: The three rotation matrices around the cartesian axes are just 2D rotations in the plane perpendicular to each cartesian axis. In the 3D representation, we extend the 2D rotation matrices with an identity-like column and row that is such that it leaves the corresponding cartesian rotation axes invariant, namely

$$R_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{pmatrix}, \quad R_2 = \begin{pmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{pmatrix}, \quad R_3 = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (5.S30)$$

Their infinitesimal versions are obtained by the replacements $\sin \varphi \rightarrow \delta\varphi$, $\cos \varphi \rightarrow 1$, subtracting the identity and dividing by $i\delta\varphi$ we find

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & +i \\ 0 & -i & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ +i & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & +i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (5.S31)$$

Therefore $(L_i)_{jk} = i\varepsilon_{ijk}$.

- d) Show that infinitesimal rotations commute with each other up to higher order terms, whereas macroscopic rotations do not commute in general.

Solution: We may write the infinitesimal rotation around the i -th axis of an angle $\delta\varphi_i$ as $R_i \sim I + i\delta\varphi_i L_i$, therefore

$$R_i R_j \sim I + i\delta\varphi_i L_i + i\delta\varphi_j L_j \sim R_j R_i, \quad (5.S32)$$

up to terms that are order $(\delta\varphi)^2$. In order to show that two finite rotations along different axes do not in general commute a counterexample is sufficient. Here we choose to compute $R_1 R_2$ and $R_2 R_1$ and to take $\varphi = \pi/2$ to make the calculation even easier. Using

$$R_1|_{\varphi=\pi/2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad R_2|_{\varphi=\pi/2} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad (5.S33)$$

we get

$$R_1 R_2|_{\varphi=\pi/2} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad R_2 R_1|_{\varphi=\pi/2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}. \quad (5.S34)$$

- e) Write down the infinitesimal rotation of angle $\delta\varphi$ around a generic unit vector $\vec{n}$ (use the fact that $\vec{n}$ is left unchanged). By performing a large number of such rotations, extend the result to macroscopic angles φ around $\vec{n}$. Show that for every rotation with $\varphi \in (0, 2\pi)$ and arbitrary $\vec{n}$, there exists another representation with different φ' , $\vec{n}'$.

Solution: We then want to find a rotation of the form $R_n = I + \sum_i i\delta\varphi_i L_i$ that leaves $\vec{n}$ invariant, which means that

$$R_n \vec{n} = \vec{n} + \sum_i i\delta\varphi_i L_i \vec{n} \stackrel{!}{=} \vec{n}. \quad (5.S35)$$

Therefore

$$\left[\sum_i i\delta\varphi_i L_i \vec{n} \right]_j = \sum_i \delta\varphi_i \varepsilon_{ijk} n_k = 0, \quad (5.S36)$$

which is obviously satisfied if $\delta\varphi_i = \delta\varphi n_i$: we conclude $R_n = I + i\delta\varphi \vec{n} \cdot \vec{L}$. The angle φ becomes infinitesimal after dividing it into m small angles and taking the limit $m \rightarrow \infty$, so that $\delta\varphi = \varphi/m$. We find

$$R_{n,\varphi} = \lim_{m \rightarrow \infty} (R_{n,\varphi/m})^m = \lim_{m \rightarrow \infty} \left(I + i \frac{\varphi}{m} \vec{n} \cdot \vec{L} \right)^m = \exp(i\varphi \vec{n} \cdot \vec{L}), \quad (5.S37)$$

where the exponential of a matrix is defined by its series. There are two equivalent rotations in this representation if $\varphi \in (0, 2\pi)$, since the one around $\vec{n}$ of angle φ is equivalent to the one around $-\vec{n}$ of angle $2\pi - \varphi$.

5.3. Magnetic field of a finite coil

Consider a wire coiled up cylindrically along the z -axis. Let R be the radius of this cylindrical coil and L its length (it starts at $z = -L/2$ and ends at $z = +L/2$). Let $n = N/L$ be the winding number per unit length and I the (constant) current flowing through the wire. You may neglect boundary effects.

Calculate the z -component of the magnetic flux density B for points on the symmetry axis. Determine the magnetic field for $L \rightarrow \infty$ at constant n .

Solution: We can parametrise the coil in cylindrical coordinates:

$$\vec{x}'(t) = \begin{pmatrix} R \cos t \\ R \sin t \\ \tilde{z}(t) \end{pmatrix} \quad \implies \quad d\vec{x}'(t) = \begin{pmatrix} -R \sin t \\ R \cos t \\ \tilde{z}'(t) \end{pmatrix} dt \quad (5.S38)$$

with

$$\tilde{z}(t) = -\frac{L}{2} + \frac{L}{2\pi N} t \quad (5.S39)$$

such that $t \in [0, 2\pi N]$. Here $N = nL$ is the total winding number. The Biot-Savart law

$$\vec{B}(x) = \frac{\mu_0 I}{4\pi} \int \frac{(\vec{x}' - \vec{x}) \times d\vec{x}'}{\|\vec{x} - \vec{x}'\|^3} \quad (5.S40)$$

then yields for all points on the z -axis:

$$\vec{B}(0, 0, z) = \frac{\mu_0 I}{4\pi} \int_0^{2\pi N} dt \frac{1}{(R^2 + (\tilde{z}(t) - z)^2)^{3/2}} \begin{pmatrix} R \cos t \\ R \sin t \\ \tilde{z}(t) - z \end{pmatrix} \times \begin{pmatrix} -R \sin t \\ R \cos t \\ \tilde{z}'(t) \end{pmatrix}. \quad (5.S41)$$

By symmetry, we are only interested in the z -component of this field because other components are due to boundary effects. The z -component simplifies to

$$B_z(0, 0, z) = \frac{\mu_0 I}{4\pi} \int_0^{2\pi N} dt \frac{R^2}{(R^2 + (\tilde{z}(t) - z)^2)^{3/2}}. \quad (5.S42)$$

Performing a substitution $x = \tilde{z}(t) - z$ we get

$$\begin{aligned} B_z(0, 0, z) &= \frac{\mu_0 I N}{2L} \int_{-L/2-z}^{L/2-z} \frac{dx R^2}{(R^2 + x^2)^{3/2}} \\ &= \left[\frac{\mu_0 I N}{2L} \frac{x}{\sqrt{R^2 + x^2}} \right]_{x=-L/2-z}^{L/2-z} \\ &= \frac{\mu_0 I N}{2L} \left(\frac{1}{\sqrt{1 + (2R/(L - 2z))^2}} + \frac{1}{\sqrt{1 + (2R/(L + 2z))^2}} \right), \end{aligned} \quad (5.S43)$$

where we used $\int dx R^2 (x^2 + R^2)^{-3/2} = x(x^2 + R^2)^{-1/2}$.

Taking the limit $L \rightarrow \infty$ for constant $n = N/L$ gives

$$B_z(0, 0, z) \rightarrow \mu_0 I n. \quad (5.S44)$$

5.4. Current in a cylindrical wire

Consider a straight cylindrical wire of radius R oriented along the z -axis. The magnitude of the current density inside this wire depends on the distance from the centre of the wire as follows:

$$j(\rho) = j_0 e^{-\rho^2/R^2} \theta(R - \rho), \quad (5.5)$$

where $\rho = \sqrt{x^2 + y^2}$ and $\theta(x)$ is the unit step function.

a) Find the current I flowing through the wire. Express j_0 through I .

Solution: Sketch of the situation



(5.S45)

The current can be calculated using $I = \int_A dx^2 \vec{n} \cdot \vec{j}$.

The cross-section area element $dx^2 \vec{n}$ can be parametrised using cylindrical coordinates

$$\vec{x} = \begin{pmatrix} \rho \cos \varphi \\ \rho \sin \varphi \\ z \end{pmatrix}, \quad (5.S46)$$

with $0 \leq \rho \leq R$, $0 \leq \varphi \leq 2\pi$, z along the length of the wire.

$$dx^2 \vec{n} = (\vec{x}_\rho \times \vec{x}_\varphi) d\rho d\varphi = \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix} \times \begin{pmatrix} -\rho \sin \varphi \\ \rho \cos \varphi \\ 0 \end{pmatrix} d\rho d\varphi = \rho d\rho d\varphi \vec{e}_z. \quad (5.S47)$$

Thus the current can be calculated using the given formula

$$\begin{aligned} I &= \int_0^{2\pi} d\varphi \int_0^R d\rho \rho j_0 e^{-\rho^2/R^2} = 2\pi j_0 \left[-\frac{1}{2} R^2 e^{-\rho^2/R^2} \right]_{\rho=0}^R \\ &= -\pi j_0 R^2 (e^{-1} - 1) = \pi j_0 R^2 \frac{e - 1}{e}. \end{aligned} \quad (5.S48)$$

Therefore:

$$j_0 = \frac{I e}{\pi R^2 (e - 1)}. \quad (5.S49)$$

- b) Find the magnetic field inside and outside the wire as a function of the total current. Sketch the field lines, paying attention to the direction. Let the current flow into the positive z -direction.

Solution: Ampère's law states

$$\oint_{\partial D} \vec{B} \cdot d\vec{x} = \mu_0 I = \mu_0 \int_D dx^2 \vec{n} \cdot \vec{j}. \quad (5.S50)$$

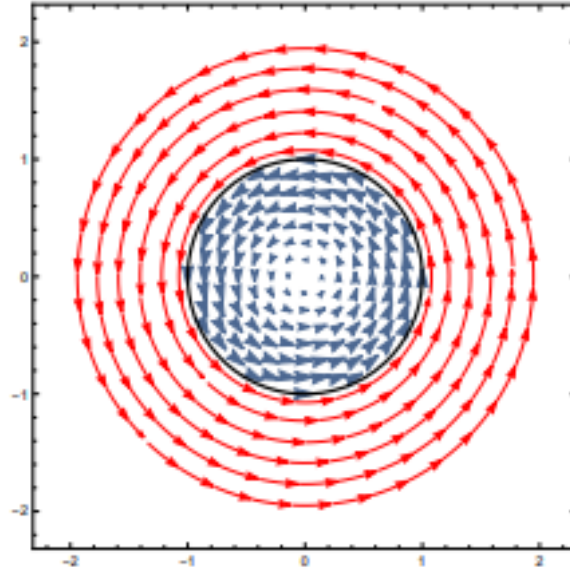
Choose D to be a disk of radius r in the xy -plane, centred in the origin and with normal pointing along the positive z axis. Since the current flows in the z -direction, the magnetic field has a non-trivial component only in the φ -direction. If $r < R$ (inside the wire), then

$$\begin{aligned} B_\varphi(r) 2\pi r &= \mu_0 \int_0^{2\pi} d\varphi \int_0^r d\rho \rho j_0 e^{-\rho^2/R^2}, \\ B_\varphi(r) &= \frac{\mu_0 j_0}{r} \int_0^r d\rho \rho e^{-\rho^2/R^2} = \frac{\mu_0 j_0}{r} \left[-\frac{1}{2} R^2 e^{-\rho^2/R^2} \right]_{\rho=0}^r = \frac{\mu_0 I e}{2\pi r (e-1)} \left(1 - e^{-r^2/R^2} \right) \\ \vec{B} &= \frac{\mu_0 j_0 R^2}{2r} \left(1 - e^{-r^2/R^2} \right) \vec{e}_\varphi = \frac{\mu_0 I e}{2\pi r (e-1)} \left(1 - e^{-r^2/R^2} \right) \vec{e}_\varphi. \end{aligned} \quad (5.S51)$$

If $r > R$ (outside the wire), then

$$B_\varphi(r) 2\pi r = \mu_0 I, \quad \vec{B} = \frac{\mu_0 I}{2\pi r} \vec{e}_\varphi. \quad (5.S52)$$

So the field is always parallel to $\vec{e}_\varphi$. One can check that both solutions match on the boundary. (Here the solid line denotes the edge of the wire. The current comes out of the sheet. Blue arrows represent the magnetic field inside the wire and red lines the field outside)



(5.S53)

6.1. Multipole expansion in spherical coordinates

We first recall the multipole expansion of the scalar potential. Given a localised distribution of charge described by the charge density $\rho(x')$, the potential $\Phi(x)$ outside the region where $\rho(x')$ is non-vanishing can be expanded as:

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{Q_{\ell,m}}{2\ell+1} \frac{1}{r^{\ell+1}} Y_{\ell,m}(\vartheta, \varphi), \quad (6.1)$$

where $Y_{\ell,m}(\vartheta, \varphi)$ denotes the spherical harmonics and $Q_{\ell,m}$ the multipole moments. The latter are defined as

$$Q_{\ell,m} = \int d^3x' Y_{\ell,m}^*(\vartheta', \varphi') r'^{\ell} \rho(x'). \quad (6.2)$$

Consider now a circular loop of radius R lying in the x,y -plane centred at the origin with linear charge density $\lambda = \lambda_0 \cos \varphi$, where φ is the azimuthal angle measured in the x,y -plane.

a) Express the charge density $\rho(x')$ given by the loop in spherical coordinates.

Solution: In spherical coordinates we have

$$\vec{x} = \begin{pmatrix} r \sin \vartheta \cos \varphi \\ r \sin \vartheta \sin \varphi \\ r \cos \vartheta \end{pmatrix}. \quad (6.S1)$$

Firstly we consider the charge dQ contained in an arc of length $r \sin \vartheta d\varphi$. Given that the circular loop has radius R and lies in the x,y -plane we can fix $r = R$, $\cos \vartheta = 0$, hence:

$$dQ = \lambda_0 \cos(\varphi) \delta(r - R) \delta(\cos \vartheta) r \sin \vartheta d\varphi dr d\vartheta. \quad (6.S2)$$

Now we can write the the charge distribution $\rho(x)$ as:

$$\rho(x) = \frac{dQ}{d^3x} = \frac{\lambda_0 \cos(\varphi) \delta(r - R) \delta(\cos \vartheta) r \sin \vartheta d\varphi dr d\vartheta}{r^2 \sin \vartheta dr d\vartheta d\varphi} = \frac{\lambda_0 \cos(\varphi)}{r} \delta(r - R) \delta(\cos \vartheta). \quad (6.S3)$$

b) Compute the multipole moments $Q_{\ell,m}$ and use them to write down the multipole expansion of $\Phi(x)$ up to $\ell = 3$ for $\|x\| > R$. Discuss the nature of the resulting multipoles.

Solution: We recall the expression of the spherical harmonics $Y_{\ell,m}(\vartheta, \varphi)$:

$$Y_{\ell,m}(\vartheta, \varphi) = \sqrt{(2\ell+1) \frac{(\ell-m)!}{(\ell+m)!}} P_{\ell,m}(\cos \vartheta) e^{im\varphi} \quad (6.S4)$$

where $P_{\ell,m}(\cos \vartheta)$ are the associated Legendre functions. For $Q_{\ell,m}$ we have:

$$\begin{aligned} Q_{\ell,m} &= \int Y_{\ell,m}^*(\vartheta', \varphi') r'^{\ell} \rho(x') d^3x' \\ &= \lambda_0 \sqrt{(2\ell+1) \frac{(\ell-m)!}{(\ell+m)!}} \\ &\quad \cdot \int_0^{2\pi} d\varphi' \int_{-1}^1 d\cos \vartheta' \int_0^{\infty} r'^2 P_{\ell,m}(\cos \vartheta') e^{-im\varphi'} r'^{\ell} \frac{\lambda_0 \cos \varphi'}{r'} \delta(r' - R) \delta(\cos \vartheta') \\ &= \lambda_0 \sqrt{(2\ell+1) \frac{(\ell-m)!}{(\ell+m)!}} R^{\ell+1} P_{\ell,m}(0) \int_0^{2\pi} d\varphi \cos \varphi e^{-im\varphi}. \end{aligned} \quad (6.S5)$$

We notice that last integral is non-vanishing only for $m = \pm 1$, indeed:

$$\int_0^{2\pi} d\varphi \cos \varphi e^{-im\varphi} = \int_0^{2\pi} d\varphi \frac{1}{2} [e^{i(1-m)\varphi} + e^{i(1+m)\varphi}] = \pi [\delta_{m,1} + \delta_{m,-1}], \quad (6.S6)$$

thus we have:

$$Q_{\ell,m} = \lambda_0 \pi \sqrt{(2\ell+1) \frac{(\ell-m)!}{(\ell+m)!}} R^{\ell+1} P_{\ell,m}(0) [\delta_{m,1} + \delta_{m,-1}]. \quad (6.S7)$$

Since $m = \pm 1$ we need $\ell > 0$ and if we consider the parity property of the $P_{\ell,m}(x)$ namely:

$$P_{\ell,m}(-x) = (-1)^{\ell+m} P_{\ell,m}(x), \quad (6.S8)$$

we need $\ell+1$ to be even. Thus we have multipoles of odd order $\ell = 1, 3, 5, \dots$. In addition we use the relation between the associated Legendre functions:

$$P_{\ell,-m}(x) = (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_{\ell,m}(x), \quad (6.S9)$$

to get $Q_{\ell,-1} = -Q_{\ell,1}$.

Before calculating the potential $\Phi(x)$, we observe that:

$$\begin{aligned} & Q_{\ell,1} Y_{\ell,1}(\vartheta, \varphi) + Q_{\ell,-1} Y_{\ell,-1}(\vartheta, \varphi) \\ &= Q_{\ell,1} Y_{\ell,1}(\vartheta, \varphi) + (-1)^2 Q_{\ell,1} Y_{\ell,1}^*(\vartheta, \varphi) \\ &= 2Q_{\ell,1} \operatorname{Re}(Y_{\ell,1}(\vartheta, \varphi)) \\ &= 2Q_{\ell,1} \sqrt{(2\ell+1) \frac{(\ell-1)!}{(\ell+1)!}} P_{\ell,1}(\cos \vartheta) \cos \varphi \\ &= 2\pi \lambda_0 R^{\ell+1} (2\ell+1) \frac{(\ell-1)!}{(\ell+1)!} P_{\ell,1}(0) P_{\ell,1}(\cos \vartheta) \cos \varphi. \end{aligned} \quad (6.S10)$$

Using the expansion for $\Phi(x)$ in $\|x\| = r > R$ we have:

$$\begin{aligned} \Phi(x) &= \frac{1}{4\pi\epsilon_0} \sum_{\ell=1, \text{odd}}^{\infty} \frac{1}{2\ell+1} \frac{1}{r^{\ell+1}} [Q_{\ell,1} Y_{\ell,1}(\vartheta, \varphi) + Q_{\ell,-1} Y_{\ell,-1}(\vartheta, \varphi)] \\ &= \frac{\lambda_0}{2\epsilon_0} \cos \varphi \sum_{n=0}^{\infty} \left(\frac{R^2}{r^2} \right)^{n+1} \frac{(2n)!}{(2n+2)!} P_{2n+1,1}(0) P_{2n+1,1}(\cos \vartheta). \end{aligned} \quad (6.S11)$$

For the first two non-vanishing terms we need:

$$\begin{aligned} P_{1,1}(x) &= -\sqrt{1-x^2}, \\ P_{3,1}(x) &= -\frac{3}{2} (5x^2 - 1) \sqrt{1-x^2}, \end{aligned} \quad (6.S12)$$

thus we have:

$$\Phi(x) = \frac{\lambda_0 \cos \varphi \sin \vartheta}{4\epsilon_0} \frac{R^2}{r^2} \left[1 + \frac{R^2}{r^2} \frac{3}{8} (1 - 5 \cos^2 \vartheta) + \dots \right]. \quad (6.S13)$$

The first dominant term is a dipole, indeed we expect no monopole contribution since the net charge on the loop is zero. The second term $\ell = 3$ is the octupole contribution.

- c) Calculate the electric field $\vec{E}(x)$ with $\|x\| > R$ of the dipole and quadrupole moments determined in part b).

Solution: The definition of the dipole moment $\vec{P}$ is

$$\vec{P} = \int dx' \rho(x') \vec{x}', \quad (6.S14)$$

but in this case it is more convenient to use the expression for the potential to read it off:

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{\vec{P} \cdot \vec{x}}{r^3} + \dots \right] \implies \vec{P} = \pi\lambda_0 R^2 \vec{e}_x. \quad (6.S15)$$

The field in $\vec{x}$ given by the dipole term is ($\vec{n} = \vec{x}/\|x\|$, $r = \|x\|$):

$$\vec{E}_{\text{dip}}(x) = \frac{3\vec{n}(\vec{P} \cdot \vec{n}) - \vec{P}}{4\pi\epsilon_0\|x\|^3} = \frac{3\lambda_0 R^2}{4\epsilon_0 r^3} \begin{pmatrix} \sin^2 \vartheta \cos^2 \varphi - \frac{1}{3} \\ \frac{1}{2} \sin^2 \vartheta \sin(2\varphi) \\ \frac{1}{2} \sin(2\vartheta) \cos \varphi \end{pmatrix}. \quad (6.S16)$$

Since there is no quadrupole contribution to the potential, the corresponding field is vanishing.

6.2. Capacitor filled with dielectric

Consider a parallel plate capacitor with quadratic plates of edge length a and distance d between the plates. It is charged to the amount $\pm Q$ and subsequently separated from the voltage source. When the charged capacitor is placed on top of a dielectric fluid (with density ρ_{fl} , permittivity ϵ_r), the fluid rises between the plates up to a maximal height h_0 .



- a) Find the electrostatic energy $W_{\text{el}}(h)$ stored in the capacitor as a function of the height of the raised fluid h and the parameters defined above.

Solution: View the setup as two capacitors, one with height h filled with the dielectric fluid and one with height $a - h$ with vacuum inbetween the plates. Since the capacity is additive we get:

$$W_{\text{el}}(h) = \frac{1}{2} \frac{Q^2}{C_{\text{ges}}} = \frac{1}{2} \frac{Q^2}{C_1 + C_2} = \frac{1}{2} \frac{Q^2}{\epsilon_0 \epsilon_r a h / d + \epsilon_0 a (a - h) / d} = \frac{d}{2\epsilon_0 a} \frac{Q^2}{(\epsilon_r - 1)h + a}. \quad (6.S17)$$

- b) Find the potential energy $W_{\text{pot}}(h)$ of the fluid between the plates as a function of h .

Solution: The potential energy of a fluid layer with mass dm at the height h' is

$$dW_{\text{pot}} = gh' dm, \quad dm = \rho_{\text{fl}} dV, \quad dV = da dh', \quad (6.S18)$$

wherefore

$$W_{\text{pot}} = \int_0^h dW_{\text{pot}} = \rho_{\text{fl}} g a d \int_0^h h' dh' = \frac{1}{2} \rho_{\text{fl}} g a d h^2. \quad (6.S19)$$

- c) Derive the defining equation for h_0 under the assumption that the total energy is minimised. Which amount of charge must be taken to the capacitor, such that the fluid rises up to half of the total height? Assume that $a = 20$ cm, $d = 5$ mm, $\epsilon_r = 3$ and $\rho_{\text{fl}} = 0.8$ g/cm³.
-

Solution: The total energy is

$$W_{\text{ges}}(h) = W_{\text{el}}(h) + W_{\text{pot}}(h) = \frac{dQ^2}{2\varepsilon_0 a} \frac{1}{(\varepsilon_r - 1)h + a} + \frac{1}{2} \rho_{\text{B}} g a d h^2. \quad (6.S20)$$

Minimising the energy depending on h ,

$$W'_{\text{ges}}(h_0) \stackrel{!}{=} 0, \quad (6.S21)$$

gives

$$\frac{dQ^2}{2\varepsilon_0 a} \frac{\varepsilon_r - 1}{[(\varepsilon_r - 1)h_0 + a]^2} = \rho_{\text{B}} g d a h_0 \quad (6.S22)$$

and

$$\frac{Q^2(\varepsilon_r - 1)}{2g\varepsilon_0\rho_{\text{B}}a^2} = h_0[(\varepsilon_r - 1)h_0 + a]^2. \quad (6.S23)$$

Then the defining equation for h_0 it follows,

$$(\varepsilon_r - 1)^2 h_0^3 + 2a(\varepsilon_r - 1)h_0^2 + a^2 h_0 - \frac{Q^2(\varepsilon_r - 1)}{2g\varepsilon_0\rho_{\text{B}}a^2} = 0. \quad (6.S24)$$

From (6.S23) it follows

$$Q^2 = \frac{2h_0[(\varepsilon_r - 1)h_0 + a]^2 g\varepsilon_0\rho_{\text{B}}a^2}{\varepsilon_r - 1}, \quad (6.S25)$$

and for the case $h_0 = a/2$ with the above data and $\varepsilon_0 = 8.854188 \cdot 10^{-12} \text{ F/m}$ we obtain,

$$Q = 6.67 \cdot 10^{-6} \text{ C}. \quad (6.S26)$$

6.3. Magnetic field of a circular loop

Consider a conducting wire forming a circle of radius R in the centre of the x,y -plane. A constant current I flows counter-clockwise through this loop.

a) Calculate the magnetic field $\vec{B}$ at some point on the z -axis.

Solution: We use the Biot-Savart law,

$$\vec{B}(x) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{x}' \times (\vec{x} - \vec{x}')}{\|\vec{x} - \vec{x}'\|^3}. \quad (6.S27)$$

In cylindrical coordinates we have

$$\vec{x}' = \begin{pmatrix} R \cos \varphi' \\ R \sin \varphi' \\ 0 \end{pmatrix} \quad \Rightarrow \quad d\vec{x}' = \begin{pmatrix} -R \sin \varphi' \\ R \cos \varphi' \\ 0 \end{pmatrix} d\varphi'. \quad (6.S28)$$

Thus for all points on the z -axis the magnetic field is

$$\vec{B}(0, 0, z) = \frac{\mu_0 I}{4\pi} \int_0^{2\pi} d\varphi' \frac{1}{(R^2 + z^2)^{3/2}} \begin{pmatrix} -R \sin \varphi' \\ R \cos \varphi' \\ 0 \end{pmatrix} \times \begin{pmatrix} -R \cos \varphi' \\ -R \sin \varphi' \\ z \end{pmatrix}. \quad (6.S29)$$

By symmetry the only nonzero component of the field $\vec{B}$ is B_z . The latter simplifies to

$$\begin{aligned} B_z(0, 0, z) &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} d\varphi' \frac{R^2}{\sqrt{R^2 + z^2}^3} \\ &= \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}. \end{aligned} \quad (6.S30)$$

- b) Now calculate the magnetic field $\vec{B}$ at an arbitrary point in the x,y -plane. You should obtain a result which is not analytically solvable, but can be expressed in terms of elliptic integrals. Check that it agrees with the result from part a) at the origin.

Solution: The position-vector of a point in the x,y -plane can be parametrised using cylindrical coordinates as follows,

$$\vec{r} = \begin{pmatrix} d \cos \varphi \\ d \sin \varphi \\ 0 \end{pmatrix}, \quad (6.S31)$$

where d is the distance between the field point and the centre of the loop. Using the Biot-Savart law we get

$$\begin{aligned} \vec{B}(d, \varphi, 0) &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} d\varphi' \frac{1}{\sqrt{R^2 + d^2 - 2Rd \cos(\varphi' - \varphi)^3}} \begin{pmatrix} -R \sin \varphi' \\ R \cos \varphi' \\ 0 \end{pmatrix} \times \begin{pmatrix} d \cos \varphi - R \cos \varphi' \\ d \sin \varphi - R \sin \varphi' \\ 0 \end{pmatrix} \\ &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} d\varphi' \frac{R^2 - Rd \cos(\varphi' - \varphi)}{\sqrt{R^2 + d^2 - 2Rd \cos(\varphi' - \varphi)^3}} \vec{e}_z. \end{aligned} \quad (6.S32)$$

The last integral is not computable analytically, but it can be expressed in terms of elliptic integrals of the first and second kind, which are defined as follows,

$$\begin{aligned} F(\varphi, k) &= \int_0^\varphi \frac{d\varphi'}{\sqrt{1 - k^2 \sin^2 \varphi'}}, \\ E(\varphi, k) &= \int_0^\varphi d\varphi' \sqrt{1 - k^2 \sin^2 \varphi'}. \end{aligned} \quad (6.S33)$$

The latter ones can be computed numerically.

We can verify our results by computing the magnetic field at the centre of the loop using the results for the magnetic field on the axis and in the x,y -plane. Indeed, setting $z = 0$ in (6.S30), and $d = 0$, $\varphi = 0$ in (6.S32), we get the same result for the magnetic field in the centre of the loop, namely

$$\vec{B}_{\text{centre}} = \frac{\mu_0 I}{2R} \vec{e}_z. \quad (6.S34)$$

6.4. Magnetic moment of a rotating spherical shell

A spherical shell of radius R and charge Q (homogeneously distributed on the surface) is rotating around its z -axis with angular velocity $\vec{\omega} = \omega \vec{e}_z$.

- a) Calculate the current density $\vec{j}(\vec{x}) = \vec{v}(\vec{x})\rho(\vec{x})$.

Solution: Assume that the vector $\vec{r}$ is given in spherical coordinates

$$\vec{r} = r \begin{pmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{pmatrix}. \quad (6.S35)$$

The charge density of the shell is $\rho(\vec{x}) = (Q/4\pi R^2)\delta(\|\vec{x}\| - R)$. The velocity of the shell element is

$$\vec{v}(\vec{x}) = \vec{\omega} \times \vec{r} = \omega \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \omega \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} = \omega r \sin(\vartheta) \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix}, \quad (6.S36)$$

which, together, gives

$$\vec{j}(\vec{x}) = \frac{Q\omega r \sin \vartheta}{4\pi R^2} \delta(r - R) \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix}. \quad (6.S37)$$

b) Calculate the magnetic moment $\vec{m} = \frac{1}{2} \int d\vec{x}^3 (\vec{x} \times \vec{j}(x))$ of the spherical shell.

Solution: We calculate the magnetic moment using a given integral

$$\begin{aligned}
 \vec{m} &= \frac{1}{2} \int d\vec{x}^3 \frac{Q\omega \sin \vartheta}{4\pi R^2} \delta(r - R) r^2 \begin{pmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{pmatrix} \times \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} \\
 &= \frac{1}{2} \frac{Q\omega}{4\pi} R^2 \int_0^{2\pi} d\varphi \int_0^\pi d\vartheta \sin^2 \vartheta \begin{pmatrix} -\cos \vartheta \cos \varphi \\ -\cos \vartheta \sin \varphi \\ \sin \vartheta \end{pmatrix} \\
 &= \frac{1}{4} Q\omega R^2 \vec{e}_z \int_0^\pi d\vartheta \sin^3 \vartheta \\
 &= \frac{Q\omega R^2}{3} \vec{e}_z.
 \end{aligned} \tag{6.S38}$$

Note that the x and y -components integrate to 0.

c) Show that the leading behaviour of the magnetic field generated by this sphere for $\|x\| \gg R$ is that of a magnetic dipole, and write down the leading term of $\vec{B}$.

Hint: Use the Biot-Savart law and keep only the leading non-vanishing terms in $R/\|x\|$.

Solution: The Biot-Savart law states

$$\vec{B}(x) = \frac{\mu_0}{4\pi} \int d\vec{x}'^3 \frac{\vec{j}(\vec{x}') \times (\vec{x} - \vec{x}')}{\|\vec{x} - \vec{x}'\|^3}. \tag{6.S39}$$

First, we expand the denominator to the first order in r'/r :

$$\begin{aligned}
 \frac{1}{\|\vec{x} - \vec{x}'\|} &= \frac{1}{\|x\|} \left(1 + \frac{\vec{x} \cdot \vec{x}'}{\|x\|^2} + \mathcal{O}(r'^2/r^2) \right), \\
 \frac{1}{\|\vec{x} - \vec{x}'\|^3} &= \frac{1}{\|x\|^3} \left(1 + \frac{3\vec{x} \cdot \vec{x}'}{\|x\|^2} + \mathcal{O}(r'^2/r^2) \right).
 \end{aligned} \tag{6.S40}$$

Plugging this into the previous equation gives

$$\vec{B}(x) = \frac{\mu_0}{4\pi} \int d\vec{x}'^3 \left(\underbrace{\frac{\vec{j}(\vec{x}') \times \vec{x}}{\|x\|^3}}_{\rightarrow 0} - \underbrace{\frac{\vec{j}(\vec{x}') \times \vec{x}'}{\|x\|^3}}_{\rightarrow -2\vec{m}/\|x\|^2} + \underbrace{\frac{3(\vec{x} \cdot \vec{x}') \vec{j}(\vec{x}') \times \vec{x}}{\|x\|^5}}_{\rightarrow I_3} - \underbrace{\frac{3(\vec{x} \cdot \vec{x}') \vec{j}(\vec{x}') \times \vec{x}'}{\|x\|^5}}_{\rightarrow \mathcal{O}(r'^2/r^2)} + \dots \right). \tag{6.S41}$$

We shall neglect the last term and all other terms of $\mathcal{O}(r'/r)^2$ or higher.

Taking into account that

$$\int_0^{2\pi} d\varphi \sin \varphi = \int_0^{2\pi} d\varphi \cos \varphi = \int_0^{2\pi} d\varphi \underbrace{\sin \varphi \cos \varphi}_{=\frac{1}{2} \sin(2\varphi)} = 0, \tag{6.S42}$$

and

$$\int_0^{2\pi} d\varphi \sin^2 \varphi = \int_0^{2\pi} d\varphi \cos^2 \varphi = \pi, \tag{6.S43}$$

the first term obviously integrates to 0, and the third term can be significantly simplified:

$$\begin{aligned}
 I_3 &= \int d\vec{x}'^3 \frac{3(\vec{x} \cdot \vec{x}') \vec{j}(\vec{x}') \times \vec{x}}{\|x\|^5} \\
 &= \frac{3Q\omega}{4\pi R^2 \|x\|^5} \int_0^\infty dr' r'^3 \delta(r' - R) \int_0^\pi d\vartheta' \sin^2 \vartheta' \int_0^{2\pi} d\varphi' (\vec{x} \cdot \vec{x}') \begin{pmatrix} -\sin \varphi' \\ \cos \varphi' \\ 0 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix}.
 \end{aligned} \tag{6.S44}$$

We use

$$(\vec{x} - \vec{x}') = r' [x \sin \vartheta' \cos \varphi' + y \sin \vartheta' \sin \varphi' + z \cos \vartheta'], \quad (6.S45)$$

and

$$\begin{pmatrix} -\sin \varphi' \\ \cos \varphi' \\ 0 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} z \cos \varphi' \\ z \sin \varphi' \\ -y \sin \varphi' - x \cos \varphi' \end{pmatrix}, \quad (6.S46)$$

and take into account first (6.S42), and then (6.S43):

$$\begin{aligned} I_3 &= \frac{3Q\omega R^2}{4\pi\|\vec{x}\|^5} \int_{-1}^1 d\cos \vartheta' \sin \vartheta' \int_0^{2\pi} d\varphi' \begin{pmatrix} xz \sin \vartheta' \cos^2 \varphi' \\ yz \sin \vartheta' \sin^2 \varphi' \\ -x^2 \sin \vartheta' \cos^2 \varphi' - y^2 \sin \vartheta' \sin^2 \varphi' \end{pmatrix} \\ &= \frac{3Q\omega R^2}{4\pi\|\vec{x}\|^5} \underbrace{\int_{-1}^1 d\cos \vartheta' (1 - \cos^2 \vartheta')}_{=4/3} \underbrace{\begin{pmatrix} xz \\ yz \\ zz - \vec{x}^2 \end{pmatrix}}_{=z\vec{x} - \vec{x}^2 \vec{e}_z} \\ &= \underbrace{\frac{Q\omega R^2}{3}}_{=|\vec{m}|} \frac{1}{\|\vec{x}\|^5} \left(3 \underbrace{(\vec{x} \cdot \vec{e}_z)}_{=z} \vec{x} - 3\|\vec{x}\|^2 \vec{e}_z \right) = \frac{3(\vec{x} \cdot \vec{m})\vec{x}}{\|\vec{x}\|^5} - \frac{3\vec{m}}{\|\vec{x}\|^3}. \end{aligned} \quad (6.S47)$$

Reinserting I_3 into (6.S41) and combining with the second term gives

$$\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} \left(\frac{3(\vec{x} \cdot \vec{m})\vec{x}}{\|\vec{x}\|^5} - \frac{\vec{m}}{\|\vec{x}\|^3} \right) + \dots = \frac{\mu_0 Q R^2}{12\pi} \left(\frac{3(\vec{x} \cdot \vec{\omega})\vec{x}}{\|\vec{x}\|^5} - \frac{\vec{\omega}}{\|\vec{x}\|^3} \right) + \dots \quad (6.S48)$$

-
- d) Now let $\vec{x}'$ be a vector such that $\vec{x}' \perp \vec{\omega}$ and $\|\vec{x}'\| \gg R$. Calculate the lowest-order contribution of the force exerted by the magnetic field from the previous part on another identical sphere placed at a point $\vec{x}'$ and rotating with angular velocity $\vec{\omega}'$ parallel to $\vec{\omega}$. Due to the large distance between the spheres you can approximate them as two point-like objects carrying some magnetic moment.
-

Solution: Due to the choice of $\vec{x}'$, only the second term in (6.S48) contributes:

$$\vec{B}(\vec{x}') = -\frac{\mu_0 Q R^2}{12\pi} \frac{\vec{\omega}}{\|\vec{x}'\|^3} + \dots \quad (6.S49)$$

The lowest-order contribution to the force a magnetic field exerts on a magnetic moment is

$$\vec{F}(\vec{x}') = -\frac{QR^2}{3} \underbrace{\frac{\mu_0 Q R^2}{12\pi} \vec{\omega}' \cdot \vec{\omega} \vec{\nabla} \frac{1}{\|\vec{x}\|^3}}_{=3\vec{x}'/\|\vec{x}'\|^5} + \dots = \frac{\mu_0 Q^2 R^4 \omega \omega'}{12\pi} \frac{\vec{x}'}{\|\vec{x}'\|^5} + \dots \quad (6.S50)$$

6.5. Vector potential of a loop of wire

A circular loop of wire of radius a and negligible thickness carries a current I . The coordinates are chosen such that the wire lies in the x,y -plane. We are interested in the vector potential of the current density and want to obtain the corresponding magnetic field.

- a) Find a vector potential for the loop using the formula

$$\vec{A} = \mu_0 \int d\vec{x}'^3 \frac{\vec{j}(\vec{x}')}{4\pi\|\vec{x} - \vec{x}'\|}. \quad (6.3)$$

Hint: Expand $1/\|\vec{x} - \vec{x}'\|$ in spherical harmonics and use that $Y_{\ell,-m} = (-1)^m Y_{\ell,m}^*$.

Solution: First, we expand $1/\|x - x'\|$ in spherical harmonics and obtain

$$\begin{aligned}\vec{A} &= \mu_0 \int dx'^3 \frac{\vec{j}(x')}{4\pi\|x - x'\|} \\ &= \mu_0 \sum_{\ell, m} \int dr' r'^2 d\vartheta' \sin \vartheta' d\varphi' \frac{\vec{j}(x')}{4\pi} \frac{1}{2\ell + 1} \frac{r_{<}^\ell}{r_{>}^{\ell+1}} Y_{\ell, m}^*(\vartheta', \varphi') Y_{\ell, m}(\vartheta, \varphi)\end{aligned}\quad (6.S51)$$

where $r_{>} = \max(r, r')$ and $r_{<} = \min(r, r')$. The current density of the ring is given in spherical coordinates by the expression

$$\vec{j}(x') = \frac{I}{r'} \delta(\cos \vartheta') \delta(r' - a) \vec{e}_{\varphi'}.\quad (6.S52)$$

We substitute this and the expression for Y^* in terms of Legendre polynomials and perform the integrals over delta-functions

$$\vec{A} = \frac{\mu_0 I a}{4\pi} \sum_{\ell, m} \int d\varphi' \frac{1}{2\ell + 1} \frac{r_{<}^\ell}{r_{>}^{\ell+1}} \sqrt{(2\ell + 1) \frac{(\ell - m)!}{(\ell + m)!}} P_\ell^m(0) e^{-im\varphi'} Y_{\ell, m}(\vartheta, \varphi) \vec{e}_{\varphi'},\quad (6.S53)$$

where now, $r_{>} = \max(r, a)$ and $r_{<} = \min(r, a)$. For the integration in the φ' variable we observe

$$\int d\varphi' e^{-im\varphi'} \vec{e}_{\varphi'} = \int d\varphi' e^{-im\varphi'} [-\sin \varphi' \vec{e}_x + \cos \varphi' \vec{e}_y] = \pi \delta_{m,1} (i\vec{e}_x + \vec{e}_y) + \pi \delta_{m,-1} (-i\vec{e}_x + \vec{e}_y).\quad (6.S54)$$

From this, together with $Y_{\ell, -m} = (-1)^m Y_{\ell, m}^*$, it follows

$$\begin{aligned}\vec{A} &= \frac{2\pi\mu_0 I a}{4\pi} \sum_{\ell} \frac{1}{2\ell + 1} \frac{r_{<}^\ell}{r_{>}^{\ell+1}} (2\ell + 1) \frac{(\ell - 1)!}{(\ell + 1)!} P_\ell^1(0) P_\ell^1(\cos \vartheta) [\cos(\varphi) \vec{e}_y - \sin(\varphi) \vec{e}_x] \\ &= \frac{\mu_0 I a}{2} \sum_{\ell} \frac{r_{<}^\ell}{r_{>}^{\ell+1}} \frac{1}{\ell(\ell + 1)} P_\ell^1(0) P_\ell^1(\cos \vartheta) \vec{e}_\varphi.\end{aligned}\quad (6.S55)$$

Note that due to the symmetry of the Legendre polynomials

$$P_n(-x) = (-1)^n P_n(x), \quad \text{and} \quad P_\ell^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x),\quad (6.S56)$$

we have that $P_\ell^1(0) = 0$ for ℓ even and hence only the odd summands survive.

- b)** Based on the above result, calculate the magnetic field at points x with $\|x\| < a$. In problem 6.3 we have seen that the magnetic field a) on the z -axis and b) in the x, y -plane of the loop always points in the z -direction. Can you recover this result?

Hint: Use that

$$P_\ell^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x).\quad (6.4)$$

Solution: In the interior of the loop, the vector potential is given by

$$\vec{A} = \frac{\mu_0 I}{2} \sum_{\ell \text{ odd}} \frac{r^\ell}{a^\ell} \frac{1}{\ell(\ell + 1)} P_\ell^1(0) P_\ell^1(\cos \vartheta) \vec{e}_\varphi.\quad (6.S57)$$

The magnetic field can be obtained from a straight-forward calculation $\vec{B} = \vec{\nabla} \times \vec{A}$. In spherical coordinates, the general expression for the rotation is given by

$$\vec{\nabla} \times \vec{A} = \frac{1}{r \sin \vartheta} \left[\frac{\partial}{\partial \vartheta} (\sin \vartheta A_\varphi) - \frac{\partial A_\vartheta}{\partial \varphi} \right] \vec{e}_r + \frac{1}{r} \left[\frac{1}{\sin \vartheta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right] \vec{e}_\vartheta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\vartheta) - \frac{\partial A_r}{\partial \vartheta} \right] \vec{e}_\varphi.\quad (6.S58)$$

We substitute the specific form $\vec{A} = A_\varphi \vec{e}_\varphi$ and find

$$\vec{B} = \frac{1}{r \sin \vartheta} \frac{\partial}{\partial \vartheta} (\sin \vartheta A_\varphi) \vec{e}_r - \frac{1}{r} \frac{\partial}{\partial r} (r A_\varphi) \vec{e}_\vartheta.\quad (6.S59)$$

To carry out the derivatives, we use

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x) \quad (6.S60)$$

to find

$$\begin{aligned} \frac{d}{dx} P_\ell^1(x) &= x(1-x^2)^{-1/2} \frac{d}{dx} P_\ell(x) + (-1)^1 (1-x^2)^{1/2} \frac{d^2}{dx^2} P_\ell(x) \\ &= -\frac{x}{1-x^2} P_\ell^1(x) - \frac{1}{\sqrt{1-x^2}} P_\ell^2(x), \end{aligned} \quad (6.S61)$$

and therefore

$$\begin{aligned} \frac{\partial}{\partial \vartheta} P_\ell^1(\cos \vartheta) &= \frac{\cos \vartheta}{\sin \vartheta} P_\ell^1(\cos \vartheta) + \frac{\sin \vartheta}{|\sin \vartheta|} P_\ell^2(\cos \vartheta) \\ &= \frac{\cos \vartheta}{\sin \vartheta} P_\ell^1(\cos \vartheta) + P_\ell^2(\cos \vartheta). \end{aligned} \quad (6.S62)$$

Review: Jackson 5.5 has a more compact expression. In total we obtain

$$\vec{B} = \frac{\mu_0 I}{2a} \sum_{\ell \text{ odd}} \frac{r^\ell - 1}{a^\ell - 1} \frac{1}{\ell} P_\ell^1(0) \left\{ \frac{1}{\ell+1} \left[2 \frac{\cos \vartheta}{\sin \vartheta} P_\ell^1(\cos \vartheta) + P_\ell^2(\cos \vartheta) \right] \vec{e}_r - P_\ell^1(\cos \vartheta) \vec{e}_\vartheta \right\}. \quad (6.S63)$$

Note that $P_\ell^m(\pm 1) = 0$ for $m > 0$ and $|P_\ell^1(\cos \vartheta)/\sin \vartheta| \rightarrow |d/dx P_\ell(x)|$ for $\vartheta \rightarrow 0, \pi$, cf. (6.S60). From this it follows that $\vec{B} \parallel \vec{e}_r$ for $\vartheta = 0, \pi$. This we have already found in problem 6.3a). On the other hand, for $\vartheta = \pi/2$ we find that $\cos \vartheta = 0$. Again, due to the symmetry of the Legendre polynomials $P_n(-x) = (-1)^n P_n(x)$ we have $P_\ell^2(0) = 0$ for ℓ odd and therefore $\vec{B} \parallel \vec{e}_\vartheta$ which coincides with problem 6.3b).

- c) Assume now that the current loop is put into an iron cavity with infinite relative permeability. What are the changes for the magnetic field? What would you have to do to solve the problem (no calculation)?

Solution: The iron cavity changes the boundary condition for the magnetic field. At the boundary of the cavity, the parallel component of the magnetic field $\vec{B}$ must vanish. This can be taken into account by adding to the vector potential $\vec{A}$ another vector potential $\vec{A}_{\text{hom}}$ which is a solution of the homogeneous equation

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}_{\text{hom}}) = 0. \quad (6.S64)$$

We would have to find the general solution to the homogeneous problem and determine the unknown coefficients by the new boundary conditions. In this way we obtain a solution for the same set of differential equations, i.e. with the same current density, but different boundary conditions.

6.6. Iron pipe in a magnetic field

An infinitely long hollow cylinder (inner radius b , outer radius a) is placed with its axis orthogonally to an initially homogeneous magnetic field $\vec{B}_0$. The hollow cylinder is made of iron (permeability μ). The initial field $\vec{B}_0$ can be assumed to be sufficiently small not to saturate the iron, and the permeability μ is constant in the region of interest.



- a) Derive the expression for $\vec{B}$ in the cavity ($r < b$).

Hint: Use the absence of free currents to describe the magnetic field H by means of a scalar potential Φ via $\vec{H} = -\vec{\nabla}\Phi$. The Laplace equation holds in all three relevant regions of space. In cylindrical coordinates (r, φ, z) the Laplacian reads

$$\Delta = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}. \quad (6.5)$$

The boundary conditions of H and B at the surfaces as well as the behaviour for $r \rightarrow 0$ and $r \rightarrow \infty$ fix the constants in the solution of the differential equation.

Solution: Since there are no free currents, the vector field $\vec{H}$ is irrotational. Hence we can introduce a scalar potential Φ , that satisfies $\vec{H} = -\vec{\nabla}\Phi$. Since $\vec{B} = \mu\vec{H}$ with $\mu = \mu_0\mu_r$, we obtain that Φ satisfies the Laplace equation $\Delta\Phi = 0$. In cylindrical coordinates (r, φ, z) we get

$$\left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2} \right\} \Phi = 0. \quad (6.S65)$$

Here we take z along the cylinder axis, and $x = r \cos \varphi$ along the original magnetic field direction, $\vec{B}_0 = B_0 \vec{e}_x$. Because of the translational symmetry along the cylinder $\partial_z \vec{B} = 0$ and we have that $\partial_z \partial \Phi = \partial \partial_z \Phi = 0$ and hence

$$\Phi(r, \varphi, z) = \Phi(r, \varphi) + az, \quad a \in \mathbb{R}. \quad (6.S66)$$

The term az gets killed by Δ and we make a separation ansatz

$$\Phi(r, \varphi) = R(r)S(\varphi) \quad (6.S67)$$

to obtain

$$\frac{1}{R} \left(r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} \right) = - \frac{1}{S} \frac{d^2 S}{d\varphi^2} = \text{const} =: m^2, \quad (6.S68)$$

where the first expression does not depend on φ , and the second does not depend on r . The equation for R is a type of Euler differential equation with the general solution $R_m(r) = c_m r^m + d_m r^{-m}$, while the equation for harmonic S has a solution $S_m(\varphi) = e_m \cos(m\varphi) + f_m \sin(m\varphi)$. Because of periodicity, $\Phi(r, \varphi + 2\pi) = \Phi(r, \varphi)$, m can only be an integer. There is the special case $m = 0$, where we find the solution $R_0(r) = c_0 \log(r/r_0)$ and $S_0(\varphi) = g_0$ (due to periodicity). Therefore, in all three regions $i = 1, 2, 3$ in the figure the potential Φ_i has the form

$$\Phi_i(r, \varphi, z) = c_0^i \log(r/r_0^i) + \sum_{m=1}^{\infty} [(g_m^i r^m + h_m^i r^{-m}) \cos(m\varphi) + (j_m^i r^m + k_m^i r^{-m}) \sin(m\varphi)] + a^i z. \quad (6.S69)$$

To have a unique potential, the solution must have the symmetry properties of the boundary condition. This means that

$$j_m^i = 0, \quad k_m^i = 0, \quad a^i = 0, \quad i \in \{1, 2, 3\}. \quad (6.S70)$$

Far away from the cylinder we have $\vec{\nabla}\Phi_3 = -\vec{H}_0$, i.e., $\Phi_3 = -H_0 x = -(B_0/\mu_0)r \cos \varphi$. Comparing this with (6.S69), we obtain

$$g_1^3 = -\frac{B_0}{\mu_0}, \quad g_{m \neq 1}^3 = 0, \quad c_0^3 = 0. \quad (6.S71)$$

In addition, Φ_1 must be finite when $r \rightarrow 0$, hence

$$c_0^1 = 0, \quad h_m^1 = 0. \quad (6.S72)$$

The remaining coefficients can be determined from the boundary conditions, which are that H_φ and B_r are continuous, i.e.,

$$\begin{aligned} \mu_0 \frac{\partial \Phi_1}{\partial r} \Big|_{r=b} &= \mu \frac{\partial \Phi_2}{\partial r} \Big|_{r=b}, & \mu \frac{\partial \Phi_2}{\partial r} \Big|_{r=a} &= \mu_0 \frac{\partial \Phi_3}{\partial r} \Big|_{r=a}, \\ \frac{\partial \Phi_1}{\partial \varphi} \Big|_{r=b} &= \frac{\partial \Phi_2}{\partial \varphi} \Big|_{r=b}, & \frac{\partial \Phi_2}{\partial \varphi} \Big|_{r=a} &= \frac{\partial \Phi_3}{\partial \varphi} \Big|_{r=a}. \end{aligned} \quad (6.S73)$$

For this, one needs the partial derivatives of Φ_i for $i = 1, 2, 3$,

$$\begin{aligned} \frac{\partial \Phi_i}{\partial r} &= c_0^i r^{-1} + \sum_{m=1}^{\infty} m [g_m^i r^{m-1} - h_m^i r^{-m-1}] \cos(m\varphi) \quad (i = 1, 2, 3), \\ \frac{\partial \Phi_i}{\partial \varphi} &= - \sum_{m=1}^{\infty} m [g_m^i r^m + h_m^i r^{-m}] \sin(m\varphi) \quad (i = 1, 2, 3), \end{aligned} \quad (6.S74)$$

As the boundary conditions must hold for all φ , we get $c_0^2 = 0$ and

$$\begin{aligned} (\mu_0 g_m^1 - \mu g_m^2) b^{2m} + \mu h_m^2 &= 0, & (\mu g_m^2 - \mu_0 g_m^3) a^{2m} - (\mu h_m^2 - \mu_0 h_m^3) &= 0, \\ (g_m^1 - g_m^2) b^{2m} - h_m^2 &= 0, & (g_m^2 - g_m^3) a^{2m} + (h_m^2 - h_m^3) &= 0. \end{aligned} \quad (6.S75)$$

Solving the two equations for $r = b$ for g_m^2 and h_m^2 gives

$$g_m^2 = \frac{\mu + \mu_0}{2\mu} g_m^1, \quad (6.S76)$$

$$h_m^2 = b^{2m} \frac{\mu - \mu_0}{2\mu} g_m^1. \quad (6.S77)$$

Now we can plug these into the equations for $r = a$, and eliminate h_m^3 to obtain

$$g_m^1 = \frac{4\mu\mu_0 a^{2m}}{a^{2m}(\mu + \mu_0)^2 - b^{2m}(\mu - \mu_0)^2} g_m^3. \quad (6.S78)$$

From (6.S71) it follows that

$$g_1^1 = \frac{-4\mu a^2 B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2},$$

$$g_m^1 = 0, \quad (m \neq 1), \quad (6.S79)$$

and for the potential we find

$$\Phi_1(r, \varphi, z) = g_1^1 r \cos \varphi. \quad (6.S80)$$

Hence we obtain the magnetic field in the cavity

$$\begin{aligned} \vec{B}_1 &= -\mu_0 \vec{\nabla} \Phi_1 = -\mu_0 \frac{\partial \Phi_1}{\partial r} \vec{e}_r - \frac{\mu_0}{r} \frac{\partial \Phi_1}{\partial \varphi} \vec{e}_\varphi \\ &= -\mu_0 g_1^1 [\cos \varphi \vec{e}_r - \sin \varphi \vec{e}_\varphi] = -\mu_0 g_1^1 \vec{e}_x \\ &= \frac{4\mu\mu_0 a^2}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \vec{B}_0. \end{aligned} \quad (6.S81)$$

For a ferromagnet such as iron $\mu \gg \mu_0$, and for leading order in μ_0/μ we obtain

$$\vec{B}_1 = \frac{4\mu_0}{\mu} \frac{\vec{B}_0}{1 - (b/a)^2}. \quad (6.S82)$$

The following lines are not requested in the problem, but they are useful for sketching the field lines in the next part.

Inserting g_m^1 into (6.S76) and (6.S77) delivers

$$\begin{aligned} g_1^2 &= \frac{-2a^2(\mu + \mu_0)B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2}, & g_m^2 &= 0, & m &\neq 1, \\ h_1^2 &= \frac{-2a^2b^2(\mu - \mu_0)B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2}, & h_m^2 &= 0, & m &\neq 1, \\ h_1^3 &= \frac{(\mu^2 - \mu_0^2)a^2(a^2 - b^2)B_0}{\mu_0(a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2)}, & h_m^3 &= 0, & m &\neq 1, \end{aligned} \quad (6.S83)$$

and inserting all coefficients into the potential Φ_i gives

$$\begin{aligned} \Phi_1 &= \frac{-4\mu a^2 B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} r \cos(\varphi), \\ \Phi_2 &= \frac{-2a^2 B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \left((\mu + \mu_0)r + (\mu - \mu_0) \frac{b^2}{r} \right) \cos(\varphi), \\ \Phi_3 &= \frac{B_0}{\mu_0} \left(-r + \frac{a^2}{r} \frac{(\mu^2 - \mu_0^2)(a^2 - b^2)}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \right) \cos(\varphi). \end{aligned} \quad (6.S84)$$

With

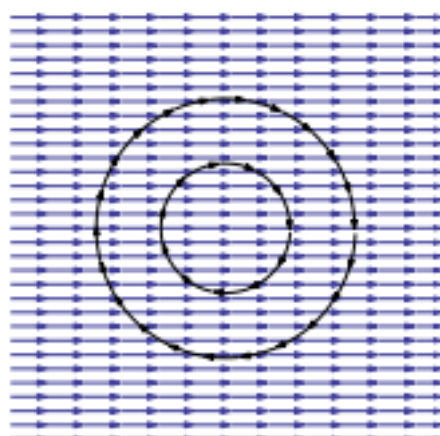
$$\vec{B}_i = \mu_i \vec{H}_i = -\mu_i \vec{\nabla} \Phi_i = -\mu_i \frac{\partial \Phi_i}{\partial r} \vec{e}_r - \frac{\mu_i}{r} \frac{\partial \Phi_i}{\partial \varphi} \vec{e}_\varphi, \quad \mu_1 = \mu_3 = \mu_0, \quad \mu_2 = \mu_r \mu_0 = \mu, \quad (6.S85)$$

the magnetic field reads

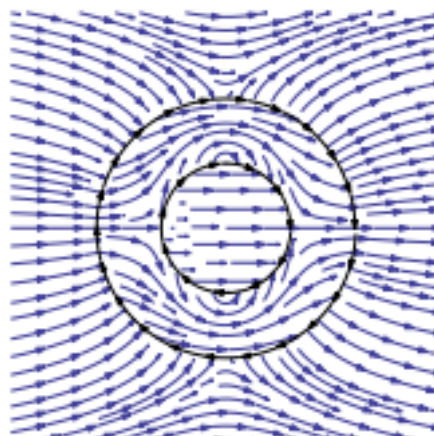
$$\begin{aligned}
 \vec{B}_1 &= \frac{4\mu\mu_0 a^2 B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \underbrace{(\cos(\varphi)\vec{e}_r - \sin(\varphi)\vec{e}_\varphi)}_{=\vec{e}_x}, \\
 \vec{B}_2 &= \frac{2\mu a^2 B_0}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \left((\mu + \mu_0) \underbrace{(\cos(\varphi)\vec{e}_r - \sin(\varphi)\vec{e}_\varphi)}_{=\vec{e}_x} - (\mu - \mu_0) \frac{b^2}{r^2} \underbrace{(\cos(\varphi)\vec{e}_r + \sin(\varphi)\vec{e}_\varphi)}_{=\cos(2\varphi)\vec{e}_x + \sin(2\varphi)\vec{e}_y} \right), \\
 \vec{B}_3 &= B_0 \left(\underbrace{(\cos(\varphi)\vec{e}_r - \sin(\varphi)\vec{e}_\varphi)}_{=\vec{e}_x} + \frac{a^2}{r^2} \frac{(\mu^2 - \mu_0^2)(a^2 - b^2)}{a^2(\mu + \mu_0)^2 - b^2(\mu - \mu_0)^2} \underbrace{(\cos(\varphi)\vec{e}_r + \sin(\varphi)\vec{e}_\varphi)}_{=\cos(2\varphi)\vec{e}_x + \sin(2\varphi)\vec{e}_y} \right). \quad (6.S86)
 \end{aligned}$$

- b) Sketch the magnetic field lines in the full region of space, before and after the cylinder has been placed in the field. Consider also the cases of a paramagnetic ($\mu_r > 1$), a diamagnetic ($\mu_r < 1$), and a superconducting ($\mu_r = 0$) cylinder.

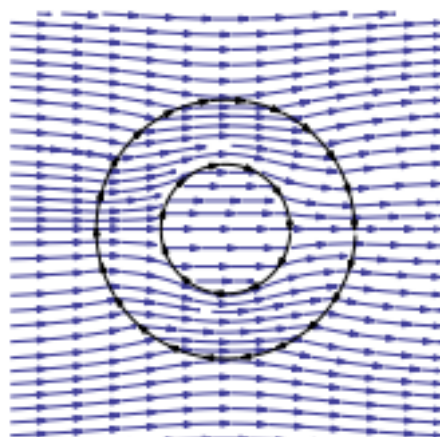
Solution: The magnetic field is homogeneous before the tube is placed in it. Since the iron as a ferromagnet ($\mu_r \gg 1$) amplifies the field in its interior, the field lines get curved towards the iron cylinder. Likewise for a paramagnet ($\mu_r > 1$). In a diamagnetic medium ($\mu_r < 1$) the field lines get curved away from the cylinder, while in the case of a superconductor ($\mu_r = 0$) they would be completely pushed out from the cylinder, and the $\vec{B}$ -field would not enter the hollow cylinder, neither the cavity inside.



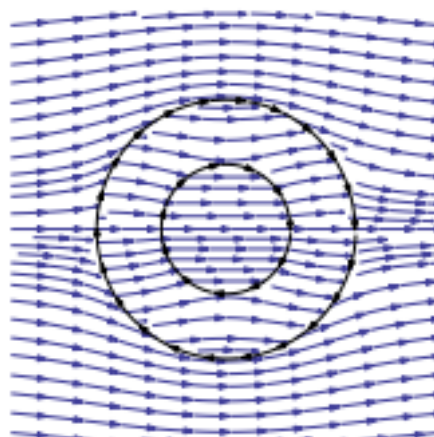
$\vec{B}$ for vacuum ($\mu_r = 1$)



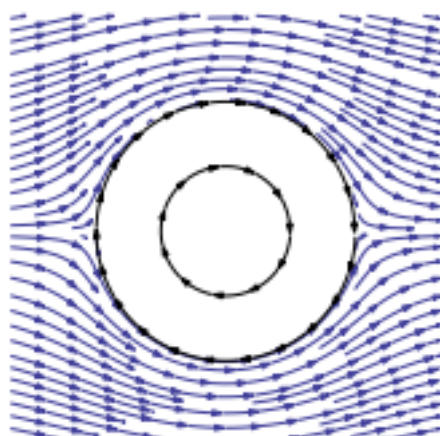
$\vec{B}$ for ferromagnetic material ($\mu_r \gg 1$)



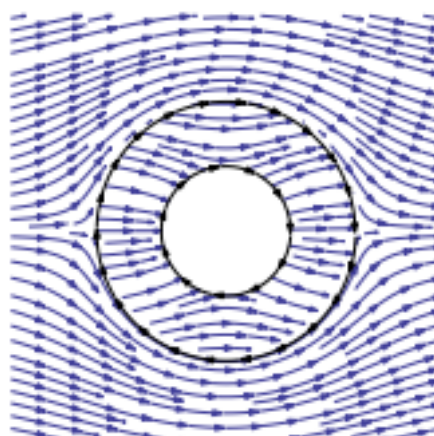
$\vec{B}$ for paramagnetic material ($\mu_r > 1$)



$\vec{B}$ for diamagnetic material ($\mu_r < 1$)



$\vec{B}$ for superconductor ($\mu_r = 0$)



$\vec{H}$ for superconductor ($\mu_r = 0$)

6.7. Charged particle in an electromagnetic field

Consider a point particle carrying charge q in an electromagnetic field described by a vector potential A and a scalar potential Φ . The Lagrangian of the particle is given by

$$L(x, \dot{x}, t) = \frac{1}{2}m\dot{x}^2 + q\dot{x} \cdot \vec{A}(x, t) - q\Phi(x, t), \quad (6.6)$$

where x is the position of the particle and m is its mass.

a) Determine the canonical momentum $\vec{p}$,

$$p_i = \frac{\partial L(x, \dot{x}, t)}{\partial \dot{x}_i}. \quad (6.7)$$

What is the relation between the canonical momentum p and the kinetic momentum $m\dot{x}$? Perform a Legendre transformation to determine the Hamiltonian,

$$H(x, p, t) = \vec{p} \cdot \dot{\vec{x}} - L(x, \dot{x}, t). \quad (6.8)$$

Solution: The canonical momenta read

$$p_i = \frac{\partial L(x, \dot{x}, t)}{\partial \dot{x}_i} = m\dot{x}_i + qA_i(x, t), \quad (6.887)$$

where $m\dot{x}$ is the kinetic momentum and $q\vec{A}$ is the electromagnetic momentum. The Hamiltonian can be determined as

$$\begin{aligned} H(x, p, t) &= \vec{p} \cdot \dot{\vec{x}} - L(x, \dot{x}, t) \\ &= \vec{p} \cdot \dot{\vec{x}} - \frac{1}{2}m\dot{\vec{x}}^2 - q\dot{\vec{x}} \cdot \vec{A}(x, t) + q\Phi(x, t) \\ &= \frac{1}{2}m\dot{\vec{x}}^2 + q\Phi(x, t) = \frac{1}{2}m^{-1}(\vec{p} - q\vec{A}(x, t))^2 + q\Phi(x, t). \end{aligned} \quad (6.888)$$

b) Use $\vec{B} = \vec{\nabla} \times \vec{A}$ to explicitly verify

$$\sum_{j=1}^3 \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right) \dot{x}_j = (\dot{\vec{x}} \times \vec{B})_i. \quad (6.9)$$

Solution: We first formulate the relation between the potential and the field in components

$$\vec{B} = \vec{\nabla} \times \vec{A} \quad \implies \quad B_i = \varepsilon_{ijk} \partial_j A_k. \quad (6.889)$$

With the relationship $\sum_i \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$ we have:

$$\begin{aligned} (\dot{\vec{x}} \times \vec{B})_i &= \varepsilon_{ijk} \dot{x}_j B_k = \varepsilon_{ijk} \dot{x}_j \varepsilon_{klm} \partial_l A_m \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \dot{x}_j \partial_l A_m = \dot{x}_m \partial_i A_m - \dot{x}_l \partial_l A_i \\ &= (\partial_i A_j - \partial_j A_i) \dot{x}_j. \end{aligned} \quad (6.890)$$

c) Start from the Hamiltonian equations

$$\dot{p}_i = - \frac{\partial H}{\partial x_i}, \quad \dot{x}_i = \frac{\partial H}{\partial p_i}, \quad (6.10)$$

and derive the equation of motion for the charged particle in an electromagnetic field

$$m\ddot{\vec{x}} = q(\vec{E} + \dot{\vec{x}} \times \vec{B}). \quad (6.11)$$

Solution: The Hamiltonian equations for this problem are

$$\begin{aligned} \dot{p}_i &= - \frac{\partial H}{\partial x_i} = \frac{1}{m} (p_i - qA_i) q \partial_i A_j - q \partial_i \Phi, \\ \dot{x}_i &= \frac{\partial H}{\partial p_i} = \frac{1}{m} (p_i - qA_i). \end{aligned} \quad (6.891)$$

From these we get

$$\begin{aligned} m\ddot{x}_i &= \frac{d}{dt} (p_i - qA_i) = \dot{p}_i - q(\partial_j A_i \dot{x}_j + \partial_t A_i) \\ &= q(-\partial_i \Phi - \partial_t A_i) - q(\partial_j A_i - \partial_i A_j) \dot{x}_j. \end{aligned} \quad (6.892)$$

Using the expressions for the $\vec{E}$ and $\vec{B}$ field in terms of the potentials, we find

$$m\ddot{\vec{x}} = q(\vec{E} + \dot{\vec{x}} \times \vec{B}). \quad (6.893)$$

7.1. Electric Field of a Moving Wire

Consider an infinitely long wire carrying a constant current I in the $\vec{e}_z$ direction, moving in the $\vec{e}_y$ direction at constant speed v . We aim to find the electric field of this configuration at the moment where the wire coincides with the z axis.

- Compute the magnetic field of the static wire in cylindrical coordinates.
- Via a coordinate transformation to Cartesian coordinates, transform the y coordinate such that the magnetic field now corresponds to the moving wire.
- Compute the curl of the electric field generated by this time-varying magnetic field.
- Use the translational symmetry along z and Maxwell's equations to solve for the electric field when the wire coincides with the z axis.

7.2. Induction in a magnetic field

A homogeneous magnetic field B is aligned along the z -axis. Within this magnetic field, a conducting wire forming a circle of radius R rotates with circular velocity $\vec{\omega}$. Its rotational axis lies in the plane of the conductor and passes through its centre. Let ϑ be the angle between the rotational axis and the field direction. There is no voltage induced in the loop at time $t = 0$.

Find the induced voltage in the conductor as a function of time.

Solution: This exercise is a simple application of Faraday's law, to show the link between a time-varying magnetic flux and the voltage that it induces.


(7.S1)

From Faraday's induction law, the induced voltage equals a negative time derivative of the magnetic flux through the conductor loop,

$$U = -\dot{\Psi}. \quad (7.S2)$$

The magnetic flux is the product of the magnitude of the magnetic field B and the projection of the surface inside the loop onto the x,y -plane, taking into account the initial conditions:

$$\Psi = B\pi R^2 \sin \vartheta \cos(\omega t), \quad (7.S3)$$

and hence

$$U = -\dot{\Psi} = \pi R^2 B \omega \sin \vartheta \sin(\omega t). \quad (7.S4)$$

More detailed: We choose the coordinate system such that $\vec{\omega}$ lies in the y,z -plane:

$$\vec{\omega} = \omega \begin{pmatrix} 0 \\ \sin \vartheta \\ \cos \vartheta \end{pmatrix}. \quad (7.S5)$$

We want the normal vector to the circle enclosed by the loop, $\vec{n}$, to always be perpendicular to $\vec{\omega}$, and rotate with angular velocity ω . Its phase ϕ is not determined a priori, we can determine it when we impose the initial condition. One thus obtains:

$$\vec{n} = \begin{pmatrix} \sin(\omega t + \phi) \\ -\cos(\omega t + \phi) \cos \vartheta \\ \cos(\omega t + \phi) \sin \vartheta \end{pmatrix}. \quad (7.S6)$$

Then

$$\begin{aligned}
\Psi &= \int_A dx^2 \vec{n} \cdot \vec{B} = B \int_A dx^2 \vec{n} \cdot \vec{e}_z \\
&= B \int_0^{2\pi} d\vartheta \int_0^R d\rho \rho \begin{pmatrix} \sin(\omega t + \phi) \\ -\cos(\omega t + \phi) \cos \vartheta \\ \cos(\omega t + \phi) \sin \vartheta \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
&= B\pi R^2 \sin \vartheta \cos(\omega t + \phi),
\end{aligned} \tag{7.S7}$$

which is the same result as in (7.S4), if $\phi = 0$.

7.3. Self-induction of a coaxial cable

A coaxial cable is represented by two coaxial conducting cylindrical shells with radii R_1 and R_2 with $R_1 < R_2$. A current I is flowing through each of the cylindrical shells along their axes in opposite directions. Compute the magnetic field of the cable and sketch the corresponding magnetic field lines. Then calculate the self-induction per unit length of this coaxial cable. Check that the formula for the magnetic energy of the cable holds:

$$W = \frac{1}{2} LI^2. \tag{7.1}$$

Hint: Determine the self-induction from the magnetic flux through a surface enclosed by the current via

$$\Psi = LI. \tag{7.2}$$

Solution: From Ampère's law,

$$\oint_{\gamma} d\vec{x} \cdot \vec{B} = \mu_0 I, \tag{7.S8}$$

we get the magnetic field of a conducting cylindrical shell of radius R (in cylindrical coordinates based on the axis of the cable)

$$\vec{B}(x) = \frac{\mu_0 I}{2\pi r} \theta(r - R) \vec{e}_{\varphi}. \tag{7.S9}$$

Thus we can compute the magnetic field of the coaxial cable as superposition of the field of a cylindrical shell with current I and radius R_1 and a cylindrical shell of radius R_2 and current $-I$,

$$\vec{B}(x) = \frac{\mu_0 I}{2\pi r} \begin{cases} 0, & r < R_1, \\ \vec{e}_{\varphi}, & R_1 < r < R_2, \\ 0, & R_2 < r. \end{cases} \tag{7.S10}$$

We can now compute the magnetic flux through a surface bounded by the current, we choose a rectangle of length ℓ coinciding with the axis of the cable and length w in the radial direction, such that $w > R_2$ (although since the field is only non-zero between R_1 and R_2 , w will not appear in our answer). We obtain:

$$\begin{aligned}
\Psi &= \int dx^2 \vec{B}(x) \cdot \vec{n} \\
&= \int_{-\ell/2}^{\ell/2} dz \int_0^w dr \frac{\mu_0 I}{2\pi r} \\
&= \frac{\mu_0 \ell}{2\pi} I \log \frac{R_2}{R_1}.
\end{aligned} \tag{7.S11}$$

From that we can identify the self-induction L as

$$L = \frac{\mu_0 I}{2\pi} \ell \log \frac{R_2}{R_1}, \tag{7.S12}$$

so that we find the self-induction per unit length as

$$\frac{L}{\ell} = \frac{\mu_0}{2\pi} \log \frac{R_2}{R_1}. \quad (7.S13)$$

Now we can compute the energy of the magnetic field W for a cable of length ℓ ,

$$\begin{aligned} W &= \frac{1}{2\mu_0} \int dx^3 \|\vec{B}(x)\|^2 \\ &= \int_{-\ell/2}^{\ell/2} dz \int_0^{2\pi} d\varphi \int_{R_1}^{R_2} dr \, r \left(\frac{\mu_0 I}{2\pi r} \right)^2 \\ &= \frac{1}{2} \frac{\mu_0 \ell}{2\pi} I^2 \log \frac{R_2}{R_1}. \end{aligned} \quad (7.S14)$$

We find that this concords with the given formula for the magnetic field energy in terms of the self-inductance.

8.1. Dynamics of an electric circuit

Consider the circuit shown in the figure below. The circuit consists of a voltage source U_0 , a resistor of resistance R , a capacitor of capacitance C , a solenoid of inductance L and a switch.



Let the switch be in position 1 initially, so the voltage source, resistor and capacitor form a circuit. Assume that the capacitor is initially uncharged.

- a) Write down a differential equation for the charge on the capacitor, solve it, and calculate the time it takes the capacitor to charge to 90% of its maximal charge.

Solution: According to Kirchhoff's first law, the sum of voltage drops on all components of a closed loop is 0. The voltage drop of the source is considered negative, it is $U_B = -U_0$. Using Ohm's law $U_R = IR$ and using the definition of capacitance $U_C = Q/C$. Hence we get

$$U_B + U_R + U_C = -U_0 + \dot{Q}R + Q/C = 0. \quad (8.S1)$$

This is an inhomogeneous first-order differential equation. The homogeneous equation has the solution $Q_{\text{hom}} = K \exp(-t/RC)$, and a particular solution of the inhomogeneous equations is simply a constant $Q_0 = U_0 C$. The total solution is $Q(t) = Q_0 + Q_{\text{hom}}$. Since $Q(0) = U_0 C + K = 0$, we get the final solution

$$Q(t) = U_0 C \left[1 - \exp\left(-\frac{t}{RC}\right) \right]. \quad (8.S2)$$

As can be seen, the capacitor will never fully charge to Q_0 . However, it will be charged to 90% if

$$1 - \exp\left(-\frac{t}{RC}\right) = 0.9 \quad \Rightarrow \quad \exp\left(-\frac{t}{RC}\right) = 0.1 \quad \Rightarrow \quad t = RC \log(10). \quad (8.S3)$$

Now the switch is flipped to position 2, and the solenoid, resistor and capacitor form a circuit. Assume that the capacitor is fully charged when the switch is set to position 2 at time $t = 0$. Correspondingly, there are no currents flowing in the circuit initially.

- b) Write down a differential equation for the charge on the capacitor in the case $R = 0$, solve it, and find the frequency of the the circuit.

Solution: The voltage drop along the solenoid is $U_L = L\dot{I}$, and at the capacitor $U_C = Q/C$. According to Kirchhoff's rule the sum of these has to vanish:

$$U_C + U_L = Q/C + L\dot{Q} = 0. \quad (8.S4)$$

This equation has solutions $Q = A \cos(t/\sqrt{LC}) + B \sin(t/\sqrt{LC})$. The capacitor is fully charged in the beginning, so $Q(0) = Q_0 = U_0 C$. Also, there is no current flowing in the beginning, so $\dot{Q}(0) = 0$. Consequently we find

$$Q(t) = U_0 C \cos(t/\sqrt{LC}), \quad (8.S5)$$

and hence the period $T = 2\pi\sqrt{LC}$ and the frequency $1/T = 1/(2\pi\sqrt{LC})$.

- c) Now set $R > 0$ instead. Write down the corresponding differential equation, solve it, and sketch the charge on the capacitor as a function of time. Discuss the three qualitatively different cases that can arise.

Solution: Now we write the equation of motion in general form

$$U_C + U_L + U_R = Q/C + \dot{Q}L + \dot{Q}R = 0. \quad (8.S6)$$

Using the ansatz $Q = Ae^{-bt}$ gives

$$b_{1,2} = \frac{R}{2L} \left(1 \pm \sqrt{1 - \frac{4L}{R^2C}} \right). \quad (8.S7)$$

Again, there is full charge and no current at $t = 0$. This device is an electromagnetic example of a damped harmonic oscillator. There are three cases depending on radicand:

Overdamping ($R > \sqrt{4L/C}$), see figure (8.S12), top curve: The solution is

$$Q(t) = Q_0 \frac{b_2 \exp(-b_1 t) - b_1 \exp(-b_2 t)}{b_2 - b_1}, \quad b_{1,2} > 0. \quad (8.S8)$$

Critical damping ($R = \sqrt{4L/C}$), see figure (8.S12), middle curve: In this special case there is only one b , and hence only one degree of freedom in our ansatz. This means that there must be another solution which is of the type $Q = Bte^{bt}$. The final solution in this case is

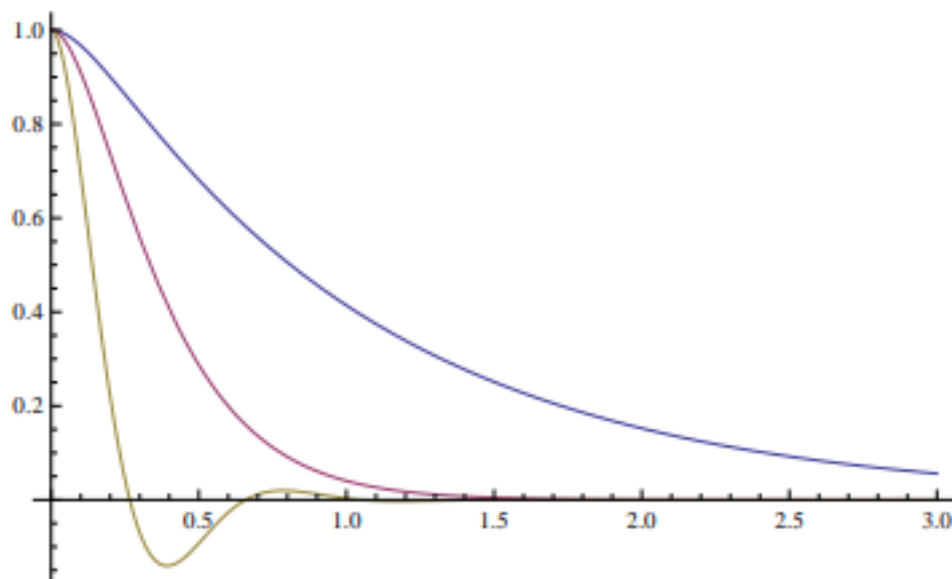
$$Q(t) = Q_0(1 + bt) \exp(-bt). \quad (8.S9)$$

Underdamping ($R < \sqrt{4L/C}$), see figure (8.S12), bottom curve: In this case both $b_{1,2}$ are complex, so we get both exponential decay and oscillatory behaviour. The solution is

$$Q(t) = Q_0 \exp(-\kappa t) \left(\cos(\omega t) + \frac{\kappa}{\omega} \sin(\omega t) \right) \quad (8.S10)$$

with

$$\kappa = \operatorname{Re} b = \frac{R}{2L} > 0, \quad \omega = \operatorname{Im} b = \pm \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}. \quad (8.S11)$$



(8.S12)

8.2. The Poynting vector

Maxwell's equations in vacuum are given by

$$\vec{\partial} \times \vec{E} = -\partial_t \vec{B}, \quad \vec{\partial} \times \vec{B} = \mu_0 \vec{j} + \epsilon_0 \mu_0 \partial_t \vec{E}, \quad (8.2)$$

and the speed of light is $c = 1/\sqrt{\epsilon_0 \mu_0}$.

a) Prove the following identity,

$$\frac{1}{2} \frac{\partial}{\partial t} (c^2 \vec{B}^2 + \vec{E}^2) = -c^2 \vec{\partial} \cdot (\vec{E} \times \vec{B}) - \frac{1}{\epsilon_0} \vec{E} \cdot \vec{j}. \quad (8.3)$$

Solution: Use the identity

$$\begin{aligned} \vec{\partial} \cdot (\vec{E} \times \vec{B}) &= \vec{B} \cdot (\vec{\partial} \times \vec{E}) - \vec{E} \cdot (\vec{\partial} \times \vec{B}) \\ &= -\vec{B} \cdot \partial_t \vec{B} - \mu_0 \vec{E} \cdot \vec{j} - \epsilon_0 \mu_0 \vec{E} \cdot \partial_t \vec{E} \\ &= -\frac{1}{2c^2} \left(c^2 \frac{\partial \vec{B}^2}{\partial t} + \frac{\partial \vec{E}^2}{\partial t} + \frac{2}{\epsilon_0} \vec{E} \cdot \vec{j} \right), \end{aligned} \quad (8.S13)$$

from which the necessary equation follows by rearranging.

b) Consider a particle with charge q moving in the electromagnetic field with velocity $\vec{v}$. Show that the time derivative of its kinetic energy is given by

$$\dot{W}_{\text{kin}} = q \vec{v} \cdot \vec{E}. \quad (8.4)$$

What is the equivalent for a continuous charge distribution?

Solution: Change of kinetic energy is power due to a force exerted on the charge,

$$\dot{W}_{\text{kin}} = P = \vec{v} \cdot \vec{F} = q \vec{v} \cdot \vec{E}. \quad (8.S14)$$

Only the Coulomb force performs work. The Lorentz force is perpendicular to the velocity and does not perform work.

To extend this to a charge distribution we can write $d^3P = dq^3 \vec{v} \cdot \vec{E} = dx^3 \vec{j} \cdot \vec{E}$, so

$$\dot{W}_{\text{kin}} = \int_V dx^3 \vec{j} \cdot \vec{E}. \quad (8.S15)$$

The Poynting vector is defined as

$$\vec{S} := \epsilon_0 c^2 \vec{E} \times \vec{B}. \quad (8.5)$$

c) Prove Poynting's theorem,

$$\frac{d}{dt} \left(\frac{1}{2} \epsilon_0 \int_V dx^3 (c^2 \vec{B}^2 + \vec{E}^2) + W_{\text{kin}} \right) = - \oint_{\partial V} dx^2 \vec{n} \cdot \vec{S}, \quad (8.6)$$

where V is some time-independent volume and ∂V its surface. Interpret the physical meaning of each of the terms.

Solution: The Poynting theorem comes from simply integrating equation (8.3) over a volume V and using the divergence theorem on the term $\vec{\partial} \cdot (\vec{E} \times \vec{B})$.

The terms on the left hand side are the total electromagnetic energy and kinetic energy. Their change with respect to time is equal to the flux of the Poynting vector over the surface, which means that the Poynting vector represents a directed energy flux density.

If we consider Poynting's theorem over the whole space, then the boundary integral is at infinity and so it vanishes. If we then integrate the resulting equation w.r.t. time, we will find that the total energy is constant – the law of conservation of energy.

- d) Consider a monochromatic electromagnetic wave with angular frequency ω . The time-averaged Poynting vector with period $T = 2\pi/\omega$ is given by

$$\langle \vec{S} \rangle = \frac{1}{T} \int_0^T dt \vec{S}(t). \quad (8.7)$$

Show that it can be written as

$$\langle \vec{S} \rangle = \frac{1}{2} \epsilon_0 c^2 \operatorname{Re} [\vec{E}_0 \times \vec{B}_0^*], \quad (8.8)$$

where $\vec{E}_0$ and $\vec{B}_0$ are the (complex) amplitudes of the electric and magnetic fields with the time dependency $e^{-i\omega t}$,

$$\vec{E}(t) = \operatorname{Re} [\vec{E}_0 e^{-i\omega t}], \quad \vec{B}(t) = \operatorname{Re} [\vec{B}_0 e^{-i\omega t}]. \quad (8.9)$$

Solution: We expand the real part of the time-dependent Poynting vector

$$\begin{aligned} \vec{S}(t) &= \epsilon_0 c^2 \vec{E}(t) \times \vec{B}(t) \\ &= \epsilon_0 c^2 \operatorname{Re} [\vec{E}_0 e^{-i\omega t}] \times \operatorname{Re} [\vec{B}_0 e^{-i\omega t}] \\ &= \frac{1}{4} \epsilon_0 c^2 [\vec{E}_0 e^{-i\omega t} + \vec{E}_0^* e^{i\omega t}] \times [\vec{B}_0 e^{-i\omega t} + \vec{B}_0^* e^{i\omega t}] \\ &= \frac{1}{4} \epsilon_0 c^2 [\vec{E}_0^* \times \vec{B}_0 + \vec{E}_0 \times \vec{B}_0^* + \vec{E}_0 \times \vec{B}_0 e^{-2i\omega t} + \vec{E}_0^* \times \vec{B}_0^* e^{2i\omega t}] \\ &= \frac{1}{2} \epsilon_0 c^2 \operatorname{Re} [\vec{E}_0 \times \vec{B}_0^* + \vec{E}_0^* \times \vec{B}_0 e^{-2i\omega t}]. \end{aligned} \quad (8.S16)$$

The second term is oscillatory, so it will integrate to 0. The first term is constant, so it is its own average. Finally,

$$\langle \vec{S} \rangle = \frac{1}{2} \epsilon_0 c^2 \operatorname{Re} [\vec{E}_0 \times \vec{B}_0^*]. \quad (8.S17)$$

9.1. Invariant distance

The Lorentz boost with arbitrary direction and velocity is given by

$$t' = \gamma t - \frac{\gamma}{c^2} \vec{x} \cdot \vec{v}, \quad \vec{x}' = \vec{x} - \gamma \vec{v} t + (\gamma - 1) \frac{\vec{x} \cdot \vec{v}}{\vec{v}^2} \vec{v}, \quad \text{where} \quad \gamma := \frac{1}{\sqrt{1 - \vec{v}^2/c^2}}. \quad (9.1)$$

- a) Find a matrix $(\Lambda^{-1})^\mu{}_\nu$ (in 1 + 3 block form) such that the above transformation takes the form $x'^\mu = (\Lambda^{-1})^\mu{}_\nu x^\nu$.

Solution: The coordinate four-vector is

$$x^\mu = \begin{pmatrix} ct \\ \vec{x} \end{pmatrix}. \quad (9.S1)$$

We can now write the given transformation as

$$\begin{aligned} x'^0 &= \gamma x^0 - \frac{\gamma}{c} \vec{v} \cdot \vec{x}, \\ x'^j &= -\frac{\gamma v_j}{c} x^0 + x^j + \frac{(\gamma - 1)v_j}{\vec{v}^2} \vec{v} \cdot \vec{x}, \end{aligned} \quad (9.S2)$$

so that the transformation can be written as

$$x'^\sigma = (\Lambda^{-1})^\sigma{}_\mu x^\mu, \quad (9.S3)$$

with

$$(\Lambda^{-1})^\sigma{}_\mu = \begin{pmatrix} \gamma & -\frac{\gamma \vec{v}^T}{c} \\ -\frac{\gamma \vec{v}}{c} & 1 + \frac{(\gamma - 1) \vec{v} \vec{v}^T}{\vec{v}^2} \end{pmatrix}. \quad (9.S4)$$

Concerning alternative expressions for the lower right term, you may note that

$$\frac{(\gamma - 1)(\gamma + 1)}{\gamma^2} = 1 - \frac{1}{\gamma^2} = \frac{\vec{v}^2}{c^2} \quad \text{i.e.} \quad \frac{\gamma - 1}{\vec{v}^2} = \frac{\gamma^2}{(\gamma + 1)c^2}. \quad (9.S5)$$

- b) Verify that the matrix $\Lambda^\mu{}_\nu$ from part a) satisfies the condition

$$\Lambda^\lambda{}_\mu \eta_{\lambda\sigma} \Lambda^\sigma{}_\nu = \eta_{\mu\nu}. \quad (9.2)$$

Hint: Without loss of generality, you may restrict to the case $\vec{v} = v \vec{e}_z$.

Solution: The transformation that transforms a contravariant vector into a covariant one reads

$$V_\mu = \eta_{\mu\nu} V^\nu, \quad (9.S6)$$

where $\eta_{\mu\nu}$ is the Minkowski metric

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1). \quad (9.S7)$$

For a Lorentz transformation $\Lambda^\sigma{}_\nu$, the following condition should be fulfilled

$$\Lambda^\lambda{}_\mu \eta_{\lambda\sigma} \Lambda^\sigma{}_\nu = \eta_{\mu\nu}. \quad (9.S8)$$

In case of $\vec{v} = v \vec{e}_z$, we get (note that $\Lambda(\vec{v}) = \Lambda^{-1}(-\vec{v})$)

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & 0 & 0 & \frac{\gamma v}{c} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{\gamma v}{c} & 0 & 0 & \gamma \end{pmatrix}, \quad (9.S9)$$

and the above condition can be easily verified by matrix multiplication (we drop the middle two rows and columns of the matrices),

$$\begin{pmatrix} \gamma & \frac{\gamma v}{c} \\ \frac{\gamma v}{c} & \gamma \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & \frac{\gamma v}{c} \\ \frac{\gamma v}{c} & \gamma \end{pmatrix} = \begin{pmatrix} -\gamma^2 \left(1 - \frac{v^2}{c^2}\right) & 0 \\ 0 & \gamma^2 \left(1 - \frac{v^2}{c^2}\right) \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (9.S10)$$

- c) The square distance between two spacetime points x_1 and x_2 is given by $s^2 = s^\mu s_\mu$ with $s^\mu := x_1^\mu - x_2^\mu$. Show that it is a scalar and, moreover, invariant under Poincaré-transformations. **Review:** It is not clear what exactly is required by “show that it is a scalar” that is not included in “invariant under Poincaré”.

Solution: By inserting the metric into the definition of the squared distance,

$$s^2 = s^\mu s_\mu = s^\mu \eta_{\mu\nu} s^\nu, \quad (9.S11)$$

and imposing the transformation property of contravariant vectors under Lorentz transformations,

$$x'^\sigma = (\Lambda^{-1})^\sigma_\mu x^\mu, \quad (9.S12)$$

the problem can be reduced to (9.2) for the Lorentz transformation Λ^{-1} ,

$$s'^2 = s'^\lambda \eta_{\lambda\sigma} s'^\sigma = ((\Lambda^{-1})^\lambda_\mu s^\mu) \eta_{\lambda\sigma} ((\Lambda^{-1})^\sigma_\nu s^\nu) = s^\mu \underbrace{((\Lambda^{-1})^\lambda_\mu \eta_{\lambda\sigma} (\Lambda^{-1})^\sigma_\nu)}_{=\eta_{\mu\nu}} s^\nu = s^2. \quad (9.S13)$$

Moreover, the distance vector s_μ is invariant under translations (of x_1 and x_2 by the same amount), and so is s^2 .

9.2. Electromagnetic field tensor

The electromagnetic field tensor is given by

$$F_{\mu\nu} = -\partial_\mu A_\nu + \partial_\nu A_\mu = \begin{pmatrix} 0 & c^{-1}E_x & c^{-1}E_y & c^{-1}E_z \\ -c^{-1}E_x & 0 & -B_z & +B_y \\ -c^{-1}E_y & +B_z & 0 & -B_x \\ -c^{-1}E_z & -B_y & +B_x & 0 \end{pmatrix}. \quad (9.3)$$

- a) Show that the electromagnetic field tensor is invariant under the gauge transformation

$$A'_\mu = A_\mu + \partial_\mu \Lambda \quad (9.4)$$

for any scalar field Λ .

Solution: Under the gauge transformation $A'_\mu = A_\mu + \partial_\mu \Lambda$, the electromagnetic field tensor $F_{\mu\nu}$ transforms as

$$F'_{\mu\nu} = -\partial_\mu A'_\nu + \partial_\nu A'_\mu = F_{\mu\nu} - \partial_\mu \partial_\nu \Lambda + \partial_\nu \partial_\mu \Lambda = F_{\mu\nu}. \quad (9.S14)$$

Therefore, $F_{\mu\nu}$ is invariant.

- b) The dual electromagnetic field tensor is defined by

$$\tilde{F}_{\mu\nu} := \frac{1}{2} \varepsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}. \quad (9.5)$$

Determine the matrix elements of $\tilde{F}_{\mu\nu}$ analogously to the above expression for $F_{\mu\nu}$.

Solution: As an example let us evaluate $\tilde{F}_{01}$ and $\tilde{F}_{23}$,

$$\begin{aligned}\tilde{F}_{01} &= \frac{1}{2}\varepsilon_{0123}F^{23} + \frac{1}{2}\varepsilon_{0132}F^{32} = (+\varepsilon_{0123})(+F^{23}) = (+\varepsilon_{0123})(+F_{23}) = +F_{23} = -B_x, \\ \tilde{F}_{23} &= \frac{1}{2}\varepsilon_{2301}F^{01} + \frac{1}{2}\varepsilon_{2310}F^{10} = (+\varepsilon_{0123})(F^{01}) = (+\varepsilon_{0123})(-F_{01}) = -F_{01} = -c^{-1}E_x.\end{aligned}\quad (9.S15)$$

This easily generalises to $\tilde{F}_{ij} = -\varepsilon_{ijk}E^k/c$ and thus

$$\tilde{F}_{\mu\nu} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & -c^{-1}E_z & +c^{-1}E_y \\ B_y & +c^{-1}E_z & 0 & -c^{-1}E_x \\ B_z & -c^{-1}E_y & +c^{-1}E_x & 0 \end{pmatrix}. \quad (9.S16)$$

- c) Verify that the homogeneous Maxwell equations $\vec{\partial} \times \vec{E} = -\partial_t \vec{B}$ and $\vec{\partial} \cdot \vec{B} = 0$ can be expressed as

$$\partial^\mu \tilde{F}_{\mu\nu} = 0. \quad (9.6)$$

Solution: First note that $\tilde{F}_{00} = 0$, $\tilde{F}_{i0} = -\tilde{F}_{0i} = B_i$ and $\tilde{F}_{ij} = -\varepsilon_{ijk}E_k/c$. The contravariant derivative is $\partial^\mu = (-\partial_t/c, \vec{\partial})$. Thus

$$\partial^\mu \tilde{F}_{\mu 0} = \partial^0 \tilde{F}_{00} + \partial^i \tilde{F}_{i0} = \partial_i B_i = \vec{\partial} \cdot \vec{B} = 0 \quad (9.S17)$$

and

$$\partial^\mu \tilde{F}_{\mu i} = \partial^0 \tilde{F}_{0i} + \partial^j \tilde{F}_{ji} = c^{-1} \partial_t B_i - c^{-1} \varepsilon_{jik} \partial_j E_k = c^{-1} [\partial_t \vec{B} + \vec{\partial} \times \vec{E}]_i = 0. \quad (9.S18)$$

Therefore the four equations $\partial^\mu \tilde{F}_{\mu\nu} = 0$ are equivalent to the four homogeneous Maxwell equations.

- d) Compute the contractions $F_{\mu\nu}F^{\mu\nu}$, $F_{\mu\nu}\tilde{F}^{\mu\nu}$ and $\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}$.

Solution: By substituting the relevant components we find

$$\begin{aligned}F^{\mu\nu}F_{\mu\nu} &= 2F^{0i}F_{0i} + F^{ij}F_{ij} = -\frac{2}{c^2}E_iE_i + \varepsilon_{ijk}\varepsilon_{ijl}B_kB_l = -\frac{2}{c^2}\vec{E}^2 + 2\vec{B}^2, \\ F^{\mu\nu}\tilde{F}_{\mu\nu} &= 2F^{0i}\tilde{F}_{0i} + F^{ij}\tilde{F}_{ij} = \frac{2}{c}E_iB_i + \frac{1}{c}\varepsilon_{ijk}\varepsilon_{ijl}B_kE_l = \frac{4}{c}\vec{E} \cdot \vec{B}, \\ \tilde{F}^{\mu\nu}\tilde{F}_{\mu\nu} &= \frac{1}{4}\varepsilon^{\mu\nu\rho\sigma}\varepsilon_{\mu\nu\kappa\lambda}F_{\rho\sigma}F^{\kappa\lambda} = -F^{\mu\nu}F_{\mu\nu}.\end{aligned}\quad (9.S19)$$

9.3. Relativistic force

Two particles at a distance d move with identical velocity v in a direction perpendicular to their separation. Each particle carries the charge q .

- a) Transform the electromagnetic fields due to the particles from the rest frame of the particles to the frame of the observer. Then calculate the force between the particles in the latter frame. **Review:** Replace “the particles” with “one of the particles”, otherwise one is tempted to sum both fields which is not what we want.

Solution: Consider the frame S' moving with velocity $\vec{v} = (v, 0, 0)$ from the observer. The particles are sitting at $\vec{x}' = (0, \pm d/2, 0)$ in the frame S' . The electric field at $(0, d/2, 0)$ due to the particle at $(0, -d/2, 0)$ is $\vec{E}' = (0, E, 0)$ with $E = q/4\pi\varepsilon_0 d^2$.

Next we have to transform the electromagnetic field from the rest frame to the frame of the observer. It is convenient to use the transformation rules in three-dimensional notation

$$\begin{aligned}\vec{E} &= \gamma\vec{E}' - \gamma\vec{v} \times \vec{B}' - \frac{\gamma^2}{c^2(1+\gamma)}(\vec{E}' \cdot \vec{v})\vec{v} = \gamma\vec{E}' = \gamma E \vec{e}_y, \\ \vec{B} &= \gamma\vec{B}' + \frac{\gamma}{c^2}\vec{v} \times \vec{E}' - \frac{\gamma^2}{c^2(1+\gamma)}(\vec{B}' \cdot \vec{v})\vec{v} = \frac{\gamma}{c^2}\vec{v} \times \vec{E}' = \frac{\gamma E v}{c^2}\vec{e}_x \times \vec{e}_y = \frac{\gamma E v}{c^2}\vec{e}_z,\end{aligned}\quad (9.S20)$$

where we used that $\vec{E}' \cdot \vec{v} = 0$ and $\vec{B}' = 0$.

Alternatively we can use four-dimensional notation to transform the fields: The field tensor $F_{\mu\nu}$ transforms to the frame of the observer according to $F_{\mu\nu} = F'_{\rho\sigma} \Lambda^\rho_\mu \Lambda^\sigma_\nu$. We use

$$F'_{\mu\nu} = \begin{pmatrix} 0 & 0 & E/c & 0 \\ 0 & 0 & 0 & 0 \\ -E/c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (9.S21)$$

Note the minus sign in Λ^μ_ν : the observer moves with speed $-v$ away from the particles. The only non-vanishing components are $F'_{02} = E/c$ and $F'_{20} = -E/c$. Then

$$F_{\mu\nu} = F'_{02} \Lambda^0_\mu \Lambda^2_\nu + F'_{20} \Lambda^2_\mu \Lambda^0_\nu = \frac{E}{c} (\Lambda^0_\mu \Lambda^2_\nu - \Lambda^2_\mu \Lambda^0_\nu). \quad (9.S22)$$

Since $F_{\mu\nu}$ must be anti-symmetric, it is now sufficient to consider indices $\mu < \nu$. The non-zero components of $F_{\mu\nu}$ are obtained for $\nu = 2$ and $\mu = 0$ or $\mu = 1$. Hence the observer sees electric field in y -direction and magnetic field in z -direction with magnitudes

$$\begin{aligned} E_y &= cF_{02} = E \Lambda^0_0 \Lambda^2_2 = \gamma E, \\ B_z &= -F_{12} = -\frac{E}{c} \Lambda^0_1 \Lambda^2_2 = \frac{\gamma v}{c} \frac{E}{c} = \frac{\gamma v E}{c^2}. \end{aligned} \quad (9.S23)$$

The total force on the particle at $(0, d/2, 0)$ is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) = q\gamma \left(1 - \frac{v^2}{c^2}\right) E \vec{e}_y = \frac{q}{\gamma} E \vec{e}_y. \quad (9.S24)$$

- b) The relativistic force four-vector is defined by $K^\mu := dp^\mu/d\tau$ with τ the proper time of the particle. Show that $d/d\tau = \gamma d/dt$, where t is the time in the frame of the observer. Use this to show that the components of four-force in the frame of the observer take the form

$$K^\mu = \gamma(\vec{F} \cdot \vec{v}/c, \vec{F}), \quad (9.7)$$

where $\vec{F}$ is the force on the particle in the frame of the observer.

Hint: Differentiate the invariant $p^\mu p_\mu$ w.r.t. t .

Solution: The proper time is the time measured in the rest frame of the particle. Transforming to the frame of the observer via $x^\mu = (\Lambda^{-1})^\mu_\nu x'^\nu$ with $x'^\nu = (c\tau, 0)$ and $x^\mu = (ct, \vec{x})$ gives $t = \gamma\tau$. Hence

$$\frac{d}{d\tau} = \frac{dt}{d\tau} \frac{d}{dt} = \gamma \frac{d}{dt}. \quad (9.S25)$$

The four-momentum $p^\mu = (p^0, \vec{p})$ of the particle is given in the frame of the observer by $p^\mu = m dx^\mu/d\tau = \gamma m dx^\mu/dt = \gamma(mc, m\vec{v})$. Differentiating the invariant $p^\mu p_\mu = -(p^0)^2 + \vec{p} \cdot \vec{p} = -m^2 c^2$ w.r.t. t gives

$$p^0 \frac{dp^0}{dt} = \vec{p} \cdot \frac{d\vec{p}}{dt}. \quad (9.S26)$$

Now we use the definition of force $\vec{F} = d\vec{p}/dt$ and the results from above $p^0 = \gamma mc$, $\vec{p} = \gamma m\vec{v}$ to get

$$\frac{dp^0}{dt} = \frac{\vec{v} \cdot \vec{F}}{c}. \quad (9.S27)$$

This gives

$$K^\mu = \frac{dp^\mu}{d\tau} = \gamma \frac{dp^\mu}{dt} = \gamma(\vec{F} \cdot \vec{v}/c, \vec{F}). \quad (9.S28)$$

- c) Write down the transformation law for K^μ between the rest frame of the particles and the frame of the observer. Verify that the force computed in part a) is consistent with the transformed force from the rest frame of the particles.

Solution: In the rest frame of the particles we have $\gamma = 1$ and $\vec{F} = qE\vec{e}_y$ so that $K^\mu = (0, 0, qE, 0)$.

The observer moves with velocity $\vec{v} = (-v, 0, 0)$ w.r.t. the frame in which the particles are at rest. The four-force transforms as $K^\mu = (\Lambda^{-1})^\mu{}_\nu K'^\nu$. So the four-force in the frame of the observer is given by

$$K^\mu = \begin{pmatrix} \gamma & \gamma v/c & 0 & 0 \\ \gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ qE \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ qE \\ 0 \end{pmatrix}. \quad (9.S29)$$

This is consistent with $K^\mu = \gamma(0, 0, F, 0)$ with $F = qE/\gamma$ as computed in part a).

9.4. Fourier transform

The Fourier transformation in a three-dimensional space and its inverse are given by

$$\tilde{f}(\vec{k}) = \int d\vec{x} f(\vec{x}) e^{-i\vec{k}\cdot\vec{x}}, \quad f(\vec{x}) = \int \frac{d\vec{k}}{(2\pi)^3} \tilde{f}(\vec{k}) e^{i\vec{k}\cdot\vec{x}}. \quad (9.8)$$

Prove the following identities for the Fourier transformation:

a) $h(x) = af(x) + bg(x) \implies \tilde{h}(k) = a\tilde{f}(k) + b\tilde{g}(k) \quad (a, b \in \mathbb{C}).$

Solution: This identity follows directly from the linearity of integration:

$$\tilde{h}(k) = a \int \frac{d\vec{x}}{(2\pi)^3} \tilde{f}(x) e^{i\vec{k}\cdot\vec{x}} + b \int \frac{d\vec{x}}{(2\pi)^3} \tilde{g}(x) e^{i\vec{k}\cdot\vec{x}} = a\tilde{f}(k) + b\tilde{g}(k). \quad (9.S30)$$

b) $\vec{h}(x) = \vec{\partial}f(x) \implies \vec{\tilde{h}}(k) = i\vec{k}\tilde{f}(k).$

Solution: introductory text

$$\begin{aligned} \vec{\tilde{h}}(k) &= \int d\vec{x}^3 (\vec{\partial}f(x)) e^{-i\vec{k}\cdot\vec{x}} \\ &= \int d\vec{x}^3 e^{-i\vec{k}\cdot\vec{x}} \vec{\partial} \left(\int \frac{d\vec{k}'}{(2\pi)^3} \tilde{f}(k') e^{i\vec{k}'\cdot\vec{x}} \right) \\ &= \int \frac{d\vec{k}'}{(2\pi)^3} i\vec{k}' \int d\vec{k}^3 e^{i(\vec{k}'-\vec{k})\cdot\vec{x}} \tilde{f}(k') \\ &= \int d\vec{k}'^3 i\vec{k}' \delta^3(k' - k) \tilde{f}(k') \\ &= i\vec{k} \tilde{f}(k), \end{aligned} \quad (9.S31)$$

where we used the following integral representation of the delta-function:

$$(2\pi)^3 \delta^3(k' - k) = \int d\vec{x}^3 e^{i(\vec{k}'-\vec{k})\cdot\vec{x}}. \quad (9.S32)$$

Alternatively we could perform a partial integration to obtain the same result:

$$\begin{aligned} \vec{\tilde{h}}(k) &= \int d\vec{x}^3 \vec{\partial}f(x) e^{-i\vec{k}\cdot\vec{x}} \\ &= f(x) e^{-i\vec{k}\cdot\vec{x}} \Big|_{\|\vec{x}\| \rightarrow \infty} + \int d\vec{x}^3 f(x) i\vec{k} e^{-i\vec{k}\cdot\vec{x}} \\ &= i\vec{k} \tilde{f}(k), \end{aligned} \quad (9.S33)$$

where the boundary term is zero if the function $f(x)$ goes to zero at infinity.

c) $h(x) = f(x)g(x) \implies \tilde{h}(k) = (2\pi)^{-3}(\tilde{f} * \tilde{g})(k) := (2\pi)^{-3} \int d^3k' \tilde{f}(k') \tilde{g}(k - k').$

Solution: introductory text

$$\begin{aligned}
 \tilde{h}(k) &= \int d^3x f(x)g(x) e^{-i\vec{k}\cdot\vec{x}} \\
 &= \int d^3x \left(\int \frac{d^3k'}{(2\pi)^3} \tilde{f}(k') e^{i\vec{k}'\cdot\vec{x}} \right) g(x) e^{-i\vec{k}\cdot\vec{x}} \\
 &= \int \frac{d^3k'}{(2\pi)^3} \tilde{f}(k') \int d^3x g(x) e^{-i(\vec{k}-\vec{k}')\cdot\vec{x}} \\
 &= \int \frac{d^3k'}{(2\pi)^3} \tilde{f}(k') \tilde{g}(k - k') \\
 &= \frac{1}{(2\pi)^3} (\tilde{f} * \tilde{g})(k).
 \end{aligned} \tag{9.S34}$$

d) $h(x) = f^*(x) \implies \tilde{h}(k) = \tilde{f}^*(-k).$

Solution: introductory text

$$\tilde{h}(k) = \int d^3x f^*(x) e^{-i\vec{k}\cdot\vec{x}} = \left(\int d^3x f(x) e^{i\vec{k}\cdot\vec{x}} \right)^* = (\tilde{f}(-k))^* = \tilde{f}^*(-k). \tag{9.S35}$$

e) $h(x) = \delta^3(x) \implies \tilde{h}(k) = 1.$

Solution: introductory text

$$\tilde{h}(k) = \int d^3x \delta^3(x) e^{-i\vec{k}\cdot\vec{x}} = e^{-i\vec{k}\cdot\vec{0}} = 1. \tag{9.S36}$$

f) $\tilde{h}(x) = \vec{\partial}\delta^3(x) \implies \tilde{h}(k) = i\vec{k}.$

Solution: From part b) and part e) we find $\tilde{h}(k) = i\vec{k}.$

10.1. Partially polarised light

An almost monochromatic electromagnetic wave is propagating in the z -direction and is described by the complex notation

$$\vec{B} = \vec{e}_z \times \vec{E}, \quad \vec{E} = \vec{E}(t - \vec{e}_z \cdot \vec{x}/c), \quad \vec{E}(t) = \vec{E}_0 e^{-i\omega_0 t} \quad (10.1)$$

with $\vec{E}_0 = (E_1, E_2, 0)$ in the x, y -plane. We assume that the characteristic timescale of $\vec{E}_0$ (called coherence time τ), over which fluctuations of $\vec{E}_0$ are correlated, is much larger than the duration of a period (timescale $2\pi/\omega_0$), but is rapidly changing compared to the timescale of optical polarisation measurements. This allows us to average out the fluctuations in polarisation measurements and to treat $\vec{E}_0$ as a random variable. We indicate averages with the symbol $\langle \cdot \rangle$. We want to show that the full information about the polarisation of the wave is contained in the hermitian matrix

$$S = \begin{pmatrix} \langle E_1 E_1^* \rangle & \langle E_1 E_2^* \rangle \\ \langle E_2 E_1^* \rangle & \langle E_2 E_2^* \rangle \end{pmatrix} = S^\dagger. \quad (10.2)$$

a) Show that S can be written in the following form

$$S = s_0 \sigma_0 + s_1 \sigma_1 + s_2 \sigma_2 + s_3 \sigma_3 := s_0 \sigma_0 + \vec{s} \cdot \vec{\sigma}, \quad (10.3)$$

where $s_i \in \mathbb{R}$ (Stokes parameters) and σ_i denotes the identity and Pauli matrices

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10.4)$$

Hint: The matrices of the form S span a vector space of real dimension 4.

Solution: The 2×2 hermitian matrices $S = S^\dagger$ have the general form

$$S = \begin{pmatrix} a & b + ic \\ b - ic & d \end{pmatrix} \quad (10.S1)$$

with $a, b, c, d \in \mathbb{R}$ and they span a vector space V of real dimension 4 (we need 4 real parameters to describe the matrix completely). The three Pauli matrices, together with the unity matrix σ_0 , are hermitian and linearly independent as vectors in V . On the other hand we can easily check that every element $S \in V$ can be written as a linear combination of the σ_i 's, i.e. that the Pauli-matrices form a basis for V . Whence:

$$S = s_0 \sigma_0 + s_1 \sigma_1 + s_2 \sigma_2 + s_3 \sigma_3 \quad (10.S2)$$

with $s_i \in \mathbb{R}$.

b) Express the Stokes parameters with the averages $\langle E_j E_k^* \rangle$.

Hint: Notice first that $S \cdot T := \frac{1}{2} \text{tr}(ST)$ defines a scalar product for matrices S and T of the form (10.3). Then check $\sigma_i \sigma_j = i \varepsilon_{ijk} \sigma_k + \delta_{ij} \sigma_0$ and use this together with the properties of the scalar product to compute the Stokes parameters.

Solution: The mapping $V \times V \rightarrow \mathbb{R}$, $(S, T) \mapsto \frac{1}{2} \text{tr}(ST)$ is a scalar product, since it is linear (matrix multiplication and the tracing are both linear), symmetric (because $\text{tr}(ST) = \text{tr}(TS)$) and positive definite (since $S \geq 0$, see part d)). The formula $\sigma_i \sigma_j = i\epsilon_{ijk} \sigma_k + \delta_{ij} \sigma_0$ can be checked with a direct calculation of the matrix products. The Pauli basis is orthonormal with respect to this scalar product, since

$$\sigma_i \cdot \sigma_j = \frac{1}{2} \text{tr}(\sigma_i \sigma_j) = \delta_{ij}, \quad (10.S3)$$

since $\text{tr} \sigma_0 = 2$ and $\text{tr} \sigma_i = 0$ ($i = 1, 2, 3$).

The Stokes parameters can be hence extracted from the linear combination of S using the scalar product:

$$s_i = \sigma_i \cdot S = \frac{1}{2} \text{tr}(\sigma_i S). \quad (10.S4)$$

In more detail:

$$\begin{aligned} s_0 &= \frac{1}{2} \langle \|E_1\|^2 + \|E_2\|^2 \rangle, \\ s_1 &= \frac{1}{2} \langle E_2 E_1^* + E_1 E_2^* \rangle = \text{Re} \langle E_2 E_1^* \rangle, \\ s_2 &= -\frac{i}{2} \langle E_2 E_1^* - E_1 E_2^* \rangle = \text{Im} \langle E_2 E_1^* \rangle, \\ s_3 &= \frac{1}{2} \langle \|E_1\|^2 - \|E_2\|^2 \rangle. \end{aligned} \quad (10.S5)$$

- c) Show that $\|\vec{s}\| = s_0$ when the electromagnetic wave is purely polarised (fixed value $\vec{E}_0$ with trivial average).

Hint: Calculate the trace of S^2 .

Solution: We have

$$S = \langle (E_1, E_2)^T (E_1^*, E_2^*) \rangle. \quad (10.S6)$$

With $E := (E_1, E_2)^T$ we can write

$$S^2 = E(E^*)^T E(E^*)^T = E[(E^*)^T E](E^*)^T = \|E\|^2 S = 2s_0 S, \quad (10.S7)$$

where we used the result from part b) (we ignored the averaging as indicated in the formulation of the question). Since

$$\frac{1}{2} \text{tr}(S^2) = (S, S) = \sum_{i=0}^3 s_i^2 = s_0^2 + \vec{s}^2 = \frac{1}{2} \text{tr}(2s_0 S) = s_0 \text{tr} S = 2s_0^2 \quad (10.S8)$$

it follows

$$s_0^2 + \vec{s}^2 = 2s_0^2 \quad \implies \quad \vec{s}^2 = s_0^2 \quad \implies \quad \|\vec{s}\| = s_0. \quad (10.S9)$$

Note that the solution $\|\vec{s}\| = -s_0$ is not valid because s_0 is a strictly positive quantity (see part b)).

- d) Show that $\|\vec{s}\| \leq s_0$ holds in general.

Hint: Show that the matrix S is positive definite and compute its eigenvalues.

Solution: The matrix S is positive definite, since for every vector u in $\mathbb{C}^2$ we have:

$$u^\dagger S u = u^\dagger \langle E E^\dagger \rangle u = \langle (u^\dagger \cdot E)(u^\dagger E)^* \rangle \geq 0. \quad (10.S10)$$

The eigenvalues of

$$S = \begin{pmatrix} s_0 + s_3 & s_1 + i s_2 \\ s_1 - i s_2 & s_0 - s_3 \end{pmatrix} \quad (10.S11)$$

are calculated with

$$0 = \det(S - \lambda \sigma_0) = \lambda^2 - 2s_0 \lambda + (s_0^2 + \vec{s}^2) = (\lambda - (s_0 + \|\vec{s}\|))(\lambda - (s_0 - \|\vec{s}\|)) \quad (10.S12)$$

to be $\lambda_{\pm} = s_0 \pm \|\vec{s}\|$ and are positive because of $S \geq 0$. Therefore we obtain $\|\vec{s}\| \leq s_0$.

- e) Find a physical meaning for the Stokes parameters. What does $\vec{s} = 0$ mean?

Hint: Calculate first the eigenvalues $\lambda_{\pm}^{(i)}$ and the eigenvectors $\vec{e}_{\pm}^{(i)}$ of the Pauli matrices. Next express s_i through the coefficients $\alpha_{\pm}^{(i)}$ in the decomposition $(E_1, E_2) = \alpha_+^{(i)} \vec{e}_+^{(i)} + \alpha_-^{(i)} \vec{e}_-^{(i)}$. Compute then S and its trace in the eigenbasis $\{\vec{e}_+^{(i)}, \vec{e}_-^{(i)}\}$.

Solution: The eigenvalues of σ_i are ± 1 because $\sigma_i^2 = 1$ and $\text{tr } \sigma_i = 0$. The normalized eigenvectors $\sigma_i \vec{e}_{\pm}^{(i)} = \pm \vec{e}_{\pm}^{(i)}$ are

$$\begin{aligned}\vec{e}_{\pm}^{(1)} &= \frac{1}{\sqrt{2}}(\vec{e}_1 \pm \vec{e}_2), \\ \vec{e}_{\pm}^{(2)} &= \frac{1}{\sqrt{2}}(\vec{e}_1 \pm i\vec{e}_2), \\ \vec{e}_{\pm}^{(3)} &= \vec{e}_1, \quad \vec{e}_{\pm}^{(3)} = \vec{e}_2,\end{aligned}\tag{10.S13}$$

and correspond to diagonal polarisation, right/left circular polarisation and horizontal/vertical polarisation. With respect to the eigenbasis $\{\vec{e}_+^{(i)}, \vec{e}_-^{(i)}\}$, σ_i is represented by the matrix $\text{diag}(1, -1)$, while S is $\langle \alpha_k \alpha_j^* \rangle_{k,j=\pm}$. Since the trace is independent of the basis choice, it follows analogously to part d)

$$\begin{aligned}s_i &= \frac{1}{2} \langle \|\alpha_+^{(i)}\|^2 - \|\alpha_-^{(i)}\|^2 \rangle, \\ s_0 &= \frac{1}{2} \langle \|\alpha_+^{(i)}\|^2 + \|\alpha_-^{(i)}\|^2 \rangle.\end{aligned}\tag{10.S14}$$

Therefore $c \cdot s_0$ is the intensity and $-1 \leq s_i/s_0 \leq 1$ describes the relative fraction of $\pm$ -polarisations with respect to the three bases. A wave with $\vec{s} = 0$ is unpolarised with respect to any basis (e.g. direct sunlight with good approximation).

10.2. Group velocity

A one-dimensional Gaussian wave packet $\phi(x, t)$ is moving in a dispersive medium (i.e. $\omega(k)$ depends non-linearly on k). At time $t = 0$ it is given by

$$\phi(x, t = 0) = \exp\left(-\frac{x^2}{2(\Delta x)^2}\right),\tag{10.5}$$

where we consider Δx as a measure for the spatial extent of the wave packet. The time dependency is given by

$$\phi(x, t) = \text{Re} \int_{-\infty}^{\infty} \frac{dk}{2\pi} \tilde{\phi}(k) e^{ikx - i\omega(k)t},\tag{10.6}$$

where $\tilde{\phi}(k)$ is the Fourier transform of $\phi(x, t = 0)$.

- a) Show by completing the square that the Fourier transformed wave packet at $t = 0$ has a Gaussian profile. What is the relation between Δx and the analogous Δk ? What does this mean?

Hint:

$$\int_{-\infty}^{\infty} dx \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) = \sqrt{2\pi} |\sigma|.\tag{10.7}$$

Solution: Completing the square in the exponent, extracting the x -independent part from the integral and applying (10.7) yields

$$\begin{aligned}
 \tilde{\phi}(k, t=0) &= \int_{-\infty}^{\infty} dx \phi(x, t=0) e^{-ikx} \\
 &= \int_{-\infty}^{\infty} dx e^{-x^2/2(\Delta x)^2} e^{-ikx} \\
 &= \int_{-\infty}^{\infty} dx \exp \left[-\frac{1}{2(\Delta x)^2} \left\{ x^2 + 2ikx(\Delta x)^2 + [ik(\Delta x)^2]^2 - [ik(\Delta x)^2]^2 \right\} \right] \\
 &= e^{-k^2(\Delta x)^2/2} \int_{-\infty}^{\infty} dx \exp \left[-\frac{1}{2(\Delta x)^2} \left\{ x + ik(\Delta x)^2 \right\}^2 \right] \\
 &= \sqrt{2\pi} \Delta x e^{-k^2(\Delta x)^2/2}.
 \end{aligned} \tag{10.S15}$$

Therefore $\Delta k = (\Delta x)^{-1}$. The narrower the wave packet is in position space, the wider it is in Fourier space and vice versa.

- b) Show that the maximum of the wave packet covers a distance of $v_g t$ in the time interval t , where the *group velocity* v_g is given by

$$v_g = \left. \frac{d\omega}{dk} \right|_{k_0}. \tag{10.8}$$

Here, k_0 denotes the wave number at the maximum of $\tilde{\phi}(k)$.

Hint: Expand $\omega(k)$ to first order in k around k_0 , and evaluate the change in the maximum of the wave packet using (10.6).

Solution: The expansion of $\omega(k)$ yields

$$\omega(k_0) =: \omega_0, \quad \omega(k) \approx \omega(k_0) + (k - k_0) \left. \frac{d\omega}{dk} \right|_{k_0} = \omega_0 + (k - k_0)v_g. \tag{10.S16}$$

Inserting this into (10.6), we find an expression for $\phi(x, t)$,

$$\begin{aligned}
 \phi(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \tilde{\phi}(k) e^{i(kx - \omega(k)t)} \\
 &\approx \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \tilde{\phi}(k) e^{i\{kx - [\omega_0 + (k - k_0)v_g]t\}} \\
 &= e^{i(k_0 v_g - \omega_0)t} \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \tilde{\phi}(k) e^{ik(x - v_g t)} \\
 &= e^{i(k_0 v_g - \omega_0)t} \phi(x - v_g t, 0).
 \end{aligned} \tag{10.S17}$$

One could see that the maximum of the wave packet moves from $x = 0$ to $x = v_g t$ in the time interval t , which explains the definition of the group velocity.

- c) What is the speed of the individual phases? Under which circumstances do phase velocity and group velocity of the wave coincide?

Solution: Each individual phase moves with the phase velocity,

$$v_p(k) := \frac{\omega(k)}{k}. \tag{10.S18}$$

Phase and group velocity coincide if $\omega(k)$ depends linearly on k , i.e. in a non-dispersive medium.

- d) Estimate how fast the wave packet is widening by finding an expression for the variation of the group velocity inside the pulse. Use the relation between Δk and Δx from part a), and interpret the result accordingly.

Hint: Estimate the variation as the difference between the group velocities at k_0 and $k_0 + \Delta k$ (analogously to (10.8)), and determine Δv_g from an expansion of $\omega(k)$ around k_0 up to the first contributing order.

Solution: With the expansion of $\omega(k)$ up to one order higher than in part a),

$$\omega(k) = \omega(k_0) + \underbrace{(k - k_0) \frac{d\omega}{dk}}_{=v_g} \Big|_{k_0} + \frac{1}{2} (k - k_0)^2 \frac{d^2\omega}{dk^2} \Big|_{k_0} + \mathcal{O}((k - k_0)^3), \quad (10.S19)$$

we find an expression for the group velocity at $k_0 + \Delta k$,

$$v_g|_{k_0+\Delta k} = \frac{d\omega}{dk} \Big|_{k_0+\Delta k} = v_g + \Delta k \frac{d^2\omega}{dk^2} \Big|_{k_0}. \quad (10.S20)$$

Using the relation from part a), $\Delta x = 1/\Delta k$, we can estimate the dependency of Δv_g as a function of Δx ,

$$\Delta v_g \approx v_g|_{k_0+\Delta k} - v_g|_{k_0} = \Delta k \frac{d^2\omega}{dk^2} \Big|_{k_0} = \frac{1}{\Delta x} \frac{d^2\omega}{dk^2} \Big|_{k_0}. \quad (10.S21)$$

If $d^2\omega/dk^2 > 0$, a narrow pulse spreads rapidly (the spread in the group velocity) because of its broad spectrum of wave numbers, and vice versa. If we combine the uncertainties in the position by taking the square root of the sum of squares, we obtain

$$\Delta x(t) \approx \sqrt{(\Delta x)^2 + (\Delta v_g t)^2} \approx \sqrt{(\Delta x)^2 + \left(\frac{d^2\omega}{dk^2} \frac{1}{\Delta x} \right)^2}. \quad (10.S22)$$

11.1. Radiation from a linear antenna

A thin linear antenna of length $2d$ is oriented along the z -axis and centred in the origin of the coordinate system. The antenna is excited such that the wavelength λ of the sinusoidal current equals the antenna length. Thus $\lambda = 2d$ and the frequency is $\omega = \pi c/d$. The amplitude of the current is I_0 .

Hint: Note that this antenna differs from the one discussed in the lecture. There, the sinusoidal current is symmetric with respect to the origin, whereas here it is antisymmetric. The final results are therefore not identical.

a) Show that the vector potential $A(x, t)$ has the form:

$$\vec{A}(x, t) = \mu_0 I_0 \vec{e}_z e^{-i\omega t} \int_{-d}^d dz' \frac{\sin(kz')}{4\pi r'} e^{ikr'}, \quad (11.1)$$

where $r' = \|x - x'\|$ with $\vec{x}' = z' \vec{e}_z$ and $k = \omega/c$ is the wave number.

Solution: Sketch of the antenna:



(11.S1)

The current density can be written as:

$$\vec{j}(x, t) = I_0 \delta(x) \delta(y) \sin(\pi z/d) e^{-i\omega t} \vec{e}_z. \quad (11.S2)$$

The retarded vector potential $\vec{A}(x, t)$ reads:

$$\vec{A}(x, t) = \mu_0 \int_V d^3x' \frac{\vec{j}(x', t - \|x - x'\|/c)}{4\pi \|x - x'\|} = \mu_0 I_0 e^{-i\omega t} \vec{e}_z \int_{-d}^d dz' \frac{\sin(kz')}{4\pi r'} e^{ikr'}, \quad (11.S3)$$

where we have used $k = \omega/c$.

For the remainder of this problem we will work in the radiation zone where $r \gg 2d = \lambda$ with $r = \|x\|$. In this zone one approximates:

$$\|x - x'\| = r - \vec{n} \cdot \vec{x}' \quad (11.2)$$

where $\vec{n}$ is a unit vector in the direction of $\vec{x}$.

b) Compute exactly the power radiated per unit solid angle and plot the angular distribution as a function of the polar angle ϑ .

Hint: You can use the following identity for computing the vector potential:

$$\int dx \sin(ax) e^{-ibx} = \frac{e^{-ibx}}{b^2 - a^2} (a \cos(ax) + ib \sin(ax)). \quad (11.3)$$

Solution: In the radiation zone we can approximate r' as:

$$r' = \|x - x'\| = \sqrt{r^2 - 2r\vec{n}\cdot\vec{x}' + \|\vec{x}'\|^2} \approx r - \vec{n}\cdot\vec{x}' = r - z' \cos \vartheta \quad \Rightarrow \quad \frac{1}{r'} = \frac{1}{r} + \mathcal{O}(1/r^2), \quad (11.S4)$$

where ϑ is the polar angle of the vector $\vec{x}$ measured from the z -axis. The vector potential in this region reads:

$$\begin{aligned} \vec{A}(x, t) &= \frac{\mu_0}{4\pi} I_0 e^{-i\omega t} \vec{e}_z \int_{-d}^d dz' \frac{\sin(kz')}{r} e^{ik(r-z' \cos \vartheta)} \\ &= \frac{\mu_0}{4\pi} \frac{I_0}{r} e^{i(kr-\omega t)} \vec{e}_z \int_{-d}^d dz' \sin(kz') e^{-ikz' \cos \vartheta}. \end{aligned} \quad (11.S5)$$

Using the identity in (11.3) with $a = k$ and $b = k \cos \vartheta$ we get:

$$\begin{aligned} \int_{-d}^d dz' \sin(kz') \exp(-ikz' \cos \vartheta) &= \frac{\exp(-ikz' \cos \vartheta)}{k^2(\cos^2 \vartheta - 1)} \left[k \cos(kz') + ik \cos \vartheta \sin(kz') \right] \Big|_{z'=-d}^{z'=d} \\ &= -\frac{\exp(-ikz' \cos \vartheta)}{k \sin^2 \vartheta} \left[\cos(kz') + i \cos \vartheta \sin(kz') \right] \Big|_{z'=-d}^{z'=d}. \end{aligned} \quad (11.S6)$$

If we now set $d = \pi/k$, we get

$$\int_{-\pi/k}^{\pi/k} dz' \sin(kz') \exp(-ikz' \cos \vartheta) = \frac{\exp(-i\pi \cos \vartheta)}{k \sin^2 \vartheta} - \frac{\exp(i\pi \cos \vartheta)}{k \sin^2 \vartheta} = -2i \frac{\sin(\pi \cos \vartheta)}{k \sin^2 \vartheta}. \quad (11.S7)$$

Plugging this into the formula for the vector potential we have:

$$\vec{A}(x, t) = -i \frac{\mu_0}{2\pi} \frac{I_0}{kr} \frac{\sin(\pi \cos \vartheta)}{\sin^2 \vartheta} e^{i(kr-\omega t)} \vec{e}_z. \quad (11.S8)$$

In the radiation zone the magnetic and electric fields can be written as:

$$\begin{aligned} \vec{B} &= ik\vec{n} \times \vec{A}, \\ \vec{E} &= ikc \left(\vec{A} - \vec{n}(\vec{n} \cdot \vec{A}) \right). \end{aligned} \quad (11.S9)$$

We recall that the power radiated per unit solid angle is:

$$\frac{d^2 P}{d^2 \Omega} = \vec{n} \cdot \langle \vec{S} \rangle r^2, \quad (11.S10)$$

where the time averaged Poynting vector for a monochromatic wave is :

$$\langle \vec{S} \rangle = \frac{\epsilon_0}{2} c^2 \operatorname{Re} [\vec{E} \times \vec{B}^*]. \quad (11.S11)$$

Plugging equation (11.S9) into (11.S11) and subsequently into (11.S10), exploiting that $\vec{n}$ is real we get:

$$\begin{aligned} \frac{d^2 P}{d^2 \Omega} &= r^2 \frac{\epsilon_0}{2} c^2 \operatorname{Re} [\vec{n} \cdot (\vec{E} \times \vec{B}^*)] \\ &= \frac{\epsilon_0 c^3}{2} (kr)^2 \operatorname{Re} \left[\vec{n} \cdot \left(\left(\vec{A} - \vec{n}(\vec{n} \cdot \vec{A}) \right) \times \left(\vec{n} \times \vec{A}^* \right) \right) \right] \\ &= \frac{\epsilon_0 c^3}{2} (kr)^2 \operatorname{Re} \left[\vec{n} \cdot \left(\vec{A} \times (\vec{n} \times \vec{A}^*) - (\vec{n} \cdot \vec{A}) \vec{n} \times (\vec{n} \times \vec{A}^*) \right) \right]. \end{aligned} \quad (11.S12)$$

The second term in square brackets vanishes because we are taking the scalar product of $\vec{n}$ with a vector orthogonal to $\vec{n}$. For the first term we use the identity:

$$\vec{n} \cdot \left(\vec{A} \times (\vec{n} \times \vec{A}^*) \right) = \|\vec{A}\|^2 - (\vec{n} \cdot \vec{A})^2. \quad (11.S13)$$

$\vec{A}$ is oriented along the z -axis hence:

$$\|\vec{A}\|^2 - (\vec{n} \cdot \vec{A})^2 = \|\vec{A}\|^2 - \|\vec{A}\|^2 (\vec{n} \cdot \vec{e}_z)^2 = \|\vec{A}\|^2 (1 - \cos^2 \vartheta) = \|\vec{A}\|^2 \sin^2 \vartheta. \quad (11.S14)$$

Thus we have for the power distribution:

$$\frac{d^2 P}{d^2 \Omega} = \frac{\varepsilon_0 c^3}{2} (kr)^2 \|A\|^2 \sin^2(\vartheta), \quad (11.S15)$$

which finally gives after using (11.S8):

$$\frac{d^2 P}{d^2 \Omega} = \frac{I_0^2}{8\pi^2} \sqrt{\frac{\mu_0}{\varepsilon_0}} \left(\frac{\sin(\pi \cos \vartheta)}{\sin \vartheta} \right)^2. \quad (11.S16)$$

Please note that the last formula differs from the one given in lecture notes. There, the antenna is taken to be centre-fed and the sinusoidal current is symmetric on the two arms of the antenna. In this case however, it is antisymmetric.



- c) Consider now an expansion in multipoles, i.e. expand the exponential in the vector potential as:

$$e^{ik\vec{n}\cdot\vec{x}'} = \sum_{m=0}^{\infty} \frac{(ik)^m}{m!} (\vec{n}\cdot\vec{x}')^m \quad (11.4)$$

and keep only the leading term in k . Within this approximation compute the power radiated per unit solid angle and plot the angular distribution as a function of ϑ .

Solution: Going back to equation (11.S5) for the vector potential in the radiation zone and using the expansion for the exponential part we have:

$$\vec{A}(x, t) = \frac{\mu_0}{4\pi} \frac{I_0}{r} e^{i(kr - \omega t)} \vec{e}_z \int_{-d}^d dz' \sin(kz') \left[1 + (-ikz' \cos \vartheta) + \frac{(-ikz' \cos^2 \vartheta)}{2} + \dots \right]. \quad (11.S18)$$

We notice that all even terms in the expansion integrate to zero because $\sin(kz')$ is odd. The first term therefore vanishes and the leading contribution is:

$$\begin{aligned} \vec{A}(x, t) &\approx -i \frac{\mu_0}{4\pi} \frac{I_0}{r} e^{i(kr - \omega t)} k \cos \vartheta \vec{e}_z \int_{-d}^d dz' z' \sin(kz') \\ &= -i \frac{\mu_0}{4\pi} \frac{I_0}{r} e^{i(kr - \omega t)} k \cos \vartheta \vec{e}_z \left[\frac{\sin(kz')}{k^2} - \frac{\cos(kz')z'}{k} \right] \Big|_{-d}^d \\ &= -i \frac{\mu_0}{4\pi} \frac{I_0}{r} e^{i(kr - \omega t)} k \cos \vartheta \frac{2\pi}{k^2} \vec{e}_z = -i \frac{\mu_0}{2} \frac{I_0}{kr} e^{i(kr - \omega t)} \cos \vartheta \vec{e}_z \end{aligned} \quad (11.S19)$$

Equation (11.S15) yields:

$$\frac{d^2 P}{d^2 \Omega} = \frac{I_0^2}{8} \sqrt{\frac{\mu_0}{\varepsilon_0}} \cos^2 \vartheta \sin^2 \vartheta \quad (11.S20)$$

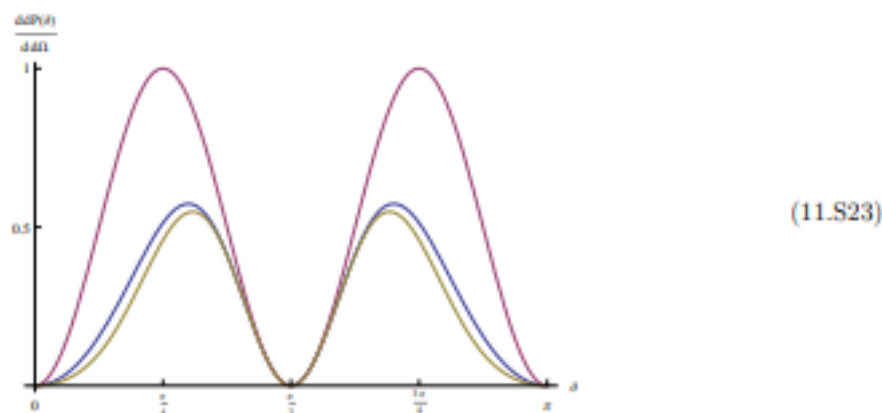


d) Compare the results obtained in part b) and part c). Discuss whether the multipole expansion approximation is valid or not and argue why.

Solution: The approximation is that of a magnetic dipole, indeed we truncated the expansion considering just the second term which is the leading one. Comparing the plots we see that the approximation is not very good except for a neighbourhood of $\vartheta = \pi/2$. The reason is that the multipole expansion is valid only in the long wavelength approximation, namely:

$$d \ll \lambda \Rightarrow k|z'| < kd \ll 1 \quad (11.S22)$$

while in our case $\lambda = d$. In the following plot we compare the exact result (blue), against the approximation computed in part c). We are retaining only the leading contribution (orange) and the approximation that one gets retaining the first two non vanishing contributions (green).



We can appreciate how the approximation improves considering one order higher in the expansion.

11.2. Dipole radiation

A thin, ideal conductor connects two metal balls at positions $\vec{x} = (0, 0, \pm a)$. The charge density is

$$\rho(x, t) = Q \delta(x) \delta(y) [\delta(z - a) - \delta(z + a)] \cos(\omega t) \quad (11.5)$$

with a , Q and ω constant. The current between the two metal balls flows along the wire.

a) Calculate the average power emitted per unit solid angle in the dipole approximation using the formula

$$\left\langle \frac{d^2 P}{d^2 \Omega} \right\rangle = \langle \|\vec{p}\|^2 \rangle \frac{\sin^2 \vartheta}{16\pi^2 \varepsilon_0 c^3}. \quad (11.6)$$

When is the dipole approximation valid?

Solution: The dipole moment is given by

$$\vec{p} = 2Qa \cos(\omega t) \vec{e}_z. \quad (11.S24)$$

The average power per solid angle at $r = \|\vec{x}\|$ is then

$$\left\langle \frac{d^2 P}{d^2 \Omega} \right\rangle = \langle \|\vec{p}\|^2 \rangle \frac{\sin^2 \vartheta}{16\pi^2 \varepsilon_0 c^3} = \frac{Q^2 a^2 \omega^4 \sin^2 \vartheta}{8\pi^2 \varepsilon_0 c^3}. \quad (11.S25)$$

We used $\langle \cos^2(\omega t) \rangle = T^{-1} \int_0^T dt \cos^2(\omega t) = \frac{1}{2}$, with $T = 2\pi/\omega$. The angle ϑ is the angle between the z -axis and $\vec{x}$, the position of the observer.

The dipole approximation is valid when $r \gg \lambda \gg a$.

- b)** Find the current density $\vec{j}(x', t_{\text{ret}})$ and find the vector potential $\vec{A}(x, t)$. Now calculate $\langle d^2 P / d^2 \Omega \rangle$ as measured by an observer far away from the wire ($\|x\| \gg a$). Show that, if $\lambda \gg a$, the leading order of your expression for $\langle d^2 P / d^2 \Omega \rangle$ matches the result obtained in part a) using the dipole approximation.

Solution: We shall work in a complex notation with implicit time-dependence of $e^{-i\omega t}$ such that the charge density is given by

$$\rho(x) = Q \delta(x) \delta(y) [\delta(z-a) - \delta(z+a)]. \quad (11.S26)$$

From the equation of continuity

$$\vec{\partial} \cdot \vec{j} = -\partial_t \rho = i\omega \rho \quad (11.S27)$$

we can find the current density between the metal balls, using a Gauß box around the ball at $z = a$

$$-\int_{\partial V} dx dy \vec{j}(x, t) \cdot \vec{e}_z = \int_V dx^3 \vec{\partial} \cdot \vec{j} = i\omega \int dx dy dz \rho. \quad (11.S28)$$

V is the region $|z| < a$ enclosing the current distribution. Thus

$$\vec{j}(x, t) = i\omega \int dz \rho \vec{e}_z = -i\omega Q \delta(x) \delta(y) \vec{e}_z. \quad (11.S29)$$

The vector potential in the radiation zone is given by ($k = \omega/c$)

$$\begin{aligned} \vec{A} &\approx \int dx'^3 \frac{\mu_0 \vec{j}(x')}{4\pi r} e^{ikr} e^{-ik\vec{a} \cdot \vec{x}'} \\ &= -\frac{\mu_0 i\omega Q \vec{e}_z}{4\pi r} e^{ikr} \int_{-a}^{+a} dz' e^{-ikz' \cos \vartheta} \\ &= \frac{-i\mu_0 c Q \vec{e}_z}{2\pi r \cos \vartheta} e^{ikr} \sin(ka \cos \vartheta). \end{aligned} \quad (11.S30)$$

For the radiation we obtain

$$\begin{aligned} \left\langle \frac{d^2 P}{d^2 \Omega} \right\rangle &= r^2 \langle \vec{n} \cdot \vec{S} \rangle = r^2 c \langle w \rangle = r^2 c \frac{k^2}{2\mu_0} (\|\vec{A}\|^2 - |\vec{n} \cdot \vec{A}|^2) \\ &= \frac{\mu_0 k^2 c^3 Q^2}{8\pi^2 \cos^2 \vartheta} (1 - \cos^2 \vartheta) \sin^2(ka \cos \vartheta) = \frac{\mu_0 k^2 c^3 Q^2}{8\pi^2} \tan^2 \vartheta \sin^2(ka \cos \vartheta) \end{aligned} \quad (11.S31)$$

This can be simplified in the case when $ka \ll 1$ using $\sin x \approx x$ to

$$\left\langle \frac{d^2 P}{d^2 \Omega} \right\rangle = \frac{\mu_0 a^2 Q^2 k^4 c^3 \sin^2 \vartheta}{8\pi^2}. \quad (11.S32)$$

This is the same result that we obtained in the dipole approximation.

11.3. Energy and momentum flux of a plane wave

Let us consider a monochromatic plane wave travelling in the positive z -direction which is linearly polarised in the x -direction and has the amplitude E_0 .

- a) Compute the Poynting vector S , and show that the intensity I of the wave reads

$$I := \langle \|S\| \rangle = \frac{1}{2} c \varepsilon_0 E_0^2. \quad (11.7)$$

Solution: The Poynting vector for this plane wave is given by

$$\vec{S} = \frac{1}{\varepsilon_0} (\vec{E} \times \vec{B}) = \varepsilon_0 c E_0^2 \cos^2(kz - \omega t + \delta) \vec{e}_z. \quad (11.S33)$$

Since the average of $\cos^2$ over a single cycle is $1/2$, the average intensity is

$$\langle \|S\| \rangle = \frac{1}{2} c \varepsilon_0 E_0^2. \quad (11.S34)$$

- b) The Maxwell stress tensor T is defined as

$$T_{ij} := \varepsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} \vec{E}^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} \vec{B}^2 \right). \quad (11.8)$$

Compute T for the given plane wave. What does it imply for the momentum flux?

Hint: Use the real representation of the plane wave.

Solution: We consider a monochromatic plane wave with fixed amplitude, direction $\vec{e}_z$, frequency ω and phase δ . For an electric field polarised in the x -direction we can write in real representation

$$\begin{aligned} \vec{E}(x, t) &= E_0 \cos(kz - \omega t + \delta) \vec{e}_x, \\ \vec{B}(x, t) &= \frac{E_0}{c} \cos(kz - \omega t + \delta) \vec{e}_y. \end{aligned} \quad (11.S35)$$

From the definition of the stress tensor and the form of the plane wave it is clear that the non-diagonal elements of T involving product of different components are zero (both $E_i E_j = 0$ and $B_i B_j = 0$ for $i \neq j$). Using $c = 1/\sqrt{\varepsilon_0 \mu_0}$ the diagonal elements are:

$$\begin{aligned} T_{xx} &= \frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) - \frac{E_0^2}{2\mu_0 c^2} \cos^2(kz - \omega t + \delta) \\ &= \frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) - \frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) = 0, \\ T_{yy} &= -\frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) + \frac{E_0^2}{2\mu_0 c^2} \cos^2(kz - \omega t + \delta) \\ &= -\frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) + \frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) = 0, \\ T_{zz} &= -\frac{\varepsilon_0 E_0^2}{2} \cos^2(kz - \omega t + \delta) - \frac{E_0^2}{2\mu_0 c^2} \cos^2(kz - \omega t + \delta) \\ &= -\varepsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \end{aligned} \quad (11.S36)$$

Thus there is no momentum flux in the x - or y -directions. The flux density of the z -momentum in the z -direction reads

$$-T_{zz} = \varepsilon_0 E_0^2 \cos^2(kz - \omega t + \delta). \quad (11.S37)$$

- c) How are momentum flux density and energy flux density related in this case?

Solution: The Poynting vector has the physical meaning of power per unit area, i.e. energy flux density, and from the previous calculation it follows

$$\langle \|S\| \rangle = -c T_{zz}. \quad (11.S38)$$

Since the wave moves with speed c , a unit area sweeps out a volume c in unit time. Thus the average energy density is $S_z/c = -T_{zz}$ and it is therefore the same as the momentum flux density.

12.1. Optics via the principle of least action

Fermat's principle states that light travelling between two points in space $\vec{x}_1$ and $\vec{x}_2$ should minimise the optical path. The latter is given by

$$S = \int_{\vec{x}_1}^{\vec{x}_2} n(\vec{x}) dl, \quad (12.1)$$

where $n(\vec{x})$ denotes the refractive index of the matter and $dl = \sqrt{dx^2 + dy^2 + dz^2}$ is the length of the infinitesimal element of the trajectory connecting $\vec{x}_1$ to $\vec{x}_2$. This can be directly interpreted as the principle of least action.

Hint: It is convenient to parametrise the trajectory for this integral with a variable t

$$S = \int_{t_1}^{t_2} n(\vec{x}(t)) \frac{dl}{dt} dt = \int_{t_1}^{t_2} n(\vec{x}(t)) \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt. \quad (12.2)$$

The choice of t does not contribute to the integral and therefore you may identify it (locally) with one of the coordinates.

a) Find the trajectory of light between two points in a homogeneous medium.

Solution: The action in this case is $S = nL$ where L is the length of trajectory. The t -dependence drops out and we can identify it with the x -coordinate (suppose that the endpoints of the path have distinct values of x and that the path does not wiggle back and forth in the x -direction; otherwise choose a different coordinate and / or shorter pieces of the path such that the conditions hold). The action is then rewritten as

$$S = nL = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2} dx. \quad (12.S1)$$

with the functional $F(y'(x), z'(x)) = n\sqrt{1 + y'^2(x) + z'^2(x)}$. The Euler-Lagrange equations are

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = \frac{\partial F}{\partial z} - \frac{d}{dx} \frac{\partial F}{\partial z'} = 0, \quad (12.S2)$$

and inserting the derivatives we find the differential equations

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2 + z'^2}} \right) = \frac{d}{dx} \left(\frac{z'}{\sqrt{1 + y'^2 + z'^2}} \right) = 0, \quad (12.S3)$$

which can be integrated to

$$\frac{y'}{\sqrt{1 + y'^2 + z'^2}} = c_1, \quad \frac{z'}{\sqrt{1 + y'^2 + z'^2}} = c_2. \quad (12.S4)$$

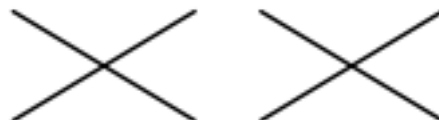
These are coupled equations that describe a straight line. To see this, let us decouple the two equations by inserting $y'^2 = c_1^2(1 + z'^2)/(1 - c_1^2)$ into the differential equation for z' . After some algebra we arrive at

$$y'^2 = \frac{c_1^2}{1 - c_1^2 - c_2^2}, \quad z'^2 = \frac{c_2^2}{1 - c_1^2 - c_2^2}, \quad (12.S5)$$

so the solutions are of the type

$$y(x) = a_1 x + b_1, \quad z(x) = a_2 x + b_2, \quad (12.S6)$$

which are parametric equations of a straight line.



$$(12.3)$$

- b) Now consider light that is reflected from a plane mirror. The light travels in vacuum from point $\vec{x}_1$ to some point $\vec{x}$ on the surface of the mirror, and then, again in vacuum, to some point $\vec{x}_2$. Minimise the action over all positions of $\vec{x}$ on the mirror, and compare the incident and emergent angles for the chosen value of $\vec{x}$.

Solution: Suppose the mirror is in the $z = 0$ plane and points $\vec{x}_1$ and $\vec{x}_2$ lie in the $y = 0$ plane. As we have already shown, light will travel on a straight line when it travels in homogeneous media, except of when it encounters the mirror. The total light path is then

$$S = n(\|\vec{x} - \vec{x}_1\| + \|\vec{x}_2 - \vec{x}\|) = n\left(\sqrt{(x_1 - x)^2 + y^2 + z_1^2} + \sqrt{(x_2 - x)^2 + y^2 + z_2^2}\right). \quad (12.S7)$$

First we find the minimum of S by first differentiating by y

$$0 \stackrel{!}{=} \frac{\partial S}{\partial y} = \frac{ny}{\sqrt{(x_1 - x)^2 + y^2 + z_1^2}} + \frac{ny}{\sqrt{(x_2 - x)^2 + y^2 + z_2^2}}. \quad (12.S8)$$

This equation is solved by $y = 0$ only because both square roots are positive. We are free to set $y = 0$ in what follows. Next we differentiate by x and

$$0 \stackrel{!}{=} \frac{\partial S}{\partial x} = \frac{n(x - x_1)}{\sqrt{(x_1 - x)^2 + z_1^2}} + \frac{n(x - x_2)}{\sqrt{(x_2 - x)^2 + z_2^2}}. \quad (12.S9)$$

From figure (12.3) we can relate the two fractions to sines of the angles α_1 and α_2

$$0 \stackrel{!}{=} n \sin \alpha_1 - n \sin \alpha_2. \quad (12.S10)$$

Since both angles are bounded by $\pm\pi/2$ we find the law of reflection $\alpha_1 = \alpha_2$.

-
- c) Finally, consider light propagating between two points in space which are located in different media with the refractive indices n_1 and n_2 , respectively. The boundary surface between the two media is a plane. Consider a light path between point $\vec{x}_1$ in the medium with n_1 and $\vec{x}_2$ in the medium with n_2 , which passes through the point $\vec{x}$ on the boundary surface. Choose $\vec{x}$ that minimises the total light path. Find the relationship between incident and emergent angles (Snell's law).

Solution: Again, within both media light will travel in straight lines, but it may change direction at the interface. We can simplify the problem to two dimensions right away, i.e. $y = 0$ will follow by the same argument as above. The total light path is then

$$S = n_1\|\vec{x} - \vec{x}_1\| + n_2\|\vec{x}_2 - \vec{x}\| = n_1\sqrt{(x_1 - x)^2 + z_1^2} + n_2\sqrt{(x_2 - x)^2 + z_2^2}. \quad (12.S11)$$

We note that the geometry of the problem obeys the same four equations calculated in the previous exercise. Extremising S we arrive at Snell's law

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2. \quad (12.S12)$$

12.2. Refraction of planar waves

A planar wave is incident perpendicularly onto a planar layer between two media. The indices of refraction of the three non-magnetic layers are n_1 , n_2 and n_3 . The thickness of the central layer is d , while the other two media each fill half spaces.

- a) Calculate the reflection and transmission coefficients (i.e. the ratio of the reflected and transmitted wave with the incoming energy flux).

Hint: The time-averaged energy-flux density of a complex wave is given by

$$\langle \vec{S} \rangle = \frac{1}{2\mu_0} \operatorname{Re}(\vec{E} \times \vec{B}^*). \quad (12.4)$$

Solution: We make an ansatz for the E -field as superposed incoming and reflected planar waves for each of the three media. The B -field follows then from $\partial_t \vec{B} = -\vec{\partial} \times \vec{E}$ and $\omega = ck_i/n_i$.

- In medium 1 with index of refraction n_1 ($z < 0$) :

$$\begin{aligned}\vec{E} &= E_i^{(1)} \vec{e}_x \exp(i(k_1 z - \omega t)) + E_r^{(1)} \vec{e}_x \exp(i(-k_1 z - \omega t)), \\ \vec{B} &= \frac{n_1}{c} E_i^{(1)} \vec{e}_y \exp(i(k_1 z - \omega t)) - \frac{n_1}{c} E_r^{(1)} \vec{e}_y \exp(i(-k_1 z - \omega t)).\end{aligned}\quad (12.S13)$$

- In medium 2 with index of refraction n_2 ($0 < z < d$) :

$$\begin{aligned}\vec{E} &= E_i^{(2)} \vec{e}_x \exp(i(k_2 z - \omega t)) + E_r^{(2)} \vec{e}_x \exp(i(-k_2 z - \omega t)), \\ \vec{B} &= \frac{n_2}{c} E_i^{(2)} \vec{e}_y \exp(i(k_2 z - \omega t)) - \frac{n_2}{c} E_r^{(2)} \vec{e}_y \exp(i(-k_2 z - \omega t)).\end{aligned}\quad (12.S14)$$

- In medium 3 with index of refraction n_3 ($z > d$) :

$$\begin{aligned}\vec{E}(\vec{x}, t) &= E_t^{(3)} \vec{e}_x \exp(i(k_3 z - \omega t)), \\ \vec{B}(\vec{x}, t) &= \frac{n_3}{c} E_t^{(3)} \vec{e}_y \exp(i(k_3 z - \omega t)).\end{aligned}\quad (12.S15)$$

We now need to satisfy the boundary conditions: $E_{\parallel}$ and $H_{\parallel}$ have to be continuous. Since we assume that $\mu_1 = \mu_2 = \mu_3 = \mu_0$, also $B_{\parallel}$ has to be continuous.

- At $z = 0$:

$$\begin{aligned}E_i^{(1)} + E_r^{(1)} &= E_i^{(2)} + E_r^{(2)}, \\ n_1(E_i^{(1)} - E_r^{(1)}) &= n_2(E_i^{(2)} - E_r^{(2)}).\end{aligned}\quad (12.S16)$$

- At $z = d$:

$$\begin{aligned}E_i^{(2)} \exp(ik_2 d) + E_r^{(2)} \exp(-ik_2 d) &= E_t^{(3)} \exp(ik_3 d), \\ n_2[E_i^{(2)} \exp(ik_2 d) - E_r^{(2)} \exp(-ik_2 d)] &= n_3 E_t^{(3)} \exp(ik_3 d).\end{aligned}\quad (12.S17)$$

Solving this linear system of equations, we obtain from the first two equations

$$\begin{aligned}E_r^{(1)} &= E_i^{(2)} + E_r^{(2)} - E_i^{(1)}, \\ \Rightarrow n_1[E_i^{(1)} - (E_i^{(2)} + E_r^{(2)} - E_i^{(1)})] &= n_2[E_i^{(2)} - E_r^{(2)}], \\ \Rightarrow E_i^{(2)} &= \frac{(n_2 - n_1)E_r^{(2)} + 2n_1 E_i^{(1)}}{n_1 + n_2}.\end{aligned}\quad (12.S18)$$

The third equation can then be written as

$$E_r^{(2)} = \frac{E_t^{(3)}(n_1 + n_2) e^{ik_3 d} - 2n_1 E_i^{(1)} e^{ik_2 d}}{(n_2 - n_1) e^{ik_2 d} + (n_1 + n_2) e^{-ik_2 d}}, \quad (12.S19)$$

and the fourth equation yields

$$E_t = \frac{4n_1 n_2 E_i e^{i(k_2 - k_3)d}}{(n_1 n_2 + n_2 n_3)(1 + e^{2ik_2 d}) + (n_2^2 + n_1 n_3)(1 - e^{2ik_2 d})}. \quad (12.S20)$$

Putting all the results back into the first equation, we find

$$E_r = E_i \frac{(n_1 n_2 - n_2 n_3)(1 + e^{2ik_2 d}) + (n_1 n_3 - n_2^2)(1 - e^{2ik_2 d})}{(n_1 n_2 + n_2 n_3)(1 + e^{2ik_2 d}) + (n_2^2 + n_1 n_3)(1 - e^{2ik_2 d})}. \quad (12.S21)$$

The time averaged energy flux density is calculated by

$$\langle S \rangle = \frac{1}{2\mu_0} \text{Re}(\vec{E} \times \vec{B}^*), \quad (12.S22)$$

so in particular

$$\begin{aligned}
|S_i| &= \frac{n_1}{\mu_0} |E_i^{(1)}|^2, \\
|S_r| &= \frac{n_1}{\mu_0} |E_r^{(1)}|^2 = |S_i| \frac{(n_1 n_2 - n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_1 n_3 - n_2^2)^2 (1 - \cos(2k_2 d))}{(n_1 n_2 + n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_2^2 + n_1 n_3)^2 (1 - \cos(2k_2 d))}, \\
|S_t| &= \frac{n_3}{\mu_0} |E_t^{(3)}|^2 = \frac{n_3}{n_1} |S_i| \frac{8n_1^2 n_2^2}{(n_1 n_2 + n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_2^2 + n_1 n_3)^2 (1 - \cos(2k_2 d))}. \quad (12.S23)
\end{aligned}$$

The reflection coefficient is then given by

$$R = \frac{|S_r|}{|S_i|} = \frac{(n_1 n_2 - n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_1 n_3 - n_2^2)^2 (1 - \cos(2k_2 d))}{(n_1 n_2 + n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_2^2 + n_1 n_3)^2 (1 - \cos(2k_2 d))}, \quad (12.S24)$$

and the transmission coefficient by

$$T = \frac{|S_t|}{|S_i|} = \frac{n_3}{n_1} \frac{8n_1^2 n_2^2}{(n_1 n_2 + n_2 n_3)^2 (1 + \cos(2k_2 d)) + (n_2^2 + n_1 n_3)^2 (1 - \cos(2k_2 d))}. \quad (12.S25)$$

- b) Let the medium with index n_1 be part of an optical system (e.g. a lens), and the medium with index n_3 be air ($n_3 = 1$). The surface of the first medium should have a layer of the medium with index n_2 of such a thickness that for a given frequency ω_0 , there is no reflected wave. Determine the thickness d and the index of refraction n_2 of this layer.

Solution: The layer in between has to be chosen such that $E_r^{(1)} = 0$, i.e.

$$e^{2ik_2 d} = \frac{n_2^2 - n_1 n_3 + n_2 n_3 - n_1 n_2}{n_2^2 + n_1 n_2 - n_2 n_3 - n_1 n_3}. \quad (12.S26)$$

The right hand side is real, therefore

$$e^{2ik_2 d} = \pm 1 \quad \implies \quad k_2 d = \frac{m\pi}{2}, \quad m \in \mathbb{Z}_+. \quad (12.S27)$$

For m even ($e^{2ik_2 d} = 1$) we find

$$\begin{aligned}
n_2^2 - n_1 n_3 + n_2 n_3 - n_1 n_2 &= n_2^2 + n_1 n_2 - n_2 n_3 - n_1 n_3, \\
\implies n_1 &= n_3.
\end{aligned} \quad (12.S28)$$

and for m odd ($e^{2ik_2 d} = -1$) we find

$$\begin{aligned}
n_2^2 - n_1 n_3 + n_2 n_3 - n_1 n_2 &= -n_2^2 - n_1 n_2 + n_2 n_3 + n_1 n_3 \\
\implies n_2 &= \sqrt{n_1 n_3}.
\end{aligned} \quad (12.S29)$$

The first solution is not a solution for our problem since $n_1 \neq n_3$. The second solution provides the requirements for the central layer. Using $n_3 = 1$, we get

$$\begin{aligned}
n_2 &= \sqrt{n_1}, \\
d &= \frac{(2m+1)\pi}{2k_2} = \frac{(2m+1)\pi}{2n_2 \omega_0}, \quad m \in \mathbb{Z}_+.
\end{aligned} \quad (12.S30)$$

12.3. Scattering of light

Classical light-scattering theory (known as Rayleigh theory) is used to describe light being scattered off small molecules (with an extension much smaller than the wavelength λ of the light). Here we consider electric and magnetic fields and the intensity of light scattered off small particles.

- a) First consider a plane monochromatic light wave propagating in x -direction, which is polarised in z -direction. This wave is scattered off a small polarisable, but non-magnetic particle at the origin. The incident wave induces a dipole moment to the particle, that is proportional to the local field, $\vec{p}(t) = \alpha \vec{E}(0, t)$, where α is its polarisability. Determine the electric and magnetic fields of the scattered wave at a far-away point $\vec{x}$, depending on the incident field E_0 , the distance from the origin r , and the angle ϑ between $\vec{x}$ and the z -axis.

Solution: The incident wave can be written as

$$\vec{E}_i(\vec{x}, t) = E_0 \vec{e}_z e^{i(kx - \omega t)} = E_0 \vec{e}_z e^{ik(x - ct)} \quad \text{with } \lambda = \frac{2\pi}{k} = \frac{2\pi c}{\omega}. \quad (12.S31)$$

At the origin we get

$$\vec{E}_i(0, t) = E_0 \vec{e}_z e^{-ikct}, \quad (12.S32)$$

and thus the time-dependent induced dipole moment in the scattering particle is

$$\vec{p}(t) = \alpha \vec{E}_i(0, t) = \alpha E_0 \vec{e}_z e^{-ikct}. \quad (12.S33)$$

Far away from the scattering particle, the magnetic field of the scattered wave has the form of a monochromatic spherical wave

$$\vec{B}_s(\vec{x}, t) = k^2 c \frac{\mu_0}{4\pi} (\vec{e}_r \times \vec{p}(t - t_0)) \frac{1}{r} e^{ikr}, \quad (12.S34)$$

where t_0 is the time when the scattering happened.

According to the Ampère-Maxwell law, the electric field of the scattered wave is

$$\vec{E}_s(\vec{x}, t) = c \vec{B}_s(\vec{x}, t) \times \vec{e}_r = \frac{k^2 c^2 \mu_0}{4\pi} (\vec{e}_r \times \vec{p}(t - t_0)) \times \vec{e}_r \frac{1}{r} e^{ikr}. \quad (12.S35)$$

Inserting $\vec{p}(t - t_0)$, replacing $\vec{e}_z = \cos \vartheta \vec{e}_r - \sin \vartheta \vec{e}_\vartheta$, and making use of $\vec{e}_\varphi = \vec{e}_r \times \vec{e}_\vartheta$, etc. these fields can be written as

$$\begin{aligned} \vec{B}_s(\vec{x}, t) &= \frac{k^2 \alpha E_0}{r} (\vec{e}_r \times (\cos \vartheta \vec{e}_r - \sin \vartheta \vec{e}_\vartheta)) e^{ik(r - c(t - t_0))} = \frac{c \mu_0 k^2 \alpha E_0 \sin \vartheta}{4\pi r} \vec{e}_\varphi e^{ik(r - c(t - t_0))}, \\ \vec{E}_s(\vec{x}, t) &= \frac{k^2 \alpha E_0 \sin \vartheta}{\varepsilon_0 c^2 r} (\vec{e}_\varphi \times \vec{e}_r) e^{ik(r - c(t - t_0))} = \frac{k^2 \alpha E_0 \sin \vartheta}{4\pi \varepsilon_0 r} \vec{e}_\vartheta e^{ik(r - c(t - t_0))}. \end{aligned} \quad (12.S36)$$

- b) Calculate the intensity of this scattered light wave, at a point $\vec{x}$ far away from the scattering particle

Hint: Use the Poynting vector.

Solution: For a monochromatic wave, the intensity can be calculated from the time average of the Poynting vector of the scattered wave,

$$I_s = \langle \|\vec{S}\| \rangle = \frac{1}{2\mu_0} |\text{Re}(\vec{E}_{s,0} \times \vec{B}_{s,0}^*)|, \quad (12.S37)$$

where $\vec{E}_{s,0}$ and $\vec{B}_{s,0}$ are the amplitudes of the electric and magnetic field after splitting off the time dependence (as done in an earlier exercise),

$$\vec{E}_s(t) = \text{Re}(\vec{E}_{s,0} e^{-ikct}), \quad \vec{B}_s(t) = \text{Re}(\vec{B}_{s,0} e^{-ikct}). \quad (12.S38)$$

Inserting the expressions for $\vec{E}_s(t)$ and $\vec{B}_s(t)$ yields

$$I_s = \frac{ck^4 \alpha^2 E_0^2 \sin^2 \vartheta}{16\pi^2 \varepsilon_0 r^2} = \frac{c\pi^2 \alpha^2 E_0^2 \sin^2 \vartheta}{\varepsilon_0 \lambda^4 r^2}. \quad (12.S39)$$

One can check that $[I_s] = \text{W} / \text{m}^2$, since $[\alpha] = \text{A}^2 \text{s}^4 / \text{kg}$, $[E_0] = \text{kg m} / \text{A s}^3$ and $[\varepsilon_0] = \text{A}^2 \text{s}^4 / \text{kg m}^3$.

- c) Use the wave-length dependence of the intensity of the scattered wave ($\propto 1/\lambda^4$) derived in the previous subproblem, to explain qualitatively the blue colour of the cloudless sky and the red colour of the sunrise and the sunset.

Solution: Apart from spectral lines, which do not play a role here, electromagnetic waves emitted from the sun cover a wide continuous spectrum with an emission maximum in the wavelength region of visible light. Sunlight is incoherent and not polarised before entering the atmosphere. This does not change the situation qualitatively as we can assume unpolarised light as a superposition of light waves polarised to equal parts in the z and y direction. The sunlight is scattered from the small particles in the atmosphere. As the brightness of the sun (at least at a cloudless sky) is indicating, most visible light waves reach the surface essentially unscattered, and we assume that the scattering probability is low enough so that we can neglect multi-particle scattering.

Light that reaches the surface from a different direction than from the sun must have been scattered in the atmosphere. As the scattered intensity depends on the wavelength as $1/\lambda^4$, we find that light at the low wavelength (blue) is much more likely to be scattered than the light at the high wavelength (red). Thus dominantly blue wavelengths reach the surface from there, and consequently the cloudless sky looks blue. At the sunset and sunrise, the sun is located just above the horizon so that the distance that light needs to travel through the atmosphere is significantly increased. Now the higher scattering probability of blue light plays the opposite role: as most of the blue light is scattered off the direct way, from the direction close to the sun, more of the red light reaches the observer on the earth surface, and sunsets and sunrises look red.

13.1. Radiation in Relativistic Circular Motion

A particle of charge q moves with constant speed $v \approx c$ in a circular orbit of radius R in the x, z -plane. At time t' it passes through the origin $\vec{x}' = 0$ with velocity $\vec{v} = v\vec{e}_z$ and is accelerated with $\vec{a} = a\vec{e}_x = (v^2/R)\vec{e}_x$.

The electric field at position $\vec{x}$ and time $t = t' + \|\vec{x} - \vec{x}'\|/c$ of a relativistic point charge at $\vec{x}'$ with velocity $\vec{v}$ and acceleration $\vec{a}$ is given by

$$\vec{E}(\vec{x}, t) = \frac{q}{4\pi\epsilon_0} \frac{\|\vec{x} - \vec{x}'\|}{((\vec{x} - \vec{x}') \cdot \vec{w})^3} \left[(1 - \beta^2) \vec{w} + \frac{1}{c^2} (\vec{x} - \vec{x}') \times (\vec{w} \times \vec{a}) \right], \quad (13.1)$$

where $\vec{w} := (\vec{x} - \vec{x}')/\|\vec{x} - \vec{x}'\| - \vec{v}/c$ and $\beta := v/c$.

- a) Using spherical coordinates and ignoring terms in (13.1) that fall off faster than $1/r$, show that the electric field measured by an observer at $\vec{x} = r\vec{e}_r$ at large distance r is

$$\vec{E}_{\text{rad}}(\vec{x}, t) = \frac{qa}{4\pi\epsilon_0 c^2 r} \frac{(\beta - \cos \vartheta) \cos \varphi \vec{e}_\vartheta + (1 - \beta \cos \vartheta) \sin \varphi \vec{e}_\varphi}{(1 - \beta \cos \vartheta)^3}. \quad (13.2)$$

Solution: Using (13.1) and ignoring contributions which fall off faster than $1/r$ we have

$$\vec{E}_{\text{rad}} = \frac{q}{4\pi\epsilon_0 c^2} \frac{\|\vec{x} - \vec{x}'\|}{((\vec{x} - \vec{x}') \cdot \vec{w})^3} [(\vec{x} - \vec{x}') \times (\vec{w} \times \vec{a})] = \frac{q}{4\pi\epsilon_0 c^2} \frac{r}{(\vec{x} \cdot \vec{w})^3} [\vec{x} \times (\vec{w} \times \vec{a})] \quad (13.S1)$$

evaluating $\vec{E}_{\text{rad}}$ at $\vec{x} - \vec{x}' = \vec{x} = r\vec{e}_r$ as the retarded position is at the origin.

Now we need to make the conversion to spherical polar coordinates. To do this we make the following change of basis vectors

$$\begin{aligned} \vec{e}_z &= \cos \vartheta \vec{e}_r - \sin \vartheta \vec{e}_\vartheta, \\ \vec{e}_x &= \sin \vartheta \cos \varphi \vec{e}_r + \cos \vartheta \cos \varphi \vec{e}_\vartheta - \sin \varphi \vec{e}_\varphi. \end{aligned} \quad (13.S2)$$

Now $\vec{w} = \vec{e}_r - \beta \vec{e}_z$. Also $\vec{a} = a\vec{e}_x$. Therefore $\vec{x} \cdot \vec{w} = r - \beta r \cos \vartheta$.

To evaluate the cross product in the radiation term, we use the identity $\vec{r} \times (\vec{w} \times \vec{a}) = \vec{w}(\vec{r} \cdot \vec{a}) - (\vec{r} \cdot \vec{w})\vec{a}$. Thus

$$\begin{aligned} \vec{r} \times (\vec{w} \times \vec{a}) &= (\vec{e}_r - \beta \vec{e}_z)(a r (\vec{e}_r \cdot \vec{e}_x)) - r(\vec{e}_r \cdot (\vec{e}_r - \beta \vec{e}_z)) a \vec{e}_x \\ &= r a \sin \vartheta \cos \varphi \vec{e}_r - \beta r a \sin \vartheta \cos \varphi (\cos \vartheta \vec{e}_r - \sin \vartheta \vec{e}_\vartheta) - a r \vec{e}_x + a \beta r \cos \vartheta \vec{e}_x \\ &= a r \sin \vartheta \cos \varphi (1 - \beta \cos \vartheta) \vec{e}_r + \beta r a \sin^2 \vartheta \cos \varphi \vec{e}_\vartheta \\ &\quad - a r (\sin \vartheta \cos \varphi \vec{e}_r + \cos \vartheta \cos \varphi \vec{e}_\vartheta - \sin \varphi \vec{e}_\varphi) \\ &\quad + (a \beta r (\sin \vartheta \cos \varphi \vec{e}_r + \cos \vartheta \cos \varphi \vec{e}_\vartheta - \sin \varphi \vec{e}_\varphi) \cos \vartheta) \\ &= a r (\beta - \cos \vartheta) \cos \varphi \vec{e}_\vartheta + a r \sin \varphi (1 - \beta \cos \vartheta) \vec{e}_\varphi \end{aligned} \quad (13.S3)$$

So using the basis vectors, velocity and acceleration in spherical coordinates we arrive at the expression

$$\vec{r} \times \vec{w} \times \vec{a} = [a r (\beta - \cos \vartheta) \cos \varphi] \vec{e}_\vartheta + [a r \sin \varphi (1 - \beta \cos \vartheta)] \vec{e}_\varphi. \quad (13.S4)$$

Now we have expressions for the scalar and cross products in the equation for $\vec{E}_{\text{rad}}$. Inserting them and simplifying gives

$$\vec{E}_{\text{rad}} = \frac{qa}{4\pi\epsilon_0 c^2 r} \frac{(\beta - \cos \vartheta) \cos \varphi \vec{e}_\vartheta + (1 - \beta \cos \vartheta) \sin \varphi \vec{e}_\varphi}{(1 - \beta \cos \vartheta)^3}. \quad (13.S5)$$

For fields of radiation, $\vec{E}_{\text{rad}}$ is perpendicular to $\vec{x} - \vec{x}'$, so the Poynting vector for the radiation component of the fields due to a point charge is

$$\vec{S}_{\text{rad}} = \frac{1}{\mu_0 c} \vec{E}_{\text{rad}}^2 \frac{\vec{x} - \vec{x}'}{\|\vec{x} - \vec{x}'\|}. \quad (13.3)$$

b) Using (13.2) and (13.3) show that the Poynting vector at the position of the observer is

$$\vec{S}_{\text{rad}} = \frac{\mu_0 q^2 v^4}{16\pi^2 c r^2 R^2} \frac{(1 - \beta \cos \vartheta)^2 - (1 - \beta^2) \cos^2 \varphi \sin^2 \vartheta}{(1 - \beta \cos \vartheta)^6} \vec{e}_r. \quad (13.4)$$

Solution: To find the Poynting vector we must find $\|E_{\text{rad}}\|^2$ in spherical polars. Explicitly, this is

$$\|\vec{E}_{\text{rad}}\|^2 = \frac{q^2 a^2 \mu_0^2}{16\pi^2 r^2} \frac{(1 - \beta \cos \vartheta)^2 \sin^2 \varphi + \cos^2 \varphi (\beta - \cos \vartheta)^2}{(1 - \beta \cos \vartheta)^6}. \quad (13.S6)$$

We simplify the numerator using $\cos^2 x + \sin^2 x = 1$,

$$\begin{aligned} & (1 - \beta \cos \vartheta)^2 \sin^2 \varphi + \cos^2 \varphi (\beta - \cos \vartheta)^2 \\ &= (1 - \beta \cos \vartheta)^2 + \cos^2 \varphi ((\beta - \cos \vartheta)^2 - (1 - \beta \cos \vartheta)^2) \\ &= (1 - \beta \cos \vartheta)^2 + \cos^2 \varphi \sin^2 \vartheta (\beta^2 - 1). \end{aligned} \quad (13.S7)$$

Therefore, inserting $(\vec{x} - \vec{x}')/\|x - x'\| = \vec{e}_r$ because the retarded position is at the origin, and using $a = v^2/R$ to get the result in terms of velocity explicitly, we have

$$\vec{S}_{\text{rad}} = \frac{\mu_0 q^2 v^4}{16\pi^2 c r^2 R^2} \frac{(1 - \beta \cos \vartheta)^2 - \cos^2 \varphi \sin^2 \vartheta (1 - \beta^2)}{(1 - \beta \cos \vartheta)^6} \vec{e}_r. \quad (13.S8)$$

c) Show that the total power radiated out towards infinity by the point charge is

$$P := \oint d^2\Omega r^2 \vec{e}_r \cdot \vec{S}_{\text{rad}} \frac{\vec{x} \cdot \vec{w}}{r} = \frac{\mu_0 q^2 v^4}{6\pi c R^2 (1 - \beta^2)^2}. \quad (13.5)$$

Note that we assume the sphere where radiation is measured to be concentric to the orbit. Therefore, there are retardation effects which are accounted for by the factor $\vec{x} \cdot \vec{w}/r$ in the above formula.

Hint: You will need the following integrals:

$$\int_{-1}^1 \frac{dx}{(1 - \beta x)^3} = \frac{2}{(1 - \beta^2)^2}, \quad \int_{-1}^1 dx \frac{1 - x^2}{(1 - \beta x)^5} = \frac{4}{3(1 - \beta^2)^3}. \quad (13.6)$$

Solution: DH: Explanation of equation 13.5 : When a particle is moving relativistically the rate at which energy passes through the observation point is not the same as the rate at which the energy left the particle. So there are retardation effects, as we want to find the power at the retarded time.

$$\frac{dW}{dt_r} = \frac{dW/dt}{\partial t_r / \partial t}. \quad (13.S9)$$

Now

$$c(t - t_r) = \|x - x'\|. \quad (13.S10)$$

Thus $c^2(t - t_r)^2 = \|x - x'\|^2 = (\vec{x} - \vec{x}') \cdot (\vec{x} - \vec{x}')$. Now we differentiate w.r.t. t to give

$$2c^2(t - t_r) \left(1 - \frac{\partial t_r}{\partial t}\right) = 2(\vec{x} - \vec{x}') \cdot \frac{\partial(\vec{x} - \vec{x}')}{\partial t} \quad (13.S11)$$

and so therefore

$$c\|x - x'\| \left(1 - \frac{\partial t_r}{\partial t}\right) = (\vec{x} - \vec{x}') \cdot \frac{\partial(\vec{x} - \vec{x}')}{\partial t}. \quad (13.S12)$$

Now using $c(t - t_r) = \|x - x'\|$,

$$\begin{aligned} \frac{\partial(\vec{x} - \vec{x}')}{\partial t} &= -\frac{\partial(\vec{x}')}{\partial t} \\ &= -\frac{\partial(\vec{x}')}{\partial t_r} \frac{\partial t_r}{\partial t} \\ &= -\vec{v} \frac{\partial t_r}{\partial t} \end{aligned} \quad (13.S13)$$

as only the retarded position has any time dependence. The position of the observer is fixed w.r.t. time. $\vec{v}$ is the velocity of the moving charge. So now putting (13.S13) into (13.S12) we obtain

$$c\|x - x'\| \left(1 - \frac{\partial t_r}{\partial t}\right) = -(\vec{x} - \vec{x}') \cdot \vec{v} \frac{\partial t_r}{\partial t}. \quad (13.S14)$$

Rearranging this we have $c\|x - x'\| = (c\|x - x'\| - \vec{v} \cdot (\vec{x} - \vec{x}')) \partial t_r / \partial t$. And so therefore

$$\begin{aligned} \|x - x'\| &= \left(\|x - x'\| - \frac{\vec{v}}{c} \cdot (\vec{x} - \vec{x}') \right) \frac{\partial t_r}{\partial t} \\ &= (\vec{x} - \vec{x}') \cdot \left(\frac{(\vec{x} - \vec{x}')}{\|x - x'\|} - \frac{\vec{v}}{c} \right) \frac{\partial t_r}{\partial t} \\ &= (\vec{x} - \vec{x}') \cdot \vec{w} \frac{\partial t_r}{\partial t}. \end{aligned} \quad (13.S15)$$

Thus

$$\frac{\partial t_r}{\partial t} = \frac{\|x - x'\|}{(\vec{x} - \vec{x}') \cdot \vec{w}} \quad (13.S16)$$

So now we can insert (13.S16) into (13.S9) to obtain

$$\frac{dW}{dt_r} = \frac{dW}{dt} \frac{(\vec{x} - \vec{x}') \cdot \vec{w}}{\|x - x'\|} \quad (13.S17)$$

DH: end derivation of funny factor

Recalling that $\vec{r} \cdot \vec{w} = r(1 - \beta \cos \vartheta)$ and using that $dx^2 = r^2 d\vartheta d\varphi \sin \vartheta$ and that $\vec{n} = \vec{e}_r$,

$$P = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \left(\int_0^\pi \int_0^{2\pi} \frac{(1 - \beta \cos \vartheta)^2 \sin \vartheta d\vartheta d\varphi}{(1 - \beta \cos \vartheta)^5} - \int_0^\pi \int_0^{2\pi} \frac{(1 - \beta^2) \sin^3 \vartheta \cos^2 \varphi d\vartheta d\varphi}{(1 - \beta \cos \vartheta)^5} \right). \quad (13.S18)$$

To perform the integration over φ we use that $\int_0^{2\pi} d\varphi \cos^2 \varphi = \pi$ which is easy to show using that $\cos 2\varphi = 2 \cos^2 \varphi - 1$.

After integrating over φ we have

$$P = \frac{\mu_0 q^2 a^2}{16\pi c} \left(2 \int_0^\pi \frac{d\vartheta \sin \vartheta}{(1 - \beta \cos \vartheta)^3} - \int_0^\pi d\vartheta \frac{(1 - \beta^2) \sin^3 \vartheta}{(1 - \beta \cos \vartheta)^5} \right). \quad (13.S19)$$

Now let $x = \cos \vartheta$. With this change of variable the expression for power now reads

$$P = \frac{\mu_0 q^2 a^2}{16\pi c} \left(2 \int_{-1}^1 \frac{dx}{(1 - \beta x)^3} - \int_{-1}^1 dx \frac{(1 - \beta^2)(1 - x^2)}{(1 - \beta x)^5} \right). \quad (13.S20)$$

Now we use the integrals given in the hint,

$$\begin{aligned} \int_{-1}^1 \frac{dx}{(1 - \beta x)^2} &= \frac{2}{(1 - \beta^2)^3} = 2\gamma^4, \\ \int_{-1}^1 dx \frac{1 - x^2}{(1 - \beta x)^5} &= \frac{4}{3} \frac{1}{(1 - \beta^2)^3} = \frac{4}{3} \gamma^6, \end{aligned} \quad (13.S21)$$

together with the identification that $1 - \beta^2 = 1/\gamma^2$, to obtain the following expression for the power radiated to infinity:

$$P = \frac{\mu_0 q^2 a^2}{16\pi c} \left(4\gamma^4 - \frac{4}{3} \gamma^4 \right) = \frac{\mu_0 q^2 a^2 \gamma^4}{6\pi c}. \quad (13.S22)$$

So finally, using $a = v^2/R$, for an observer in the far field we have,

$$P = \frac{\mu_0 q^2 v^4 \gamma^4}{6\pi R^2 c} \quad (13.S23)$$

explicitly in terms of the velocity of the particle.

- d) Calculate the energy radiated to infinity during one complete circle. For an electron ($m_e = 511 \text{ keV}/c^2$) travelling with $\beta = 0.8$ and $R = 20 \text{ m}$, how much energy is radiated in this one orbit? Compare this value to the total relativistic energy of the electron.

Solution: As v is constant the time to complete one orbit is $T = 2\pi R/v$. The energy radiated during one orbit is therefore given by

$$W = PT = \frac{\mu_0 q^2 v^3 \gamma^4}{3Rc} \quad (13.S24)$$

Inserting the numerical values $q = e = 1.60 \cdot 10^{-19} \text{ C}$, $c = 3 \cdot 10^8 \text{ ms}^{-1}$ and $\mu_0 = 4\pi \cdot 10^{-7} \text{ JA}^{-2}\text{m}^{-1}$ as well as $v = 0.8c$, we find

$$W_{\text{rad}} = 1.91 \cdot 10^{-28} \text{ J} = 1.19 \cdot 10^{-9} \text{ eV} \quad (13.S25)$$

The total relativistic energy of the electron is $W_{\text{rel}} = \gamma mc^2 = \gamma m_e c^2 = 851667 \text{ eV}$ using that $m_e c^2 = 511 \text{ keV}$. Therefore the value for the ratio is

$$\frac{W_{\text{rad}}}{W_{\text{rel}}} = 1.40 \cdot 10^{-15}. \quad (13.S26)$$

13.2. Liénard–Wiechert potential

Consider a charge moving straight along the positive z -axis with a uniform velocity v starting at $z = 0$ at $t = 0$. Show that its potential is given by

$$\Phi(\vec{x}, t) = \frac{q}{4\pi\epsilon_0} \frac{1}{\sqrt{(z - vt)^2 + (1 - v^2/c^2)(x^2 + y^2)}}. \quad (13.7)$$

- a) Calculate the potential directly in the observer frame.

Solution: The scalar potential at position $\vec{x}$ and time t is

$$\Phi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{d\vec{x}' dt' \rho(\vec{x}', t')}{\|\vec{x} - \vec{x}'\|} \delta\left(t - \frac{\|\vec{x} - \vec{x}'\|}{c} - t'\right) \theta(t > t'), \quad (13.S27)$$

where the charge density is $\rho(\vec{x}', t') = q\delta(\vec{x}' - \vec{v}t')$. Using the identity

$$\begin{aligned} & \delta\left((t - t')^2 - \frac{\|\vec{x} - \vec{x}'\|^2}{c^2}\right) \theta(t > t') \\ &= \delta\left(\left[(t - t') - \frac{\|\vec{x} - \vec{x}'\|}{c}\right] \left[(t - t') + \frac{\|\vec{x} - \vec{x}'\|}{c}\right]\right) \theta(t > t') \\ &= \delta\left(\left[t - \frac{\|\vec{x} - \vec{x}'\|}{c} - t'\right] 2 \frac{\|\vec{x} - \vec{x}'\|}{c}\right) \theta(t > t') \\ &= \frac{c}{2\|\vec{x} - \vec{x}'\|} \delta\left(t - \frac{\|\vec{x} - \vec{x}'\|}{c} - t'\right) \theta(t > t'), \end{aligned} \quad (13.S28)$$

where we also have used $\delta(ax) = |a|^{-1}\delta(x)$, the potential becomes:

$$\begin{aligned} \Phi(\vec{x}, t) &= \frac{1}{4\pi\epsilon_0} \int d\vec{x}' dt' q \delta^3(\vec{x}' - \vec{v}t') \frac{2}{c} \delta\left((t - t')^2 - \frac{\|\vec{x} - \vec{x}'\|^2}{c^2}\right) \theta(t > t') \\ &= \frac{q}{2\pi\epsilon_0 c} \int dt' \delta\left((t - t')^2 - \frac{\|\vec{x} - \vec{v}t'\|^2}{c^2}\right) \theta(t > t'). \end{aligned} \quad (13.S29)$$

The roots of the argument of the delta function are given by

$$t'^2 - 2tt' + t^2 - \frac{(z - vt')^2}{c^2} - \frac{x_\perp^2}{c^2} = 0, \quad (13.S30)$$

where $x_{\perp}$ is the component of $\vec{x}$ perpendicular to $\vec{v}$. This equation leads to

$$\left(1 - \frac{v^2}{c^2}\right)t'^2 - 2t'\left(t - \frac{zv}{c^2}\right) + t^2 - \frac{z^2 + x_{\perp}^2}{c^2} = 0. \quad (13.S31)$$

We introduce

$$\tau^2 = t^2 - \frac{z^2 + x_{\perp}^2}{c^2}, \quad (13.S32)$$

and define the so-called boosted variables

$$z_b = \frac{z - vt}{\sqrt{1 - v^2/c^2}}, \quad t_b = \frac{t - vz/c^2}{\sqrt{1 - v^2/c^2}}. \quad (13.S33)$$

One can show that the τ^2 has the same form in boosted variables. Thus we arrive at the following equation,

$$\frac{t'^2}{\gamma^2} - \frac{2t'}{\gamma} t_b + \tau^2 = 0. \quad (13.S34)$$

The solutions for this quadratic equation are:

$$t_{\pm} = \frac{2t_b/\gamma \pm 2r_b/\gamma c}{2/\gamma^2} = \gamma\left(t_b \pm \frac{r_b}{c}\right), \quad (13.S35)$$

where $r_b^2 = z_b^2 + x_{\perp}^2$ is the boosted distance. We substitute this result into the integrand of (13.S29)

$$\begin{aligned} & \delta\left((t - t')^2 - \frac{\|\vec{x} - \vec{x}'\|^2}{c^2}\right) \theta(t > t') \\ &= \delta\left(\frac{(t' - t_+)(t' - t_-)}{\gamma^2}\right) \theta(t > t') \\ &= \frac{\gamma^2 \delta(t' - t_-)}{|2t' - (t_+ + t_-)|_{t'=t_-}} \theta(t > t') \\ &= \frac{\gamma^2 \delta(t' - t_-)}{|t_+ - t_-|} \theta(t > t') \\ &= \frac{c\gamma^2 \delta(t' - t_-)}{2r_b\gamma} \theta(t > t'). \end{aligned} \quad (13.S36)$$

And, finally, we get for the potential

$$\begin{aligned} \Phi(\vec{x}, t) &= \frac{q}{2\pi\epsilon_0 c} \int_{-\infty}^{\infty} dt' \delta(t' - t_-) \theta(t > t') \frac{c\gamma}{2r_b} \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{\sqrt{(z - vt)^2 + (1 - v^2/c^2)(x^2 + y^2)}}. \end{aligned} \quad (13.S37)$$

b) Calculate the potential first in the rest frame of the charge. Then transform it back into the frame of the observer.

Solution: We denote all quantities in the rest frame of the particle with primes. In the rest frame the current is zero. The vector potential is therefore zero and the electric potential takes the simple form

$$\Phi'(\vec{x}', t') = \frac{1}{4\pi\epsilon_0} \frac{q}{\|\vec{x}'\|}. \quad (13.S38)$$

Before we transform into the frame of the observer we need to write the potential and the coordinates in a covariant way

$$A'^{\mu} = \begin{pmatrix} \Phi'/c \\ \vec{A}' \end{pmatrix} = \begin{pmatrix} \Phi'/c \\ 0 \end{pmatrix}, \quad x'^{\mu} = \begin{pmatrix} ct' \\ \vec{x}' \end{pmatrix}. \quad (13.S39)$$

Now we need the boost to the observer frame, i.e. into a frame that moves with velocity $-v\vec{e}_z$

$$A(-v\vec{e}_z)^{-1} = \begin{pmatrix} \gamma & 0 & 0 & \beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \beta\gamma & 0 & 0 & \gamma \end{pmatrix}. \quad (13.S40)$$

The potential is then

$$\begin{pmatrix} \Phi/c \\ \vec{A} \end{pmatrix} = A = A^{-1} A' = \begin{pmatrix} \gamma\Phi'/c \\ 0 \\ 0 \\ \beta\gamma\Phi'/c \end{pmatrix}. \quad (13.S41)$$

So we find

$$\Phi(x', t') = \gamma\Phi'(x', t'), \quad \vec{A}(x', t') = v\Phi'(x', t')\vec{e}_z. \quad (13.S42)$$

While we have transformed the fields themselves we still express them in terms of rest frame coordinates. Therefore we also need to transform the coordinates. However notice that we want to express the rest frame coordinates in terms of observer coordinates, so we need to boost in the opposite direction

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = x' = Ax = \begin{pmatrix} \gamma ct - \beta\gamma z \\ x \\ y \\ -\beta\gamma ct + \gamma z \end{pmatrix}. \quad (13.S43)$$

Inserting this into our formula we find

$$\Phi(\vec{x}, t) = \gamma\Phi'(\vec{x}', t') = \frac{1}{4\pi\epsilon_0} \frac{\gamma q}{\sqrt{x^2 + y^2 + \gamma^2(z - vt)^2}} = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(1 - \beta^2)(x^2 + y^2) + (z - vt)^2}}. \quad (13.S44)$$

Also we get the vector potential at no cost

$$\vec{A}(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \frac{qv}{\sqrt{(1 - \beta^2)(x^2 + y^2) + (z - vt)^2}} \vec{e}_z. \quad (13.S45)$$

13.3. Mirage

On a hot day one can observe reflections above hot surfaces, e.g. above a street. These reflections are a consequence of the non-constant index of refraction of the air $n(z)$ which will be a function of the height z above the ground. To model this effect we assume that n is constantly $n_0 > 1$ above a certain height z_0 and reduces quadratically in the hot layer below (with reasonably small $a > 0$)

$$n(z) = \begin{cases} n_0 & \text{for } z \geq z_0, \\ n_0 - a(z - z_0)^2 & \text{for } 0 < z < z_0. \end{cases} \quad (13.8)$$

Find the path γ that minimises the optical length

$$S = \int_{x_1}^{x_2} \|\mathrm{d}\gamma\| n(\gamma). \quad (13.9)$$

between two endpoints $\vec{x}_1, \vec{x}_2$ with $z_1, z_2 \geq z_0$ that are widely separated horizontally, $|x_1 - x_2| \gg z_0$.

Hint: Parametrise the path as a function of x and assume that $\partial z/\partial x$ is small enough that you can neglect it.

Solution: We parametrise the path by the x -coordinate

$$S = \int_{x_1}^{x_2} \mathrm{d}x n(\gamma(x)) \|\gamma'(x)\|. \quad (13.S46)$$

The Euler-Lagrange equations are

$$0 = \frac{\partial n}{\partial \gamma^i} \|\gamma'\| - \frac{\mathrm{d}}{\mathrm{d}x} \left(n \frac{\partial \|\gamma'\|}{\partial \gamma^i} \right) = \frac{\partial n}{\partial \gamma^i} \|\gamma'\| - \frac{\mathrm{d}}{\mathrm{d}x} \left(n \frac{\gamma^i}{\|\gamma'\|} \right). \quad (13.S47)$$

W.l.o.g. we can assume that $\gamma(z) = (x, 0, z(x))$. The Euler-Lagrange equations for the x - and z -component are then

$$\begin{aligned} \frac{d}{dx} \frac{n(z(x))}{\sqrt{1+z'^2}} &= 0, \\ \frac{\partial n}{\partial z} \sqrt{1+z'^2} - \frac{d}{dx} \left(n(z(x)) \frac{z'}{\sqrt{1+z'^2}} \right) &= 0. \end{aligned} \quad (13.S48)$$

The former equation leads to $n(z(x))/\sqrt{1+z'^2} =: C = \text{const.}$ The second equation then simplifies to

$$\frac{\partial n}{\partial z} \sqrt{1+z'^2} - Cz'' = 0. \quad (13.S49)$$

Inserting the explicit form of $n(z)$ we find

$$\begin{aligned} Cz'' &= 0, \quad z > z_0, \\ -2a(z - z_0)\sqrt{1+z'^2} - Cz'' &= 0, \quad z \leq z_0. \end{aligned} \quad (13.S50)$$

Above z_0 the solution is a straight line, below we rewrite

$$-2 \frac{a}{C} (z - z_0) \sqrt{1+z'^2} = z''. \quad (13.S51)$$

We assume now that the angle of the light ray to the ground, i.e. z' , is small and can be neglected

$$-2 \frac{a}{C} (z - z_0) = (z - z_0)'' \quad \Rightarrow \quad z(x) = z_0 + A \cos\left(\sqrt{\frac{2a}{C}} x\right) + B \sin\left(\sqrt{\frac{2a}{C}} x\right). \quad (13.S52)$$

To find an explicit solution we say that the light ray enters the hot layer at $x = 0$, i.e. $z(0) = z_0$, and define its angle $z'(0) = m \lesssim 0$.

$$\begin{aligned} z_0 = z(0) &= z_0 + A \cos\left(\sqrt{\frac{2a}{C}} x\right), \quad \Rightarrow \quad A = 0, \\ m = z'(0) &= \sqrt{\frac{2a}{C}} B, \quad \Rightarrow \quad B = m \sqrt{\frac{C}{2a}}. \end{aligned} \quad (13.S53)$$

Finally, remembering the definition of the constant C we find

$$C = \frac{n(z_0)}{\sqrt{1+m^2}} = \frac{n_0}{\sqrt{1+m^2}} \approx n_0. \quad (13.S54)$$

The final solution is thus

$$z(x) = z_0 + m \sqrt{\frac{n_0}{2a}} \sin\left(\sqrt{\frac{2a}{n_0}} x\right), \quad (13.S55)$$

which is valid of course only for half a period as long as $z < z_0$. In total the path looks like this:



$$(13.S56)$$

14.1. Rectangular waveguide

Consider a waveguide extended infinitely along the z -axis with a rectangular basis $0 < x < d_x$ and $0 < y < d_y$. Its surfaces are ideal conductors. Due to the geometry of the problem, you can make the following ansatz for propagating electromagnetic waves,

$$\begin{aligned}\vec{E}(x, y, z, t) &= \text{Re}(\vec{E}_0(x, y) e^{i(kz - \omega t)}), \\ \vec{B}(x, y, z, t) &= \text{Re}(\vec{B}_0(x, y) e^{i(kz - \omega t)}).\end{aligned}\quad (14.1)$$

- a) The 3D-vectors split into 2D-vectors (here: x - and y -components) and scalars (z -component). From the Maxwell equations, derive equations for the x - and y -components of $\vec{E}_0$ and $\vec{B}_0$ in terms of their z -components, and show that the following equations hold for the z -components,

$$\begin{aligned}\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] E_z &= 0, \\ \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] B_z &= 0.\end{aligned}\quad (14.2)$$

Solution: Let us start with the general Maxwell equations in vacuum

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad (14.S1)$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}, \quad (14.S2)$$

$$\vec{\nabla} \cdot \vec{B} = 0, \quad (14.S3)$$

$$\vec{\nabla} \times \vec{B} = \epsilon_0 \mu_0 \partial_t \vec{E} + \mu_0 \vec{j}. \quad (14.S4)$$

Inside the cavity of the conductor the equations (14.S1) and (14.S4) reduce to

$$\vec{\nabla} \cdot \vec{E} = 0, \quad (14.S5)$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \partial_t \vec{E}. \quad (14.S6)$$

We now use the ansatz $\vec{E}(x, y, z, t) = \vec{E}_0(x, y) e^{i(kz - \omega t)}$ and $\vec{B}(x, y, z, t) = \vec{B}_0(x, y) e^{i(kz - \omega t)}$ where we write the amplitudes in components

$$\begin{aligned}\vec{E}_0 &= E_x(x, y) \vec{e}_x + E_y(x, y) \vec{e}_y + E_z(x, y) \vec{e}_z, \\ \vec{B}_0 &= B_x(x, y) \vec{e}_x + B_y(x, y) \vec{e}_y + B_z(x, y) \vec{e}_z.\end{aligned}\quad (14.S7)$$

Plugging this ansatz into equation (14.S2) we obtain

$$\frac{\partial}{\partial y} E_x - ik E_y = i\omega B_z, \quad (14.S8)$$

$$ik E_x - \frac{\partial}{\partial x} E_z = i\omega B_y, \quad (14.S9)$$

$$\frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x = i\omega B_z, \quad (14.S10)$$

and equation (14.S6) becomes

$$\frac{\partial}{\partial y} B_z - ik B_y = -\frac{i\omega}{c^2} E_x, \quad (14.S11)$$

$$ik B_x - \frac{\partial}{\partial x} B_z = -\frac{i\omega}{c^2} E_y, \quad (14.S12)$$

$$\frac{\partial}{\partial x} B_y - \frac{\partial}{\partial y} B_x = -\frac{i\omega}{c^2} E_z. \quad (14.S13)$$

Adding $(k/\omega)(14.S8) + (14.S12)$ leads to

$$\begin{aligned} 0 &= \frac{k}{\omega} \frac{\partial}{\partial y} E_z - i \frac{k^2}{\omega} E_y - \frac{\partial}{\partial x} B_z + \frac{i\omega}{c^2} E_y \\ \Rightarrow E_y &= \frac{i}{\omega^2/c^2 - k^2} \left(k \frac{\partial}{\partial y} E_z - \omega \frac{\partial}{\partial x} B_z \right), \end{aligned} \quad (14.S14)$$

and adding $(\omega/kc^2)(14.S8) + (14.S12)$ leads to

$$\begin{aligned} 0 &= \frac{\omega}{kc^2} \frac{\partial}{\partial y} E_z - i \frac{\omega^2}{kc^2} B_x - \frac{\partial}{\partial x} B_z + ikB_x \\ \Rightarrow B_x &= \frac{i}{\omega^2/c^2 - k^2} \left(k \frac{\partial}{\partial x} B_z - \frac{\omega}{c^2} \frac{\partial}{\partial y} E_z \right). \end{aligned} \quad (14.S15)$$

Analogously $-(k/\omega)(14.S9) + (14.S11)$ leads to

$$\begin{aligned} 0 &= -i \frac{k^2}{\omega} E_x + \frac{k}{\omega} \frac{\partial}{\partial x} E_z + \frac{\partial}{\partial y} B_z + \frac{i\omega}{c^2} E_x \\ \Rightarrow E_x &= \frac{i}{\omega^2/c^2 - k^2} \left(\omega \frac{\partial}{\partial y} B_z + k \frac{\partial}{\partial x} E_z \right), \end{aligned} \quad (14.S16)$$

and $-(\omega/kc^2)(14.S9) + (14.S11)$ leads to

$$\begin{aligned} 0 &= \frac{\omega}{kc^2} \frac{\partial}{\partial x} E_z + i \frac{\omega^2}{kc^2} B_y + \frac{\partial}{\partial y} B_z - ikB_y \\ \Rightarrow B_y &= \frac{i}{\omega^2/c^2 - k^2} \left(k \frac{\partial}{\partial y} B_z + \frac{\omega}{c^2} \frac{\partial}{\partial x} E_z \right). \end{aligned} \quad (14.S17)$$

We succeeded in writing the x - and y -components of the fields in terms of the z -components. The so far unused equations (14.S3) and (14.S5) yield the differential equations for the z -components of the fields:

$$\begin{aligned} \vec{\partial} \cdot \vec{E} &= \frac{i}{\omega^2/c^2 - k^2} \left(\omega \frac{\partial^2}{\partial x \partial y} B_z + k \frac{\partial^2}{\partial x^2} E_z + k \frac{\partial^2}{\partial y^2} E_z - \omega \frac{\partial^2}{\partial y \partial x} B_z \right) + ikE_z = 0, \\ &\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] E_z = 0, \end{aligned} \quad (14.S18)$$

$$\begin{aligned} \vec{\partial} \cdot \vec{B} &= \frac{i}{\omega^2/c^2 - k^2} \left(k \frac{\partial^2}{\partial x^2} B_z + k \frac{\partial^2}{\partial y^2} B_z \right) + ikB_z = 0, \\ &\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] B_z = 0. \end{aligned} \quad (14.S19)$$

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- b)** Express the boundary conditions $E_{\parallel} = B_{\perp} = 0$ as conditions for the z -components of the fields.
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Solution: From $E_{\parallel} = 0$ we immediately obtain $E_z = 0$. For the B -field let us consider the $x = 0$ surface. The boundary condition is $B_x(0, y) = 0$. Equation (14.S12) together with $E_y(0, y) = 0$ leads then to $\partial_x B_z(0, y) = 0$. A similar argument for the $y = 0$ surface yields $\partial_y B_z(x, 0) = 0$. In general, the boundary condition can be written as $\vec{n} \cdot \vec{\partial} B_z = 0$.

- c)** Determine the solutions for so-called transverse magnetic waves (TM-waves), which satisfy $B_z = 0$.
-

Solution: We have to determine E_z . The general solutions of the differential equation

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \lambda^2 \right] f(x, y) = 0 \quad (14.S20)$$

are given by the ansatz $f_\lambda(x, y) = e^{i(\alpha x + \beta y)}$ with $\alpha^2 + \beta^2 = \lambda^2$. By taking linear combinations of such solutions, we can write a general solution also as

$$f_\lambda(x, y) = (A \sin(\alpha x) + B \cos(\alpha x))(C \sin(\beta y) + D \cos(\beta y)), \quad (14.S21)$$

with some undetermined coefficients A, B, C, D , which is more suitable to adjust the boundary conditions. For the field component E_z we have $\lambda^2 = \omega^2/c^2 - k^2$ and the boundary conditions

$$E_z(0, y) = E_z(d_x, y) = E_z(x, 0) = E_z(x, d_y) = 0. \quad (14.S22)$$

The solution then reduces to

$$E_z(x, y) = E_0 \sin\left(\frac{m\pi x}{d_x}\right) \sin\left(\frac{n\pi y}{d_y}\right), \quad m, n > 0, \quad (14.S23)$$

and the dispersion relation becomes

$$k(\omega) = \sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{d_x}\right)^2 - \left(\frac{n\pi}{d_y}\right)^2}. \quad (14.S24)$$

For a given angular frequency ω there is only a finite number of excitations that lead to an undamped propagating wave solution (k real).

- d) Show that no transverse electromagnetic (TEM) waves (i.e. waves with $E_z = B_z = 0$) exist in an ideal waveguide.

Hint: Use Gauß' theorem and Faraday's law as well as the boundary condition for $E_{\parallel}$ to show that there are no TEM-waves in this waveguide.

Solution: With $E_z = 0$ we obtain

$$\vec{\partial} \cdot \vec{E} = \frac{\partial}{\partial x} E_x + \frac{\partial}{\partial y} E_y = 0, \quad (14.S25)$$

and with $B_z = 0$ and (14.S10) it follows

$$\frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x = 0. \quad (14.S26)$$

Since the E -field has zero curl and zero divergence in this case, we can write it as the gradient of a scalar potential which satisfies the Laplace equation. The boundary condition $\vec{E}_{\parallel} = 0$ implies that the boundary is an equipotential surface. The Laplace equation, however, does not permit local extrema and therefore the potential has to be constant. Consequently the electric field vanishes, which proves that there are no TEM-waves.

14.2. Coaxial waveguide

An electromagnetic wave propagates along the z -direction between two coaxial, cylindrical conductors with radii $r_2 > r_1$ centred at the z -axis.

- a) Show that it is possible to have a TEM-mode by finding such a solution explicitly.

Solution: We make the ansatz

$$\begin{aligned}\vec{E}(x, y, z, t) &= \vec{E}(x, y) e^{i(kz - \omega t)}, \\ \vec{B}(x, y, z, t) &= \vec{B}(x, y) e^{i(kz - \omega t)},\end{aligned}\quad (14.S27)$$

where $k = \omega/c$. We split vectors $\vec{V} = \vec{V}_\perp + V_z \vec{e}_z$ into a component perpendicular to, and along the z -direction. We are looking for TEM-waves, i.e. $E_z = 0$ and $B_z = 0$. The Maxwell equation

$$\partial_t \vec{B} + \vec{\partial} \times \vec{E} = 0 \quad (14.S28)$$

splits into the two equations

$$\begin{aligned}-i\omega t \vec{B}_\perp + ik \vec{e}_z \times \vec{E}_\perp &= 0, \\ \vec{\partial}_\perp \times \vec{E}_\perp &= 0.\end{aligned}\quad (14.S29)$$

The latter one implies that there exists a potential $\phi(x, y)$ such that

$$\vec{E}_\perp = -\vec{\partial}_\perp \phi. \quad (14.S30)$$

There are no charges between the conductors, thus

$$\vec{\partial} \cdot \vec{E} = \vec{\partial}_\perp \cdot \vec{E}_\perp = 0 \quad \implies \quad \Delta_\perp \phi = 0. \quad (14.S31)$$

Due to the symmetry of the problems the potential can only be a function of the radius $\phi(x, y) = \phi(r)$. The Laplace equation therefore simplifies to

$$\frac{1}{r} \partial_r (r \partial_r \phi(r)) = 0, \quad (14.S32)$$

which is solved by

$$r \partial_r \phi(r) = a \quad \implies \quad \phi = a \log r + b \quad (14.S33)$$

for constants a and b . The electric field is then

$$\vec{E}_\perp = -\frac{a}{r} \vec{e}_r. \quad (14.S34)$$

The magnetic field follows from (14.S29)

$$\vec{B}_\perp = \frac{k}{\omega} \vec{e}_z \times \vec{E}_\perp = -\frac{a}{cr} \vec{e}_\varphi. \quad (14.S35)$$

b) Is the frequency bounded for this mode? What is the dispersion relation?

Solution: The frequency of the TEM-mode can be arbitrary as seen in part a). The dispersion relation is simply $\omega(k) = ck$.

c) Calculate the average transport of power along the cylinder axis.

Solution: The Poynting vector is given by

$$\langle \vec{S} \rangle = \frac{1}{2\mu_0} \text{Re}(\vec{E} \times \vec{B}^*) = \frac{1}{2\mu_0 c} \frac{a^2}{r^2} \vec{e}_z. \quad (14.S36)$$

To get the total amount of power transported we need to integrate over a section of the coaxial wave guide

$$P = \int_0^{2\pi} d\varphi \int_{r_1}^{r_2} dr r \langle S \rangle = \sqrt{\frac{\epsilon_0}{\mu_0}} \pi a^2 \log \frac{r_2}{r_1}. \quad (14.S37)$$